

The Equity Constraint Channel of Monetary Policy*

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Abstract

We show that financially constrained firms that rely on equity at the margin endure larger stock-price declines and deeper cuts in capital expenditures, R&D, and patenting after contractionary monetary policy shocks. We explain these findings with a model where equity and debt markets are partially segmented and contractionary shocks raise the cost of equity more than debt. Price-sensitive investors withdraw from the equity market after tightening, and amplification concentrates in constrained firms with greater ex-ante ownership by price-sensitive investors. Our findings suggest that equity frictions are central to understanding the transmission of monetary policy to the corporate sector.

KEYWORDS: Equity financing, financial constraint, investment, investor demand, monetary policy.

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1. Introduction

The role of financial constraints in the transmission of monetary policy to the corporate sector has been extensively researched in economics and finance.¹ To address the unobservable nature of financial constraints, most of this literature relies on proxies such as leverage, collateral, credit risk, and debt maturity, which largely revolve around debt financing. Influenced by frameworks in which negative shocks tighten financial conditions (Bernanke et al., 1999), existing studies have emphasized the “debt channel” with mixed results. For instance, Jungherr et al. (2024) find that firms with a higher maturing bond share reduce investment by more after contractionary shocks, while Ottonello and Winberry (2020) show that firms with low debt burdens and high distance to default are the most responsive.

The starting point for our paper is that more financially constrained firms often turn to equity financing rather than debt to access liquidity.² Thus, the focus on debt financing frictions may be the reason for the mixed results in the literature on monetary policy and constraints. To measure equity constraints, we leverage a text-based measure of financial constraint that separates whether the firm primarily uses equity- or debt-financing sources (Hoberg and Maksimovic, 2015; Linn and Weagley, 2023). A firm is thereby defined as equity-focused constrained if it mentions in its 10-K filing that investment is delayed due to liquidity issues and mainly relies on equity financing. Debt-focused constrained firms are defined similarly, for firms that rely on debt financing.

We provide novel evidence that equity constraints are the primary amplifier of monetary policy transmission to firms. Following contractionary shocks, equity-focused constrained firms experience substantially larger stock price declines, implement deeper and more persistent cuts in capital expenditures and R&D, and file fewer patents. These firms decrease financing almost exclusively along the equity margin, by sharply cutting equity and SEO issuance while barely adjusting debt issuance. In contrast, we find that debt-focused constraints do not persistently magnify the impact of monetary policy shocks.

¹Examples include Gertler and Gilchrist (1994); Jeenas (2024); Ottonello and Winberry (2020); Cloyne et al. (2023); Jungherr et al. (2024); Perez-Orive et al. (2024).

²The pecking order theory and its extensive empirical foundations (Myers and Majluf, 1984; Leary and Roberts, 2010) suggest that equity finance is the least preferred form of finance for firms.

To understand these results and derive new insights, we develop a parsimonious model in which equity and corporate debt markets are partially segmented, following [Kojen and Yogo \(2019\)](#) and [Vayanos and Vila \(2021\)](#): each market is served by a price-sensitive marginal investor alongside a clientele of non-price-sensitive institutions. We assume that the fraction of price-sensitive investors is higher in the equity market, which is confirmed by Fed Flow of Funds sectoral holdings data: price-sensitive investors on average hold 53% of corporate equity but only 36% of corporate debt in our sample (Figure 1). In the model, contractionary monetary policy reduces the risk-bearing capacity of the price-sensitive investor. Because equity carries greater payoff risk than debt, and the price-sensitive investor’s equity exposure is larger, the premium wedge of equity financing rises more in response to a contractionary shock than that of debt financing. The investment of equity-focused constrained firms is therefore more responsive to the shock because their cost of capital is more sensitive to monetary policy shocks.

The model delivers three testable predictions. First, at the market level, a contractionary shock induces the price-sensitive investor to reduce equity holdings by more than corporate debt holdings. Second, at the firm level, the withdrawal of price-sensitive investors mostly affects firms for which equity is the marginal source of financing, such as equity-constrained firms. Third, equity-focused constrained firms cut investment by more than other firms, and this amplification is concentrated in firms whose investor base is tilted toward price-sensitive investors.

To test these predictions, we combine aggregate sectoral holdings from the Flow of Funds, institutional ownership from 13F filings, daily stock returns from CRSP, firm policies from COMPUSTAT and SDC Platinum, and patent filings from [Kogan et al. \(2017\)](#), with the monetary policy shocks of [Jarociński and Karadi \(2020\)](#), which separate the “pure” policy shock from the central bank information effect.³ We then use instrumental-variable local projections, an identification strategy that combines instrumental variables (IV) with a time series setting using local projections ([Jordà, 2005](#)), to empirically estimate how a monetary policy shock impacts firm outcomes.

³Recent studies emphasize that the information effect can bias estimates of monetary transmission; see [Nakamura and Steinsson \(2018\)](#) and [Miranda-Agrippino and Ricco \(2021\)](#). Our results are robust to directly controlling for information shocks and to the alternative shocks of [Bauer and Swanson \(2023\)](#).

We start our empirical analysis by testing the model’s asset-market prediction. We show that, following a contractionary shock, the share of price-sensitive investors in the equity market significantly contracts by 0.4 percentage points per 25 basis points at its peak. The response of the share of price-sensitive in the corporate debt market is statistically insignificant at all horizons, indicating that the marginal investors withdraw relatively more from equity. These findings provide direct, market-level evidence for the first link of the mechanism: monetary policy operates on the demand for corporate equity.

We next investigate the model’s second prediction at the firm level. [Choi et al. \(2025\)](#) show that fluctuations in investor demand can alter firms’ marginal cost of capital and distort capital allocation, highlighting the importance of investor composition for firms’ financing conditions and real decisions. Building on this insight, we ask whether monetary policy shocks shift firms’ ownership composition, and whether these shifts disproportionately affect equity-focused constrained firms. Although such firms are characterized by long-term growth potential, their sensitivity to macroeconomic shocks and limited internal liquidity make them attractive to short-horizon, performance-sensitive investors during periods of high risk-taking.

We identify the price-sensitive marginal investors empirically as “transient” institutions in the sense of [Bushee \(1998, 2001\)](#) and [Borochin and Yang \(2017\)](#): institutions with short investment horizons, reflected in high portfolio turnover and diversified holdings. We show that equity-focused constrained firms experience significantly greater outflows from transient investors following contractionary shocks. This investor flight is the firm-level evidence of the aggregate demand contraction: it loads precisely onto the firms that depend on equity at the margin, offering a demand-based amplification mechanism consistent with our model and the frameworks of [Kojen and Yogo \(2019\)](#) and [Choi et al. \(2025\)](#).

We complement our analysis by tracing out the consequences for stock prices. Following a 25 bps contractionary monetary policy shock, equity-focused constrained firms experience an average return that is 0.745% lower than that for unconstrained firms on the day of the FOMC announcement, while debt-focused constrained firms have an average realized return that is 0.458% lower than that for unconstrained firms. The heterogeneous impact is persistent. The cumulative five-day stock price response is substantially larger for firms with equity-focused constraints (2%) than for those with

debt-focused constraints (0.64%), and the difference between the two estimates is statistically significant. We also show that, consistent with [Jarociński and Karadi \(2020\)](#), the “information effect” has the opposite but transitory impact on the heterogeneous stock price response, vis-à-vis the “pure” monetary policy shocks. This decrease in returns is consistent with the model’s mechanism, whereby the cost of capital is more sensitive to shocks for equity-constrained firms.

We uncover significant real effects of monetary policy on firms. In response to a 25 bps increase in the 1-year Treasury rate, CAPX and R&D significantly drop by 5.2% and 1.3% after 16 quarters, respectively. Equity-focused constraints again significantly amplify these effects of monetary policy shocks. A two-standard-deviation increase in the equity-focused constraint measure amplifies a firm’s average investment (CAPX) response by 10% and it magnifies the R&D response by 34%. Debt-focused constraints amplify the effect of monetary policy shocks to a much smaller extent. For example, a two-standard-deviation increase in the debt-focused constraint measure significantly increases a firm’s average investment (CAPX) response to a 25 bps higher 1-year rate by 3.6% after four quarters, with no significant effect thereafter. The difference between the equity- and debt-focused amplification effects is again statistically significant for most horizons, confirming that the equity constraint channel of monetary policy is quantitatively significant.

The decrease in R&D expenses in response to contractionary shocks translates into lower innovation output. In response to a 25 bps increase in the 1-year Treasury rate, the number of patent filings drops by 1.69% on average after 17 quarters. A two-standard-deviation increase in the equity-focused constraint measure amplifies the responses by 31.46 bps, corresponding to approximately 18.6% of the average response. We do not find such an amplification effect for debt-focused constrained firms.

We provide consistent evidence that the contraction in investment runs through the equity margin by studying financing responses. We show that equity-focused constrained firms cut equity issuance by 6.9% and SEO issuance by 10% relative to their means, but debt issuance by only 0.9%. They are also more reluctant than unconstrained firms to draw down cash reserves, consistent with a rising shadow value of internal funds when external equity becomes more expensive.

We conclude our analysis by providing direct evidence for the asset demand mechanism. If the amplification operates through the withdrawal of price-sensitive investors, it should be concentrated in equity-focused constrained firms whose investor base is ex-ante tilted toward these investors. We sort firms by predetermined transient institutional ownership, and find that the amplification of investment and R&D responses is present only among high-transient-ownership firms and absent otherwise, even though the two groups are comparable on financial fragility and cash-flow duration, among other observable measures. This suggests that the equity constraint channel operates mainly where the price-sensitive marginal investor is present.

Overall, our findings suggest a novel channel through which monetary policy shapes the corporate sector. Monetary policy moves the risk-bearing capacity of price-sensitive investors; their withdrawal raises the cost of equity capital; and the firms that depend on equity at the margin bear the real consequences. To the best of our knowledge, we are the first to document a constraint channel of monetary policy operating through the composition and behavior of the firm’s investor base. Our evidence implies that changes in equity financing conditions, not only debt conditions, are central to the transmission of monetary policy to investment and innovation.

The remainder of the paper proceeds as follows. Section 2 discusses the related literature. Section 3 presents the conceptual framework. Section 4 describes the data and the financial constraint measures. Section 5 presents the empirical strategy and main results, from the aggregate evidence to the firm-level transmission. Section 6 concludes.

2. Related Literature

Our paper relates to several strands of literature. The first studies the investment channel of monetary policy and the role of financial frictions. Building on the financial accelerator (Bernanke et al., 1999), a large empirical literature measures financial constraints through balance-sheet proxies such as cash flows (Fazzari et al., 1988; Oliner and Rudebusch, 1992), size (Gertler and Gilchrist, 1994), bank debt (Ippolito et al., 2018), leverage (Lakdawala et al., 2026; Ottonello and Winberry, 2020), liquidity (Jeenas, 2024), covenants (Perez-Orive et al., 2024), age and dividend status (Durante et al., 2022; Cloyne

et al., 2023), interest burdens (Cao et al., 2023), and debt maturity (Deng and Fang, 2022; Jungherr et al., 2024; Oliveira et al., 2024).

These papers mostly focus on one or two proxies that are potentially correlated with the unobservable financial constraint.⁴ These characteristics are also highly related to the debt channel. In contrast, we use text-based measures that separately identify equity- and debt-focused constraints (Hoberg and Maksimovic, 2015; Linn and Weagley, 2023) to provide novel evidence that equity-focused constraints are quantitatively the more important amplifier of monetary policy for investment, innovation, and stock prices.⁵

Our paper also draws on a literature that investigates the response of the stock market to monetary policy shocks. Since the seminal work of Bernanke and Kuttner (2005), recent theoretical work has provided a rationale for the negative stock market response to an unexpected increase in the federal funds rate around FOMC announcements (Bianchi et al., 2022; Pflueger and Rinaldi, 2022; Kekre and Lenel, 2022). Cross-sectional evidence on constrained firms is scarce and conflicting: Lamont et al. (2001) find no differential response, Ozdagli (2017) finds constrained firms respond less, and Chava and Hsu (2020) find they respond more with a delay. This discrepancy might stem not only from their different empirical design and sample, but also from the measures of financial constraint they use.⁶ Borrowing from recent advancements in the measurement of financial constraint and information-effect-purged shocks (Jarociński and Karadi, 2020; Miranda-Agrippino and Ricco, 2021), we show that both equity- and debt-focused constrained firms experience disproportionately lower returns following contractionary monetary policy shocks, and that impacts are far higher for equity-focused constrained firms.

Our paper is related to recent work on the importance of equity financing in the aftermath of monetary policy shocks. Jeenas and Lagos (2024) exploit cross-sectional variation in stock turnover to instrument policy-induced changes in Tobin’s q , and show that higher valuations cause equity issuance and investment, particularly among firms with low balance-sheet liquidity. Beyhaghi et al. (2026) find that private firms reduce

⁴The findings in this literature are generally consistent with theories that predict stronger reactions of financially constrained firms to monetary policy (Bernanke and Gertler, 1989). However, the use of different proxies inevitably makes cross-validation difficult.

⁵Hsu et al. (2023) show that the Fed information effect matters for investment sensitivity. Our estimates use the “pure” monetary policy shocks of Jarociński and Karadi (2020), and are robust to directly controlling for information shocks (Internet Appendix).

⁶See, for example, Farre-Mensa and Ljungqvist (2016).

debt, equity, and real assets. In contrast, public firms also reduce debt, but simultaneously raise equity, allowing them to maintain real assets. [Abreu et al. \(2026\)](#) show how monetary policy affects venture capital investments across early and late-stage start-ups.

We contribute to this literature by showing that public firms that report equity-focused constraints are highly sensitive to contractionary shocks because they are unable to offset the impact by issuing equity. We provide evidence that this heightened sensitivity is driven by investor demand. In particular, we are the first to document a constraint channel operating through the composition and behavior of the firm’s investor base rather than through firms’ balance sheets.

We also contribute to the literature linking monetary policy, intangibles, and innovation. [Döttling and Ratnovski \(2023\)](#) and [Caggese and Pérez-Orive \(2022\)](#) show that intangible-intensive firms respond less to monetary policy; [Ma and Zimmermann \(2023\)](#) document aggregate declines in R&D and patenting after tightening. [Datsenko and Holm \(2026\)](#) show that the R&D response to contractionary policy is concentrated in high-growth firms in Norway. We find consistent average results and show that the equity constraint channel, operating through investor demand, plays a critical role in the cross-section, especially for R&D expenses and the number of patent filings.

3. Conceptual Framework

We introduce a parsimonious two-period model that relates firm investment to equity and bond premia in the context of monetary policy. We show that in response to a contractionary shock, the cost of capital increases more for firms that rely on equity financing, leading to a larger decline in their investment.

The model not only delivers testable predictions about heterogeneous responses of firm investment to monetary policy shocks, but also provides further insights into how monetary policy changes investor demand. The structure is deliberately stylized to focus on the core mechanism. We aim to propose a simple framework that is consistent with the price impact of monetary policy, rather than incorporate all potential mechanisms of monetary policy transmission. To validate our proposed mechanism, we take the predictions from the model to data and test them in [Section 5](#).

3.1 Setup

Environment and timing. The model has two periods. At $t = 0$, firms choose investment and financing. Between $t = 0$ and $t = 1$, output is produced. At $t = 1$, the state of the world realizes, output is determined, debt holders are paid, and equity holders receive the residual. The monetary policy shock ϵ^{MP} is realized and observed at $t = 0$, before firms make their decisions. Following the shock, investors rebalance their portfolios and the equilibrium required returns on the firm's financing sources adjust. Firms then choose investment and financing taking these equilibrium prices as given.

Throughout, the firm's unlevered cost of capital r consists of the risk-free rate and the risk premium on the firm's underlying assets. s_E and s_B are financing wedges above the benchmark r , arising from the segmented structure of equity and corporate debt markets described below. Firms that finance the investment internally pay the unlevered cost of capital and no wedge; firms that access external markets additionally pay the wedge demanded by the marginal investor in the corresponding market. From the perspective of the investors, s_E and s_B are the additional compensation required for absorbing equity and corporate debt supply, respectively; from the firm's perspective, they are financing wedges above the unlevered benchmark cost of capital.⁷

Risk structure. The conditional covariance matrix of the two risky claims is

$$\Sigma \equiv \begin{pmatrix} \sigma_E^2 & \sigma_{EB} \\ \sigma_{EB} & \sigma_B^2 \end{pmatrix}, \quad \Sigma \succ 0. \quad (1)$$

Corporate debt is exposed to aggregate default and recovery risk, so $\sigma_B^2 > 0$. Because equity is the residual claim whereas debt is senior, we assume that aggregate equity carries greater payoff risk than aggregate corporate debt:

$$\sigma_E^2 > \sigma_B^2 > 0. \quad (2)$$

⁷We use the two terms interchangeably.

We treat Σ as a primitive of the model and hold it fixed in the local monetary-policy comparative statics:

$$\frac{\partial \Sigma}{\partial \epsilon^{MP}} = 0. \quad (3)$$

Thus, monetary policy changes the prices of risky claims in the model, rather than the primitive covariance structure of their payoffs.

Investor demand. We model equity and corporate debt markets as partially segmented, following the preferred-habitat and demand-based asset pricing literatures (Vayanos and Vila, 2021; Koijen and Yogo, 2019). Different investor clienteles absorb different shares of equity and debt supply because of regulatory mandates, liability-matching needs, and other institutional considerations. At the same time, a common class of price-sensitive investors can hold both assets.

Price-sensitive investors. Let

$$\mathbf{S} \equiv \begin{pmatrix} S_E \\ \tilde{S}_B \end{pmatrix}$$

denote price-sensitive (PS) investors' holdings of equity and corporate debt. The PS investors have a mean-variance objective

$$\max_{\mathbf{s}} \left\{ \mathbf{S}' \mathbf{s} - \frac{\gamma(\epsilon^{MP})}{2} \mathbf{S}' \Sigma \mathbf{S} \right\}, \quad (4)$$

where $\mathbf{s} \equiv (s_E, s_B)'$ stacks the financing wedges and $\gamma(\epsilon^{MP}) > 0$ captures the effective tightness of the investors' risk-bearing capacity. The first-order condition is

$$\mathbf{s} = \gamma(\epsilon^{MP}) \Sigma \mathbf{S}. \quad (5)$$

Equivalently, the demand of PS investors is

$$\mathbf{D}^{PS}(\mathbf{s}; \gamma) = \frac{1}{\gamma} \Sigma^{-1} \mathbf{s}. \quad (6)$$

Non-price-sensitive investors. In addition to PS investors, each market contains a clientele of relatively price-insensitive investors. We refer to these investors as non-price-sensitive (NPS), although their demand is allowed to have a small positive response to expected returns. NPS investors are regulated intermediaries whose allocations are shaped by liability-matching and capital requirements—most notably insurance companies, the largest institutional holder of US corporate bonds, with portfolio composition shaped primarily by NAIC risk-based capital regulation (Becker and Ivashina, 2015)—together with behaviorally inelastic asset managers (Kojen and Yogo, 2019). Unlike the demand of PS investors, the demand of NPS investors is not derived from mean-variance optimization; it represents the reduced-form behavior of investors whose allocations are largely predetermined by non-price considerations. We assume

$$D_E^{NPS}(s_E) = \bar{D}_E + \eta_E s_E, \quad (7)$$

$$D_B^{NPS}(s_B) = \bar{D}_B + \eta_B s_B, \quad (8)$$

where $\eta_E, \eta_B > 0$ are small. The parameters $\bar{D}_E$ and $\bar{D}_B$ capture target holdings associated with institutional mandates and clientele preferences, while η_E and η_B govern the price elasticity of NPS demand. In vector notation,

$$\mathbf{D}^{NPS} = \bar{\mathbf{D}} + H\mathbf{s}, \quad H \equiv \begin{pmatrix} \eta_E & 0 \\ 0 & \eta_B \end{pmatrix}, \quad \bar{\mathbf{D}} \equiv \begin{pmatrix} \bar{D}_E \\ \bar{D}_B \end{pmatrix}. \quad (9)$$

Market clearing. We treat firms as atomistic relative to aggregate capital markets. In the local comparative statics, outstanding aggregate equity and debt supplies are predetermined. Let aggregate supplies of equity and corporate debt be

$$\bar{\mathbf{S}} \equiv \begin{pmatrix} \bar{S}_E \\ \bar{S}_B \end{pmatrix}.$$

We normalize the aggregate supply of each asset-market portfolio to one. Accordingly, S_E and $\tilde{S}_B$ denote the fractions of aggregate equity and corporate debt, respectively, that

are borne by PS investors. Market clearing requires

$$\mathbf{S} + \mathbf{D}^{NPS} = \bar{\mathbf{S}}. \quad (10)$$

We define the residual supply before the price response of NPS investors as $\mathbf{R} \equiv \bar{\mathbf{S}} - \bar{\mathbf{D}}$, and equilibrium financing wedges are

$$\mathbf{s} = [I + \gamma \Sigma H]^{-1} \gamma \Sigma \mathbf{R}. \quad (11)$$

Because of the assumption (2) and the condition that PS investors' equilibrium exposure to equity exceeds their exposure to corporate debt, which we verify empirically in Section 5.1, the equilibrium wedges satisfy $s_E > s_B > 0$.⁸

Firm problems and regimes. At $t = 0$, firm i chooses investment $I_i \geq 0$, internal funds used N_i , debt issuance B_i , and equity issuance E_i to maximize expected profits,

$$\max_{I_i, N_i, B_i, E_i \geq 0} f_i(I_i) - (1+r)N_i - (1+r+s_B)B_i - (1+r+s_E)E_i, \quad (12)$$

subject to the budget identity $I_i = N_i + B_i + E_i$, the internal-funds constraint $N_i \leq W_i$, and a collateral constraint on debt $B_i \leq \mu K_i$ following Kiyotaki and Moore (1997), where W_i is internal wealth and K_i is collateralizable capital. Production $f_i(I_i)$ satisfies $f'_i > 0$ and $f''_i < 0$. The firm takes r , s_B , and s_E as given when making the decisions.

Because $r < r + s_B < r + s_E$, the firm exhausts the cheaper source before turning to the more expensive one: internal funds first, then debt, then equity at the margin. This is the pecking order of Myers and Majluf (1984), generated here by the segmented structure of investor demand rather than by asymmetric information: external finance is more expensive than internal funds because the marginal investor in each external market must be compensated for absorbing additional risk (Leary and Roberts, 2010). The pecking order delivers three regimes:

Unconstrained. Firms with sufficient internal wealth: $I_i^* \leq W_i$, $B_i^* = E_i^* = 0$. First-order condition: $f'_i(I_i^*) = 1 + r$.

⁸See Appendix A for the formal conditions.

Debt-focused constrained (FCD). Firms accessing debt at the margin but not exhausting collateral: $N_i^* = W_i$, $0 < B_i^* < \mu K_i$, $E_i^* = 0$. First-order condition: $f_i'(I_i^*) = 1 + r + s_B$.

Equity-focused constrained (FCE). Firms that have exhausted debt capacity: $N_i^* = W_i$, $B_i^* = \mu K_i$, $E_i^* > 0$. First-order condition: $f_i'(I_i^*) = 1 + r + s_E$.

The regime is endogenous to firm primitives (W_i, K_i, f_i) and the wedges: firms with high internal wealth relative to investment opportunities remain unconstrained; firms with intermediate wealth and sufficient collateral operate at the debt margin; firms with low collateral relative to investment opportunities exhaust debt capacity and move to the equity margin.

Let ν_W and ν_B denote the multipliers on the internal-funds constraint $N_i \leq W_i$ and the collateral constraint $B_i \leq \mu K_i$, respectively. Both FCD and FCE firms are financially constrained in the sense that their internal-funds constraint binds with a strictly positive shadow price ($\nu_W = s_B$ and $\nu_W = s_E$, respectively), so internal and external finance are not perfect substitutes and an extra dollar of internal wealth raises firm value. FCE firms, however, are additionally rationed on debt: the collateral constraint binds ($\nu_B = s_E - s_B > 0$), which is precisely what pushes them onto the most expensive financing margin. This structure is consistent with the evidence in [Hoberg and Maksimovic \(2015\)](#) and [Linn and Weagley \(2023\)](#) that equity-focused constrained firms tend to be more constrained than debt-focused constrained firms, and with the summary statistics in Section 4.

3.2 Comparative Statics

We now derive comparative statics with respect to a contractionary monetary policy shock. Following [Drechsler et al. \(2018\)](#), we assume that contractionary monetary policy reduces the risk-bearing capacity of PS investors:

$$\gamma_\epsilon \equiv \frac{\partial \gamma}{\partial \epsilon^{MP}} > 0. \quad (13)$$

The covariance matrix Σ , aggregate asset supplies, and the parameters governing NPS demand are held fixed in the local comparative static. Monetary policy therefore affects

equilibrium risk premia by changing the willingness of PS investors to bear risk and, through market clearing, the allocation of risky assets across PS and NPS investors.

Sensitivity of the financing wedge. Let $\rho_E \equiv \partial s_E / \partial \epsilon^{MP}$ and $\rho_B \equiv \partial s_B / \partial \epsilon^{MP}$ denote the equilibrium responses of the two financing wedges to the shock.⁹ When NPS demand is only weakly elastic, each response is governed by the risk that the corresponding asset adds to the portfolio of the PS investors.

We show in Appendix A that a sufficient condition for the equity risk premium to respond more strongly than the debt risk premium is

$$(\sigma_E^2 - \sigma_{EB}) S_E > (\sigma_B^2 - \sigma_{EB}) \tilde{S}_B. \quad (14)$$

This condition states that equity contributes more incremental portfolio risk than corporate debt to the balance sheet of the common PS investors. Lemma 1 in Appendix A establishes that, under the risk asymmetry (2) and the exposure condition $S_E > \tilde{S}_B$, condition (14) holds for any admissible covariance σ_{EB} .

The exposure condition $S_E > \tilde{S}_B$, that the price-sensitive marginal investors' equilibrium exposure to equity exceeds their exposure to corporate debt, is an equilibrium outcome. We verify it directly in the data, using Fed Flow of Funds sectoral holdings. We classify holder sectors as price-sensitive or non-price-sensitive and show in Figure 1 that the price-sensitive share of the equity market exceeds the corresponding share of the corporate debt market in every quarter of our sample period, with a gap of 15-25 percentage points (see Section 5.1). Finally, $\rho_B > 0$ requires $\sigma_{EB} > -\sigma_B^2 \tilde{S}_B / S_E$, which is satisfied whenever equity and debt payoffs are not strongly negatively correlated. We conclude that

$$\rho_E > \rho_B > 0. \quad (15)$$

⁹We derive these responses in Appendix A.

Prediction 1: demand asymmetry. The corresponding responses of PS investors' holdings follow directly from market clearing:

$$\frac{\partial S_E}{\partial \varepsilon^{MP}} = -\eta_E \rho_E, \quad \frac{\partial \tilde{S}_B}{\partial \varepsilon^{MP}} = -\eta_B \rho_B. \quad (16)$$

When risk premia rise following a contractionary shock, relatively price-insensitive investors increase their holdings in response to the improved expected return, allowing PS investors to reduce their exposure. The price response is first-order in the shift of γ , while the quantity response is proportional to η_a : if NPS demand were perfectly inelastic in a given market ($\eta_a = 0$), the quantity held by PS investors in that market would be locally fixed and the entire adjustment would occur through the risk premium.

Under the additional mild assumption that the NPS clientele in the debt market is at least as inelastic as the counterpart in the equity market, $\eta_B \leq \eta_E$, which is consistent with the rigidity of insurance and pension mandates in corporate debt relative to the more flexible buy-and-hold layers in equity, equations (15) and (16) imply

$$\left| \frac{\partial S_E}{\partial \varepsilon^{MP}} \right| > \left| \frac{\partial \tilde{S}_B}{\partial \varepsilon^{MP}} \right|. \quad (17)$$

The price-sensitive marginal investor reduces equity holdings by more than corporate debt holdings after a contractionary monetary policy shock, with the debt-side response near zero when η_B is small. Both objects, the risk-premium asymmetry (15) and the demand-quantity asymmetry (17), are coordinates of the same equilibrium response to the shift of γ . We test Prediction 1 directly in Section 5.1.

Monetary policy transmission to investment. Totally differentiating each regime's first-order condition with respect to ε^{MP} , the investment response in each regime is

$$\left| \frac{dI_i^*}{d\varepsilon^{MP}} \right|_g = \frac{|r' + \rho_g|}{|f_i''(I_i^*)|}, \quad g \in \{\text{Unc}, \text{FCD}, \text{FCE}\}, \quad (18)$$

where $r' \equiv \partial r / \partial \varepsilon^{MP}$ and ρ_g denotes the response of the wedge at the regime's financing margin: $\rho_{\text{Unc}} = 0$, $\rho_{\text{FCD}} = \rho_B$, and $\rho_{\text{FCE}} = \rho_E$. The numerator cap-

tures the shift in the marginal cost of capital; the denominator captures the curvature of the production technology at the firm's optimum. Monetary policy transmits to unconstrained firms only through the unlevered cost of capital, to FCD firms additionally through the credit-spread response ρ_B , and to FCE firms additionally through the equity-premium response.

Proposition 1. *Suppose (2), (13), $S_E > \tilde{S}_B$, $\sigma_{EB} \geq 0$, and small NPS elasticities hold, so that $\rho_E > \rho_B > 0$ (Appendix A). Suppose further that the contractionary shock weakly raises the unlevered cost of capital, $r' \geq 0$, and that production curvature is comparable across regimes at the respective optima. Then the magnitude of the investment response to a contractionary monetary policy shock satisfies*

$$\left| \frac{dI_i^*}{d\varepsilon^{MP}} \right|_{FCE} > \left| \frac{dI_i^*}{d\varepsilon^{MP}} \right|_{FCD} > \left| \frac{dI_i^*}{d\varepsilon^{MP}} \right|_{Unc}.$$

Proof. See Appendix A. □

Prediction 2: transient investors and FCE firms. The price-sensitive marginal investor in the equity market corresponds empirically to transient institutional investors in the sense of Bushee (1998, 2001): short-horizon, high-turnover institutions that adjust holdings in response to price signals and macroeconomic conditions. Prediction 1 states that these investors reduce aggregate equity exposure after a contractionary shock. At the firm level, FCE firms are differentially exposed to this contraction because equity is their marginal source of financing: a withdrawal of price-sensitive equity demand raises their marginal cost of capital one-for-one, whereas firms at the debt margin or with sufficient internal funds are insulated from the equity-market wedge. The model therefore predicts that transient-investor outflows following contractionary shocks are concentrated in FCE firms. We test this prediction in Section 5.2.

Prediction 3: investment response. FCE firms reduce investment more sharply following a contractionary monetary policy shock, and the effect is concentrated in firms with high ex-ante transient ownership. The first part of the statement is Proposition 1: because FCE firms transmit monetary policy through the equity-premium response

$\rho_E > \rho_B$, their investment falls by more than that of FCD or unconstrained firms after a contractionary shock. The second part follows from combining Proposition 1 with Prediction 2. The mechanism behind ρ_E operates through the withdrawal of the price-sensitive marginal investor from the equity market. At the firm level, this demand contraction raises the marginal cost of equity capital most for firms whose investor base is heavily weighted toward these investors: within the FCE regime, firms with high ex-ante transient institutional ownership bear the full force of the demand-side mechanism, while FCE firms with little ex-ante transient ownership have limited exposure to it. The model therefore predicts that the amplified investment response of FCE firms is concentrated in the high-transient-ownership subsample. We test both parts of this prediction in Section 5.3.

4. Data

We employ a comprehensive dataset covering the 1991-2019 period. We obtain both annual and quarterly data on firm characteristics from COMPUSTAT. Daily stock returns data from CRSP are merged with COMPUSTAT using the linking table from WRDS. We obtain public SEOs from SDC Platinum. Following the literature, we exclude financial firms, regulated utilities, and government. We require firms to provide valid and positive information on their total assets and sales. We also exclude very small firms with physical capital under \$5 million, missing capital expenditures (CAPX), and negative R&D. Section C in the Appendix lists the definitions of firm-level variables in our analysis.

4.1 Financial Constraint Measures

Financial constraints are unobservable to the econometrician and hence hard to measure (Farre-Mensa and Ljungqvist, 2016). The monetary economics literature typically measures financial constraints using balance-sheet variables (age, size, leverage, etc.) or accounting-based indices (Kaplan and Zingales, 1997; Whited and Wu, 2006; Hadlock and Pierce, 2010); despite their widespread use, Farre-Mensa and Ljungqvist (2016) show that these measures do not reliably identify constrained behavior in either the debt or equity market.

We instead rely on text-based measures. [Hoberg and Maksimovic \(2015\)](#) exploit mandated disclosures in the Management’s Discussion and Analysis (MD&A) section of the 10-K, where firms discuss liquidity issues and intended financing sources for investment.¹⁰ They first identify a small training sample of confidently constrained firms by counting instances in which words such as *delay*, *abandon*, and *postpone* appear close to mentions of investment (*construction, expansion, acquisition, etc.*); a firm’s constraint score is the cosine similarity between its 10-K text and that of the training sample. Importantly, they use the financing sources discussed in the filing to separate equity- from debt-focused firms: a firm-year is equity-focused constrained if the firm reports being at risk of delaying investment due to liquidity issues and relies mainly on equity financing, and debt-focused constrained analogously.

Because 10-K sections are often missing or cannot be machine-parsed,¹¹ we use the measures of [Linn and Weagley \(2023\)](#), who map accounting variables to the [Hoberg and Maksimovic \(2015\)](#) measure with a random forest. This substantially increases time-series and cross-sectional coverage while inheriting the realistic behavior of the original measure; the authors further show that the measures behave consistently with theoretical intuitions and that firms indeed face constraints in the relevant source of financing (see [Linn and Weagley \(2023\)](#) for further discussion).

We thus use the measures *FCE* and *FCD* as firm-level, annual proxies for equity- and debt-focused constraints, respectively. Each year we sort firms into terciles on each dimension (nine groups) and focus on equity-focused constrained firms (top *FCE*, bottom *FCD* tercile) and debt-focused constrained firms (top *FCD*, bottom *FCE* tercile). We do not specifically focus on the firms that are in the top tercile of both *FCD* and *FCE*, since we aim to separate the roles of equity and debt financing constraints in monetary transmission. Our baseline comparison group is unconstrained firms (bottom tercile of both measures), which exhibit little similarity to constrained issuers and thus do not appear to face meaningful external financing frictions in either market.

Tables 1 and 2 report summary statistics. Relative to unconstrained firms, equity-focused constrained firms invest more in CAPX and R&D, hold more cash,

¹⁰[Buehlmaier and Whited \(2018\)](#) follow a similar approach. [Bodnaruk et al. \(2015\)](#) parse the universe of 10-K filings and measure tone as the percentage of constraining words, such as *required* and *obligations*.

¹¹[Hoberg and Maksimovic \(2015\)](#) classify 42%-68% of U.S. domestic firms in Compustat each year.

have lower cash flow, higher Q, and longer cash flow duration, and are smaller and younger; debt-focused constrained firms invest less in R&D, hold less cash, and have lower Q. Relative to equity-focused constrained firms, debt-focused constrained firms have higher book leverage (0.306 versus 0.149, consistent with [Hoberg and Maksimovic \(2015\)](#)), long-term leverage, long-term debt, and maturity, and they invest less in CAPX and R&D and hold less cash.

4.2 Macroeconomic Variables

The main source is FRED. We use the following macroeconomic variables: 1-year Treasury (Interest Rate on 1-year U.S. Treasuries), CPI (Consumer Price Index), Employment Ratio (Employment-Population Ratio), Industrial Production (Industrial Production Index), GDP Growth (Change in Real Gross Domestic Product), and Excess Bond Premium (Excess bond premium of [Gilchrist and Zakrajšek \(2012\)](#)).

4.3 Flow of Funds Holdings

To measure the composition of investor holdings across the corporate equity and corporate debt markets, we use the Financial Accounts of the United States (“Z.1,” commonly known as the Flow of Funds), published quarterly by the Federal Reserve Board. We obtain market-value holdings by institutional sector from Table F.51.1.s for corporate equities (formerly Table L.223) and Table F.3.5.s for corporate and foreign bonds (formerly Table L.213), covering our estimation window 1991–2019. For each instrument, these tables report the dollar holdings of every institutional sector in the U.S. economy together with the total amount outstanding, so that sectoral holdings aggregate to the market total. Our data cover holder composition in both the equity and the corporate debt markets, which is crucial for studying the cross-market comparisons in Section 5.1.

We classify each holder sector as *price-sensitive* or *non-price-sensitive*. Price-sensitive sectors comprise households (which in the Flow of Funds include hedge funds), the rest of the world, security brokers and dealers, closed-end funds, exchange-traded funds, and finance companies. Non-price-sensitive sectors comprise life insurance companies, property-casualty insurance companies, private and public pension funds, depository institutions, credit unions, and mutual funds.

The classification of insurance companies as non-price-sensitive holders of corporate debt is grounded in [Becker and Ivashina \(2015\)](#), who document that insurance companies are the largest institutional holder of U.S. corporate bonds, with portfolio composition shaped primarily by NAIC risk-based capital regulation. The classification of mutual funds as non-price-sensitive in our baseline follows the demand elasticity estimates in [Koijen and Yogo \(2019\)](#), who find that mutual funds, like banks, insurance companies, and pension funds, exhibit substantially less elastic demand than households and investment advisors. Pension funds and depository institutions are classified as non-price-sensitive on the basis of liability-matching mandates and held-to-maturity accounting treatment, respectively. Money market funds, government-sponsored enterprises, and the Federal Reserve are excluded from both categories.

5. Empirical Strategy and Main Results

This Section tests the model’s predictions in the order in which the mechanism operates. We begin at the market level: Section [5.1](#) verifies the exposure condition $S_E > \tilde{S}_B$ and tests Prediction 1. We then move to the firm level: Section [5.2](#) identifies transient institutional investors as the empirical counterpart of the price-sensitive marginal investor and shows that their outflows are concentrated in equity-focused constrained firms (Prediction 2). Section [5.3](#) then analyzes the consequences for these firms (Prediction 3).

5.1 Share of Price-Sensitive Investors in Equity and Bond Markets

In this section, we first document the composition of the two markets, verifying the exposure condition $S_E > \tilde{S}_B$ of Section [3.2](#). We then estimate the dynamic response of each market’s price-sensitive share to monetary policy shocks, testing Prediction 1.

Composition of holdings. Figure [1](#) plots the quarterly share of the equity and corporate debt markets held by price-sensitive investors, computed as aggregate price-sensitive sector holdings divided by total holdings in each market. The two markets have different investor bases. Over 1991–2019, price-sensitive investors hold on average 53.3% of corporate equity but only 36.1% of corporate debt; non-price-sensitive

investors hold 40.5% of equity but 57.1% of debt. The price-sensitive share of the equity market exceeds the corresponding share of the debt market in every quarter of the sample, with a gap of 15–25 percentage points. This provides strong empirical support for the exposure condition $S_E > \tilde{S}_B$.¹²

Dynamic response to monetary policy. We next estimate how the share of price-sensitive investors in each market responds to a monetary policy shock using the local projection

$$\Delta \text{Share}_{t+h,t-1}^a = \alpha^{a,h} + \beta^{a,h} \hat{y}_t + X'_{t-1} \Gamma^{a,h} + \varepsilon_t^{a,h}, \quad h = 1, \dots, 20, \quad (19)$$

separately for the equity market ($a = E$) and the corporate debt market ($a = B$). $\Delta \text{Share}_{t+h,t-1}^a$ is the change in the share of price-sensitive investors in the market a from quarter $t - 1$ to quarter $t + h$, $\hat{y}_t$ is the instrumented 1-year Treasury rate (see Section 5.3.2 for details), and X_{t-1} contains the macroeconomic controls (log CPI, log industrial production, the excess bond premium, and the employment ratio). The model predicts that both shares decline, with the equity contraction dominating: $|\beta^{E,h}| > |\beta^{B,h}|$, and a debt-side response close to zero when the non-price-sensitive debt clientele is highly inelastic.

Figure 2 reports the two impulse responses from the estimation: Panel A for the equity market and Panel B for the corporate debt market. The share of price-sensitive investors in the equity market declines following a contractionary shock, reaching approximately -0.4 percentage points per 25 basis points at $h = 17$, statistically significant from $h \approx 7$ onward. In contrast, the response of the share of price-sensitive investors in the corporate debt market is not statistically significant at all horizons. The asymmetry is exactly the pattern of Prediction 1: the marginal investors withdraw relatively from equity, while the investors in the debt market, anchored by the inelastic clientele, barely rebalance.

¹²The asymmetry is robust to the alternative classification schemes described in Section 4.3: treating mutual funds and exchange-traded funds as price-sensitive yields $S_E/\tilde{S}_B \approx 4.7$, and treating both as non-price-sensitive yields ≈ 4.2 . Under all schemes, the price-sensitive share of the equity market exceeds that of the debt market in every quarter.

5.2 Transient Investor Holdings at the Firm Level

We now move to the firm level and ask two questions posed by Prediction 2: which investors constitute the price-sensitive marginal investor, and are their outflows concentrated in financially constrained firms that depend on equity at the margin? Recent work by [Choi et al. \(2025\)](#) documents the role of investor demand, through changes in the cost of capital, in shaping firm valuation, financing, and investment. This evidence suggests that who owns a firm’s equity matters for how investment responds to macroeconomic shocks. If investor demand shifts are large enough to alter a firm’s ownership composition, this can lead to persistent changes in expected returns and financing conditions.

Building on this idea, we examine whether contractionary monetary policy shocks induce shifts in firms’ investor bases, and whether such changes disproportionately affect equity-focused constrained firms. Although equity-focused constrained firms are typically characterized by long-term growth potential, their high sensitivity to macroeconomic shocks and limited internal cash flows may make them more appealing to short-horizon, performance-sensitive investors during periods of risk-seeking. If these investors exit more aggressively in response to monetary tightening, equity-focused constrained firms face higher required returns and a rising cost of equity, amplifying their financing constraints and dampening investment.¹³ While we do not observe the cost of equity directly, investor composition provides a testable channel through which monetary policy can influence constrained firms, consistent with the demand-based asset pricing literature and with the mechanism of Section 3.

We collect data from LSEG Institutional (13F) Holdings, the primary source for institutional holdings data. We classify institutional investor types based on the combination of portfolio turnover and holdings concentration from [Bushee \(1998, 2001\)](#), and [Bushee and Noe \(2000\)](#). [Bushee \(1998, 2001\)](#) and [Borochin and Yang \(2017\)](#) categorize institutional investors as “transient,” “quasi-indexer,” or “dedicated” based on their investment horizons and portfolio concentration. Investors are classified as “transient” if they have short investment horizons reflected by high portfolio turnover and highly diversified portfolio holdings. Analogously, “dedicated” investors have long investment horizons reflected by low portfolio turnover and focused portfolio holdings. The third

¹³For example, [Yan and Zhang \(2009\)](#) show that short-term institutional ownership is positively related to future stock returns.

class, “quasi-indexers,” are long-horizon, low-turnover investors that are highly diversified. Transient institutions are thus the sharpest empirical counterpart of the price-sensitive marginal investor in the model: short-horizon, price-responsive, and quick to adjust holdings to macroeconomic conditions.¹⁴

We define our key variable, *Percent Owned by Transient Institutional Investors*, as the number of shares held by transient investors divided by the total number of institutional shares outstanding, measured quarterly. We then estimate our baseline differential specification, Equation (23) introduced in Section 5.3.2, using this measure as the outcome variable, examining whether equity-focused constrained firms experience a larger decline in transient ownership following contractionary monetary policy shocks relative to other firms. A larger outflow of transient investors from equity-focused constrained firms would indicate that the aggregate demand contraction of Section 5.1 loads precisely onto the constrained firms that rely on equity at the margin.

Figure 3 shows that transient institutional ownership declines significantly more in equity-focused constrained firms following contractionary monetary policy shocks. Short-horizon investors, who are more sensitive to risk, liquidity, and near-term performance, withdraw capital more aggressively from firms that rely heavily on external equity financing. This shift in investor composition serves as an amplification mechanism through which monetary policy shocks disproportionately affect equity-focused constrained firms.

Our empirical design uses the aggregate data to establish the cross-market asymmetries in Section 5.1, and the firm-level data to identify the marginal investor and trace the transmission to firms.¹⁵ To verify that the two measures capture the same investors where their coverage overlaps, we aggregate transient holdings across firms into a transient share of the equity market, using the identical denominator as Section 5.1, and esti-

¹⁴Table H.1 in Appendix H reports the list of transient, dedicated, and quasi-indexer institutional investors by average portfolio size, and Figure H.1 shows the average fraction of managers in each investor category accounted for by each institutional type, following the classification in [Kojen and Yogo \(2019\)](#).

¹⁵The Flow of Funds measure of Section 5.1 and the transient-ownership measure differ. The Flow of Funds covers holder composition in both the equity and corporate debt markets, which is essential for the cross-market comparison (the exposure condition $S_E > \tilde{S}_B$ and the differential share responses of Prediction 1). However, it contains only sector aggregates, without a firm-level dimension. The 13F data provide firm-level ownership information, who holds which firm, which is crucial to test Predictions 2 and 3. However, 13F filings report only Section 13(f) securities; thus corporate bond holdings are unobservable, so the cross-market test cannot be run with the firm-level measure.

mate the same local projection (19). The aggregate transient share declines significantly following contractionary shocks, mirroring the share of price-sensitive investors in the equity market; relative to its mean, it contracts roughly twice as strongly, consistent with transient institutions being the most price-sensitive segment of the marginal investor.¹⁶

5.3 The Differential Effect on Equity-Focused Constrained Firms

The previous two sections established the demand-side mechanism: price-sensitive investors withdraw from the equity market after contractionary shocks, and their withdrawal is concentrated in equity-focused constrained firms. This section traces the consequences for those firms (Prediction 3). We proceed in the order in which the effects materialize. We first show that the stock prices of equity-focused constrained firms fall significantly more on FOMC announcement days (Section 5.3.1). Then we demonstrate that their investment and R&D decline most sharply over the following quarters (Section 5.3.2), with the losses extending to innovation output (Section 5.3.3). Consistent with the investor demand mechanism, the financing contraction runs through the equity margin: these firms cut equity and SEO issuance far more than debt issuance (Section 5.3.4), and they hold on to precautionary cash rather than drawing it down (Section 5.3.5). Finally, Section 5.3.6 delivers the sharpest test of the mechanism: the amplification is concentrated in equity-focused constrained firms with high ex-ante transient ownership.

5.3.1 Stock Price Response

We rely on the high-frequency identification of monetary policy surprises to assess the stock price response to monetary policy (Kuttner, 2001; Bernanke and Kuttner, 2005; Gürkaynak et al., 2005).¹⁷ The surprise component is constructed from price changes of Federal funds rate futures contracts in the 30-minute window around FOMC announcements. The identifying assumption is that all public information is already incorporated into the prices at the beginning of the narrow window and therefore contains no other news affecting interest-rate expectations.

¹⁶Appendix I details the construction and reports the impulse response.

¹⁷One advantage of studying monetary policy shocks rather than other economic shocks (e.g., credit spreads) is that it allows us to analyze the impact of unanticipated monetary policy changes on the cross-section of U.S. equity returns and firm-level investment, while relying on a strong identification strategy.

However, as recent studies have shown, this methodology might capture the “information effect” of monetary policy, which could introduce bias into the estimation of monetary policy transmission (Nakamura and Steinsson, 2018). The idea is, for example, that an unexpected monetary easing might lead to pessimism among the market participants about economic fundamentals. Therefore, central banks could potentially convey information about their assessment of the economic state to the investors, through various communication tools.¹⁸ Arguably, the “information effect” could be an important factor for understanding how stock prices respond to monetary policy, especially when the financial constraint is also at play.

We use monetary policy shocks from the work of Jarociński and Karadi (2020), which separates the “pure” monetary policy effect and “information effect” by imposing sign restrictions in a Bayesian structural VAR framework. According to a broad range of models, a “pure” monetary policy tightening leads to lower stock market valuation. The empirical separation comes from identifying a shock that leads to a negative comovement between interest rate and stock price changes (monetary policy shock), versus a shock that increases both stock market prices and interest rate simultaneously (information shock), in a narrow window around an FOMC announcement. The shock series covers all FOMC announcements from 1990 to 2019.¹⁹ The mean of the shocks is negative 1 bp, while the standard deviation is 6 bps.

We first analyze whether monetary policy shocks affect firm-level returns using a panel regression of event window returns around the FOMC announcements during the period of 1990-2019. We estimate the following regression equation:

$$r_{ij,t} = \alpha + \beta mps_t + Z_{ij,t-1} + \alpha_{j,y} + e_{ij,t}, \quad (20)$$

where $r_{ij,t}$ is the return for stock i of industry j on the day of the FOMC announcement t . mps_t is the standardized monetary policy shock of that announcement, from Jarociński and Karadi (2020). The analysis is conducted at the firm-announcement level, which allows us to control for firm-level characteristics, including size, book-to-market ratio,

¹⁸Recently, Bauer and Swanson (2022) provide evidence of a “Fed response to news” channel, i.e., the incoming public economic news causes the Fed to adjust the monetary policy, is potentially at play. We show that our results are robust to the monetary policy shocks constructed in Bauer and Swanson (2023).

¹⁹We thank Jarociński and Karadi (2020) for providing the data.

leverage, operating profitability, and cash holdings. All regressions include industry-year fixed effects ($FE_{j,y}$). Standard errors are robust and clustered at the firm level. We consider multiple event windows around the FOMC announcements as monetary policy information may not be fully reflected immediately for all the firms, which is one of the main analyses of [Chava and Hsu \(2020\)](#).

Table 3 reports the coefficient estimates that are consistent with the literature. Column (1) reports the results when we use the daily return on the day of the FOMC announcements as the dependent variable. Column (2) reports the results when we use the daily return 1 day after the FOMC announcements. Column (3)-(5) report the results when we use the 1-, 2-, and 5-day cumulative returns since the FOMC announcements, respectively. On average, when there is a positive 25 bps monetary policy surprise, firms experience a significantly negative 2.14% return on the day of the FOMC announcements. The negative relation between monetary policy shock and realized stock price persists in a five-day cumulative window, with a negative 4.37% response of stock price. The results are quantitatively similar when firm fixed effects are included.

We then assess heterogeneous stock price responses around FOMC announcements over the same time period. We regress daily firm returns on the financial constraint indicator, the monetary policy shock, and their interaction using the following specification:

$$\begin{aligned} r_{ij,t} = & \beta mps_t + \gamma_1 I_{ij,t}^{FCE} + \gamma_2 I_{ij,t}^{FCD} + \delta_1 [mps_t \times I_{ij,t}^{FCE}] \\ & + \delta_2 [mps_t \times I_{ij,t}^{FCD}] + Z_{ij,t-1} + \alpha_{j,y} + e_{ij,t}, \end{aligned} \quad (21)$$

where $r_{ij,t}$ is the return for stock i of industry j on the day of the FOMC announcement t , and mps_t is the monetary policy shock. The financial constraint indicators, represented by $I_{ij,t}^{FCE}$ and $I_{ij,t}^{FCD}$, take the value of one for our primary groups of interest: the equity-focused constrained group (firms in the top tercile of FCE and bottom tercile of FCD) and the debt-focused constrained group (firms in the top tercile of FCD and bottom tercile of FCE), respectively.

To contrast equity-focused and debt-focused constrained firms with unconstrained firms, the unconstrained group of firms (those in the bottom tercile of both FCE and FCD) serves as the omitted baseline category. Therefore, the coefficients on our vari-

ables of interest are measured relative to these unconstrained firms. To ensure our estimates are not confounded by the intermediate groups generated by our double sort into terciles (see Section 4.1), the regression includes indicator variables for the remaining six groups and their interactions with mps_t . For brevity, the coefficients for these intermediate groups are suppressed in our reported tables.

$Z_{ij,t-1}$ are lagged firm-level controls, such as size, book-to-market ratio, leverage, operating profitability, and cash holding, and their interactions with mps_t . We include industry and year fixed effects $FE_{j,y}$ in all the regressions. Standard errors are robust and clustered at the firm level. We also consider multiple event windows around the FOMC announcements.

Table 4 presents the estimates of Equation (21). We report the results of the cumulative return window 1 day after, 2 days after, and 5 days after the FOMC announcements. We find that equity-focused constrained firms respond more to an unexpected change in monetary policy rate than unconstrained firms, as the coefficient on the interaction term is significantly negative. The heterogeneous impact of equity-focused constraint remains significant for a cumulative return window of several days. For example, Column (2) shows that for the holding period of 2 days after the FOMC announcement, equity-focused constrained firms have an average realized return that is 1.21% lower than that of unconstrained firms for a 25 bps surprise increase in the monetary policy rate, representing 37.9% of the average response in a 2-day window. Debt-focused constrained firms, on the other hand, have an average realized return that is 0.76% lower, which is 23.8% of the average response.

The magnitudes of the amplification in stock price responses go up when we look at a cumulative return window of 5 days after the FOMC announcements, confirming that the effect is not temporary. More importantly, the amplification of equity-focused constraint is quantitatively larger. Over a cumulative return window of 5 days after the FOMC announcements, the equity-focused constraint causes a 2% lower realized return, while the debt-focused constraint causes a 0.63% lower realized return. A Wald test rejects the null hypothesis that the two coefficients are equal ($\chi^2 = 17.327$), indicating a statistically significant difference between the estimated effects. Table E.1 in the Appendix reports consistent results using alternative event windows (one and two days after the FOMC announcements) as the basis for the dependent variable. Collectively,

our findings demonstrate that the equity constraint channel is important in explaining the heterogeneous response of stock price to monetary policy shocks.²⁰

To quantify stock price sensitivity to monetary policy shocks through the transient investor demand channel, we implement a simple back-of-the-envelope calculation. First, we estimate the impact of monetary policy shocks on the *Percent Owned by Transient Institutional Investors* to obtain the change in transient ownership induced by these shocks. Second, we estimate the sensitivity of returns to transient ownership. Finally, we multiply the two coefficients to obtain the implied price sensitivity to monetary policy shocks operating through the transient investor demand channel.

The implied stock price sensitivity for equity-focused constrained firms is more than twice as large as that for unconstrained firms: in response to a 25 bps contractionary shock, equity-focused constrained firms experience a -36.21 bps return through the transient investor demand channel, compared with -17.3 bps for unconstrained firms. To the best of our knowledge, this form of constraint channel, operating through investor behavior, is novel in the literature. It sets up the central question of the next section: whether the contraction in equity-market demand that loads onto equity-focused constrained firms translates into disproportionate declines in their investment and innovation.

5.3.2 Real Effects

We study the dynamic real effects (i.e., firm investment policies) of monetary policy. We measure the monetary policy stance as the 1-year U.S. Treasury rate. The adjustment of investment is slow-moving, with long and uncertain lags, and is measured at a quarterly frequency. As a result, the 1-year Treasury rate appropriately captures the gradual adjustment of firm investment and R&D, while also better reflecting interest rate variations during the unconventional monetary policy environment in the latter part of our sample period.

Since monetary policy is endogenous to macroeconomic conditions, we instrument the treasury rate using cumulative high-frequency “pure” monetary policy shocks from [Jarociński and Karadi \(2020\)](#) as a level measure of monetary policy

²⁰We use raw returns of firms at daily frequency in the baseline results. We also estimate using CAPM-adjusted returns. See Section [G.14](#) for further discussion.

surprises (as in [Bu et al. \(2021\)](#), [Cloyne et al. \(2023\)](#) and [Döttling and Ratnovski \(2023\)](#)), while controlling for key lagged macroeconomic variables.²¹ Importantly, [Nakamura and Steinsson \(2018\)](#) show that following an interest rate hike, nominal and real rates move nearly one-for-one several years out along the term structure, whereas expected inflation responds only modestly. This evidence supports the interpretation that these high-frequency policy surprises primarily capture real interest rate movements rather than shifts in expected inflation.²²

We use instrumental-variable local projections ([Jordà, 2005](#)) to trace out the dynamic impact of monetary policy on firm investment policies. Following [Cloyne et al. \(2023\)](#) and others, we conduct our analysis up to $h = 20$ (five years). Specifically, for each horizon h , we estimate the regression specification:

$$y_{i,t+h} - y_{i,t-1} = \beta_1^h \hat{y}t_t + \gamma_1^{h'} X_{t-1} + \gamma_2^{h'} Z_{i,t-1} + \alpha_i + \mu_{fq} + \epsilon_{i,t}, \quad (22)$$

where $y_{i,t}$ is the outcome variable (CAPX/Assets and R&D/Assets in logarithm) and $\hat{y}t$ is the instrumented 1-year Treasury rate. X_{t-1} is a vector of lagged macroeconomic control variables (log CPI, log industrial production, the excess bond premium, and the employment ratio). $Z_{i,t-1}$ is a vector of firm controls, which includes Q, leverage, size, cash flow, cash holdings, age, dividend, and the interaction of each control with the instrumented 1-year Treasury rate. We also include firm fixed effects and fiscal-quarter fixed effects. Note that we cannot include time fixed effects in Equation (22) because the time series variation in $\hat{y}t$ would be absorbed.

Figure 4 shows the impulse response function (IRF) for the response of CAPX and R&D, estimated using Equation (22). In response to a 25 bps higher 1-year Treasury rate, CAPX and R&D significantly drop by 5.2% and 1.3% after 16 quarters, respectively. This is in line with the literature analyzing the average investment response of U.S. COMPUSTAT firms.

²¹We follow [Döttling and Ratnovski \(2023\)](#) and construct the cumulative “pure” monetary policy shocks by first creating a quarterly series that accounts for the timing of FOMC announcements within a quarter. We then cumulate this quarterly series to obtain a level measure.

²²The Internet Appendix confirms the validity of our approach: the first-stage regression shows that cumulative “pure” monetary policy shocks are a strong instrument for the 1-year Treasury rate (Figure D.1 and Table D.1), and our results are quantitatively unchanged when instrumenting without the lagged macroeconomic controls or when using the cumulative shocks of [Bauer and Swanson \(2023\)](#) instead.

Our local-projection estimates reveal that the response of realized investment to monetary policy shocks peaks only after several quarters, suggesting that the transmission of monetary policy operates with a lag. This is consistent with the literature and potentially due to production frictions, meaning realized investment moves as projects start or finish due to constraints like time-to-build and irreversibility (Selgrad and Siani, 2025). We also find that R&D reacts with longer delays, which aligns with the idea that intangible capital tends to incur a higher adjustment cost (Belo et al., 2023).

To investigate the real effects of monetary policy and the role of financial constraint, we run the following specification:

$$y_{ij,t+h} - y_{ij,t-1} = \beta_1^h FCE_{ij,t-1} + \beta_2^h FCE_{ij,t-1} \times \hat{y}t_t + \gamma_1^{h'} Z_{ij,t-1} \times \hat{y}t_t + \gamma_2^{h'} Z_{ij,t-1} + \alpha_i + \eta_{jt} + \mu_{fq} + \epsilon_{ij,t}, \quad (23)$$

where $y_{ij,t}$ is the outcome variable (CAPX/Assets and R&D/Assets in logarithm) and $\hat{y}t$ is the instrumented 1-year Treasury rate. FCE is the proxy for equity-focused constraint. The proxy for debt-focused constraint, FCD , and its interaction with $\hat{y}t$ are included in the regression but not shown for brevity. $Z_{ij,t-1}$ is a vector of firm-level controls, which includes Q, leverage, size, cash flow, cash holdings, age, and dividend. We also include firm fixed effects, fiscal-quarter fixed effects, and industry $\times$ time fixed effects. Therefore, by controlling for the firm level characteristics that have been shown to affect the heterogeneous response of firm investment policies, and especially, the debt channel, we quantitatively capture and isolate the differential effect of monetary policy shocks on equity-focused constrained firms, i.e., the equity constraint channel of monetary policy.

Figure 5 shows the differential effect of monetary policy shocks for equity-focused constrained firms, estimated using Equation (23). A two-standard-deviation increase in the FCE measure significantly increases a firm's investment (CAPX) response to a 25 bps higher 1-year Treasury rate by 52.28 bps, representing approximately 10% of the average CAPX response of 5.2%. The amplification effect is even larger on the response of R&D. A two-standard-deviation increase in the FCE measure significantly increases a firm's R&D response to a 25 bps higher 1-year Treasury rate by 45 bps, corresponding approximately 34% of the average R&D response of 1.3%. The amplification is estimated after

controlling for the debt-focused constraint and other debt-related characteristics, and the difference between the equity- and debt-focused amplification effects is statistically significant for most horizons ($p < 0.01$ for CAPX and $p < 0.05$ for R&D).

Figure 6 shows the differential effect of monetary policy shocks for debt-focused constrained firms, estimated using Equation (23). A two-standard-deviation increase in the FCD measure significantly increases a firm's investment (CAPX) response to a 25 bps higher 1-year Treasury rate by 16.38 bps. This magnitude represents only 3.62% of the average CAPX response of 4.52% after four quarters. The heterogeneous effect of debt-focused constraint is not statistically significant for R&D, implying that the amplification of the debt-focused constraint is less economically significant than that of the equity-focused constraint.

Our results show that monetary policy shocks have significant real effects on both equity-focused and debt-focused constrained firms, but the equity channel appears to be quantitatively and statistically more important than the debt channel. This ordering of responses, strongest for equity-focused constrained firms, intermediate for debt-focused constrained firms, weakest for unconstrained firms, is precisely the prediction of Proposition 1: firms transmit monetary policy through the financing wedge at their marginal source of funds, and the equity wedge responds more than the debt wedge. These results underscore the equity constraint channel of monetary policy.

5.3.3 Impact on Innovation

Building on the evidence that monetary policy shocks have more pronounced effects through the equity constraint channel, we further explore how these shocks impact firms' innovation outputs, such as patent filings. We hypothesize that contractionary monetary policy shocks can significantly lower innovation, especially among equity-focused constrained firms, given that these firms are typically R&D-intensive (see Table 1). To test this hypothesis, we utilize patent data from Kogan et al. (2017). We construct the variable *Patents Count*, defined as the logarithm of the number of patents filed, and estimate the impact of monetary policy shocks on *Patents Count* using Equations (22) and (23). Additionally, we control for lagged R&D over assets to address potential correlations between R&D intensity and financial constraints.

Figure 7 illustrates the average effect of monetary policy on patent filings. A 25 bps increase in the 1-year Treasury rate leads to a significant reduction in the number of patents filed by 1.69% after 17 quarters. Table 5 further examines the amplification effect of financing constraints. We choose horizons $h = 17$ and $h = 20$, where the impulse response functions show the strongest responses (see Figure 7). Our results reveal that equity-focused constrained firms exhibit a much larger reduction in patent filings in response to a contractionary shock compared to unconstrained firms. Conversely, we find no evidence of amplification effects among debt-focused constrained firms at either 17 or 20 quarters, highlighting the role of equity constraints in shaping the transmission of monetary policy to long-run innovative activity.

5.3.4 Debt and Equity Issuance

If the equity constraint channel of monetary policy is quantitatively important (as shown above), we expect to observe that equity-focused constrained firms react to monetary policy shocks by cutting equity issuance more than other firms. To test this, we estimate the following equation:

$$\begin{aligned} \Delta y_{ij,t} = & \beta_1 \hat{y}_t + \beta_2 I_{ij,t}^{FCE} + \beta_3 I_{ij,t}^{FCD} + \beta_4 [I_{ij,t}^{FCE} \times \hat{y}_t] + \beta_5 [I_{ij,t}^{FCD} \times \hat{y}_t] \\ & + \gamma_1 Z_{ij,t-1} + \gamma_2 X_{t-1} + \alpha_i + \mu_{fq} + \lambda_{qj} + \epsilon_{it}, \end{aligned} \quad (24)$$

where $y_{ij,t}$ is the outcome variable (equity issuance, repurchases, public SEO issuance, and debt issuance) and $\hat{y}_t$ is the instrumented 1-year Treasury rate. The financial constraint indicators, represented by $I_{ij,t}^{FCE}$ and $I_{ij,t}^{FCD}$, take the value of one for our primary groups of interest: the equity-focused constrained group (firms in the top tercile of FCE and bottom tercile of FCD) and the debt-focused constrained group (firms in the top tercile of FCD and bottom tercile of FCE), respectively.

To contrast equity-focused and debt-focused constrained firms with unconstrained firms, the unconstrained group of firms (those in the bottom tercile of both FCE and FCD) serves as the omitted baseline category. Therefore, the coefficients on our variables of interest are measured relative to these unconstrained firms. To ensure our estimates are not confounded by the intermediate groups generated by our double sort into

terciles (see Section 4.1), the regression includes indicator variables for the remaining six groups and their interactions with $\hat{y}_t$. For brevity, the coefficients for these intermediate groups are suppressed in our reported tables.

$Z_{ij,t-1}$ is a vector of firm controls defined as in Equation (23). We also include firm fixed effects, fiscal-quarter fixed effects, and sector-quarter fixed effects. Finally, X_{t-1} is a vector of lagged macroeconomic control variables defined as in Equation (22).

One concern when using equity issuance data from Compustat is that employee-initiated and firm-initiated share issues are commingled. Therefore, we follow [McKeon \(2015\)](#) and construct the variable *Equity Issuance Adjusted* to exclude equity issuances resulting from the exercise of employee stock options. *Equity Issuance Adjusted* is then defined as gross equity issuance (COMPUSTAT's sstk) divided by lagged assets when gross equity issuance exceeds 3% of market equity, and zero otherwise (see [McKeon \(2015\)](#) for details).

Table 6 shows the results of estimating Equation (24) for equity issuance and repurchases. In response to a 25 bps higher 1-year treasury rate, equity-focused constrained firms reduce equity issuance by 0.0012. This effect represents approximately a 6.9% ($-0.0012/0.0172$) drop relative to the mean equity issuance, suggesting that the results are economically significant. We also show that in response to a 25 bps increase in the 1-year treasury rate, unconstrained firms reduce repurchases by 1.7% relative to the mean. We find no significant amplification effect for equity-focused constrained firms. This result is intuitive: as higher interest rates increase costs, firms become less likely to repurchase shares. However, since equity-focused constrained firms already repurchase relatively few shares, the effect is likely to be insignificant for them. In contrast, unconstrained firms are typically larger and repurchase more shares than constrained firms. In fact, the mean repurchase-to-assets ratio for unconstrained firms is nearly three times greater than that for constrained firms. As a result, unconstrained firms significantly reduce repurchases following an interest rate hike.

Table 7 shows the effect of monetary policy shocks on firms' SEO issuance and debt issuance. In response to a 25 bps higher 1-year treasury rate, equity-focused constrained firms significantly cut SEO issuance by -0.00075. This magnitude represents a 10% ($-0.00075/0.0075$) decrease relative to the mean public SEO issuance. Equity-focused constrained firms also significantly cut debt issuance by -0.0003.

However, the drop is only 0.87% ($-0.0003/0.0342$) relative to mean debt issuance. Therefore, the equity issuance response by equity-focused constrained firms is quantitatively larger than the debt issuance response.

Overall, we show that equity-focused constrained firms reduce both equity and debt issuance after a contractionary shock, but the effect is much smaller for debt issuance. The drop in equity issuance is approximately eight times greater than the drop in debt issuance. Even if we take into account that the mean of debt issuance is twice as large as the mean of equity issuance, the drop in equity issuance is still four times greater, which is economically substantial. This composition is consistent with the model: equity-focused constrained firms have exhausted debt capacity, so their external-financing adjustment runs through the equity margin, while debt, pinned at the collateral limit, barely moves. These results support and explain why equity-focused constrained firms are strongly affected by monetary policy shocks.

5.3.5 Cash Holdings

Previous research suggests that firms save cash that is raised by equity issuance for precautionary reasons (McLean, 2011). It is natural to expect that a contractionary monetary policy shock, for example, would lead firms to draw down cash holdings, in order to mitigate the increase in financing costs. Our previous results suggest that equity-focused constrained firms reduce equity issuance and real investment by more than other firms, in the aftermath of contractionary policy shocks. However, it is not necessarily optimal for equity-focused constrained firms to use up all of their cash precisely because they need to hold cash for precautionary reasons. In fact, Table 1 shows that equity-focused constrained firms hold more cash than other firms.

In Table 7, we look at the effect of monetary policy shocks on cash holdings. Our results suggest that firms do cut cash reserves in response to contractionary shocks. However, the effect is significantly smaller for equity-focused constrained firms. In particular, the response of equity-focused constrained firms is half as large as that of their unconstrained counterparts. These results suggest that equity-focused constrained firms are more reluctant to run down cash holdings following negative shocks, possibly due to the increased difficulty in raising new equity to replenish cash.

This behavior also clarifies why the equity constraint channel can operate even for firms that issue equity infrequently. While the model is static and does not feature a cash-management decision, its shadow-price logic speaks directly to the value of internal funds: for equity-focused constrained firms, the shadow price of internal funds equals the equity financing wedge ($\nu_W = s_E$), so a contractionary shock raises the marginal value of internal funds one-for-one with the wedge. To the extent that firms anticipate financing future investment at the equity margin, the same logic extends to quarters without issuance activity: the value of retained cash rises with the cost of replacing it externally, leading these firms to protect cash reserves rather than draw them down, and to cut investment instead.

5.3.6 The Role of Ex-Ante Transient Ownership

Our model predicts that the mechanism operates through the price-sensitive marginal investor in equity markets. When monetary policy tightens, risk-bearing tightness rises, equity demand from transient investors contracts, and the equity financing wedge increases disproportionately. Firms with greater ex-ante transient ownership are more exposed to this demand contraction through their cost of capital. The FCE amplification should therefore be concentrated in firms with high transient exposure. This prediction also serves as an identification check, since alternative explanations based on firm fundamentals or discount-rate exposure do not predict that the amplification should line up with the composition of the investor base. To test this prediction, we sort firms each quarter by their predetermined (ex-ante) transient institutional ownership and classify a firm as *High Exposure* if it is above the median and *Low Exposure* otherwise.

Before turning to the results, we verify that the split does not proxy for differences in firm fundamentals. Table 8 reports means and standard deviations for High and Low Exposure firms within the equity-focused constrained group (top tercile of *FCE* and bottom tercile of *FCD*). High Exposure firms are not weaker on any fragility dimension: cash flow, cash holdings, and leverage all go in a direction that would make them look less fragile. Cash-flow duration is similar across the two groups (log duration 4.17 vs. 4.08), ruling out the duration channel as an explanation for the amplification.

The two groups differ in terms of size, age, *Q*, and dividend, reflecting the well-documented preference of transient institutional investors for smaller, younger,

higher-growth firms (Bushee, 1998). We control for all four characteristics interacted with the shock, so the amplification result is estimated conditional on these differences. Overall, these findings support the interpretation that within-FCE variation in ex-ante transient exposure is likely orthogonal to the confounds most likely to threaten identification of the equity-supply channel.

We then estimate Equation (23) separately for each subsample (High- and Low-Exposure). Figure 8 shows that the amplification effect of the equity-focused constraint is present only among firms with greater ex-ante exposure to transient institutional investors: for both CAPX and R&D, the interaction coefficient is negative and significant in the High Exposure subsample, and statistically insignificant in the Low Exposure subsample. This finding is consistent with our model: the FCE amplification of the monetary policy response is concentrated where the price-sensitive marginal investor is present in the firm’s investor base, indicating that the equity constraint channel operates through investor demand.

Overall, the evidence in this section traces a single mechanism from the market level to the firm level. The sharper investment declines among equity-focused constrained firms following monetary tightening reflect a contraction in the supply of equity financing, transmitted through the withdrawal of price-sensitive investors, rather than solely differences in the demand for financing, such as those emphasized by Ottonello and Winberry (2020). While our focus is on the equity supply channel, the evidence is also consistent with a broader mechanism in which fluctuations in investor demand directly affect firm valuations and thereby investment decisions, even in the absence of new equity issuance or repurchases (see, e.g., Polk and Sapienza, 2009).

5.3.7 Additional Results

The Internet Appendix (Section G) presents additional results and robustness checks. Our findings are robust to alternative channels correlated with the FCE measure such as cash-flow duration, credit risk, refinancing constraints, intangible intensity, adjustment costs, cyclicity, and differential equity reliance, and to the identification strategy: controlling for central bank information shocks, using the alternative shocks of Bauer and Swanson (2023), measuring the policy stance with the instrumented 2-year Treasury rate, excluding the effective-lower-bound period,

testing for pre-trends, and splitting expansionary and contractionary shocks. The appendix also reports complementary evidence that equity-focused constrained firms' investment responds to equity-market but not debt-market financing shocks, and on the transmission of monetary policy to M&A activity.

6. Conclusion

This paper provides new evidence that equity-focused financial constraints play a central role in the transmission of monetary policy. While much of the existing literature has emphasized debt financing frictions, we show that firms relying on equity at the margin experience substantially stronger valuation declines, sharper reductions in investment and R&D, and larger drops in innovation output following contractionary shocks. We explain these findings with a model in which equity and corporate debt markets are partially segmented: contractionary monetary policy reduces the risk-bearing capacity of price-sensitive investors, raising the cost of equity financing more than the cost of debt, so that firms whose marginal source of financing is equity bear the largest response.

The evidence supports the mechanism at every level. At the market level, the price-sensitive share of the equity market contracts significantly after contractionary shocks, while the corresponding share of the corporate debt market does not significantly move. At the firm level, outflows of short-horizon, price-sensitive institutions concentrate in equity-focused constrained firms, raising the cost of external equity and tightening financing conditions precisely when these firms need liquidity the most. And the amplification of investment and R&D responses is present only among firms with high ex-ante ownership by these investors, indicating a constraint channel that operates through the composition of the firm's investor base rather than through its balance sheet. Importantly, this channel suppresses innovation, a key driver of long-run growth, thereby linking monetary policy not only to short-run business cycles but also to long-term productivity dynamics.

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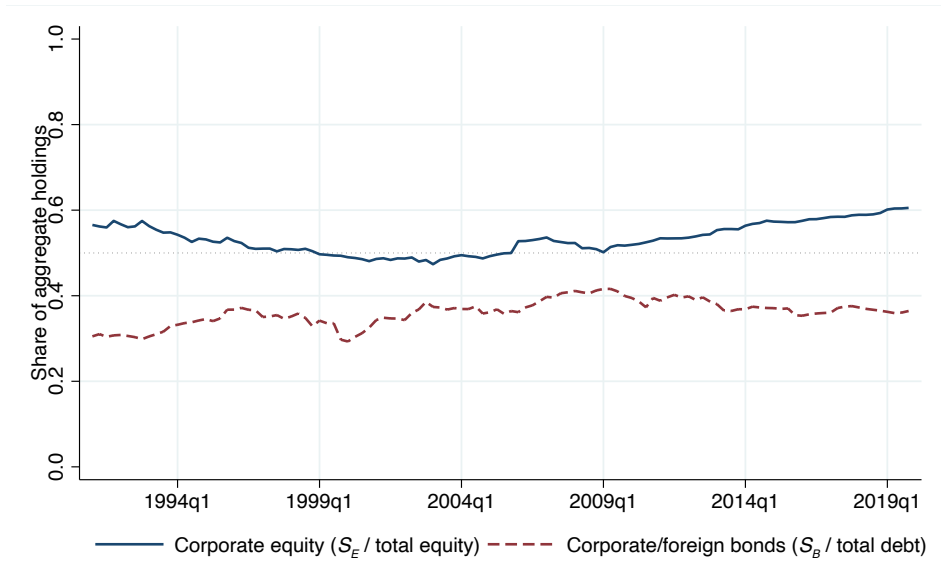
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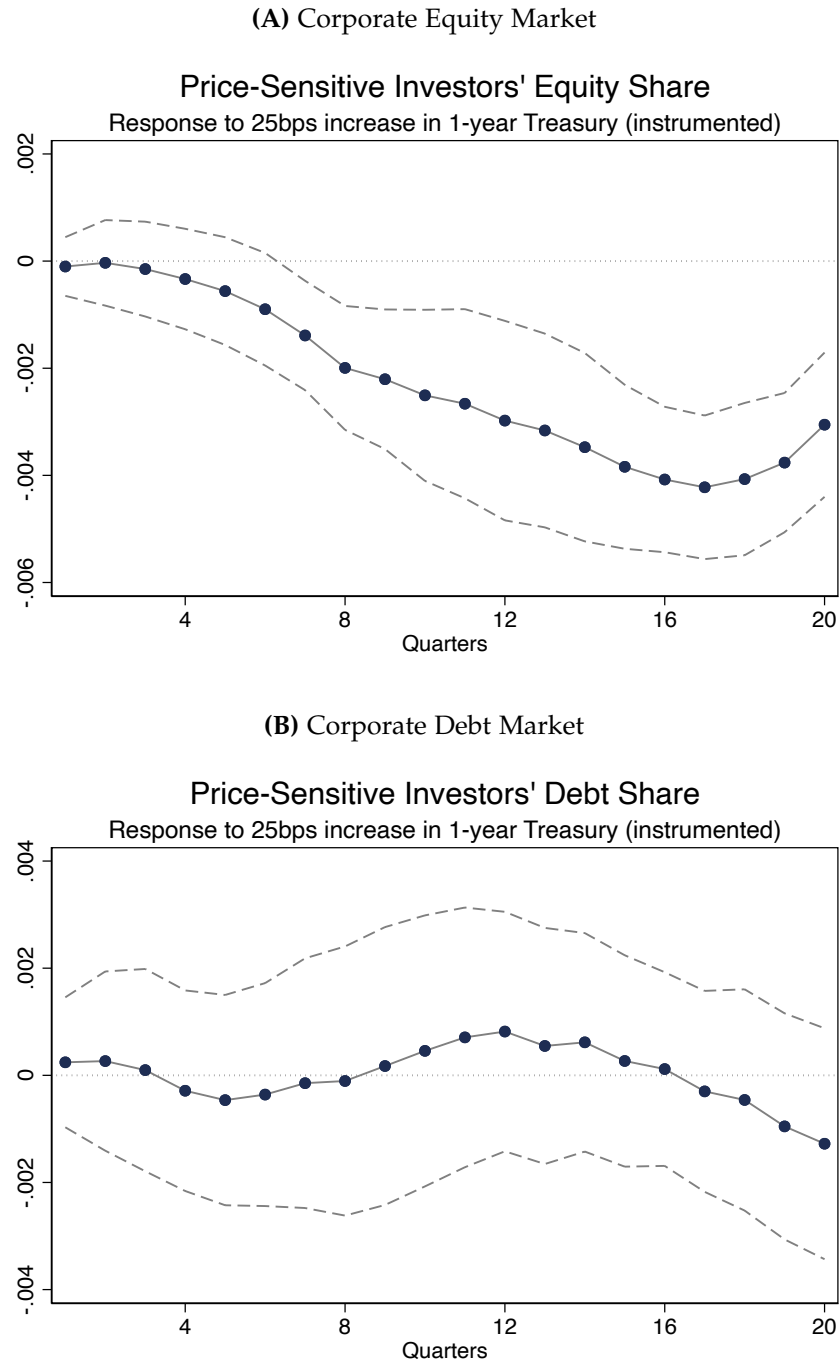
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Figure 1. Price-Sensitive Share of Aggregate Holdings: Equity vs. Corporate Debt



This figure plots the price-sensitive share of aggregate holdings for corporate equity (solid navy) and corporate and foreign bonds (dashed maroon), quarterly from 1991:Q1 to 2019:Q4. Data are from Fed Flow of Funds Tables F.51.1.s (corporate equities, formerly L.223) and F.3.5.s (corporate and foreign bonds, formerly L.213). Sector classification follows Section 4.3. The dotted horizontal line indicates the 50% mark.

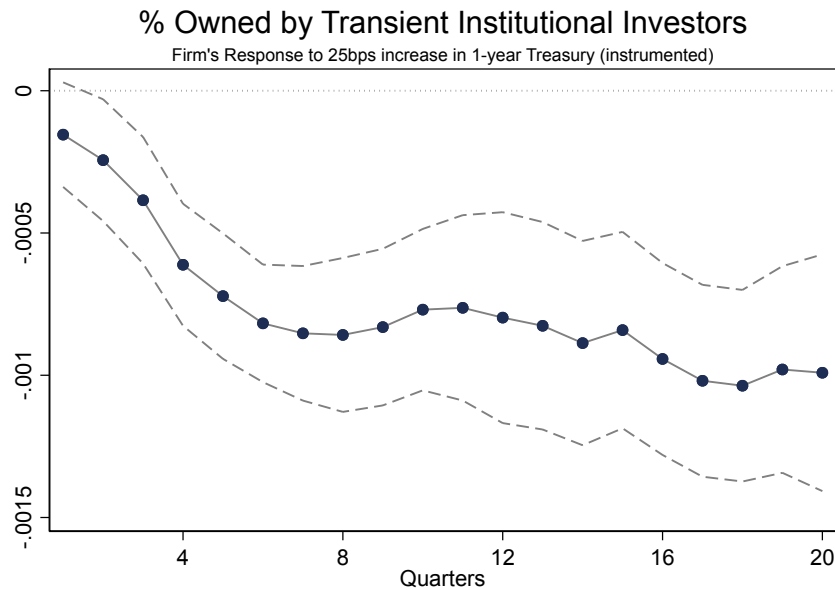
Figure 2. Response of the Price-Sensitive Share of the Equity and Corporate Debt Markets to Monetary Policy



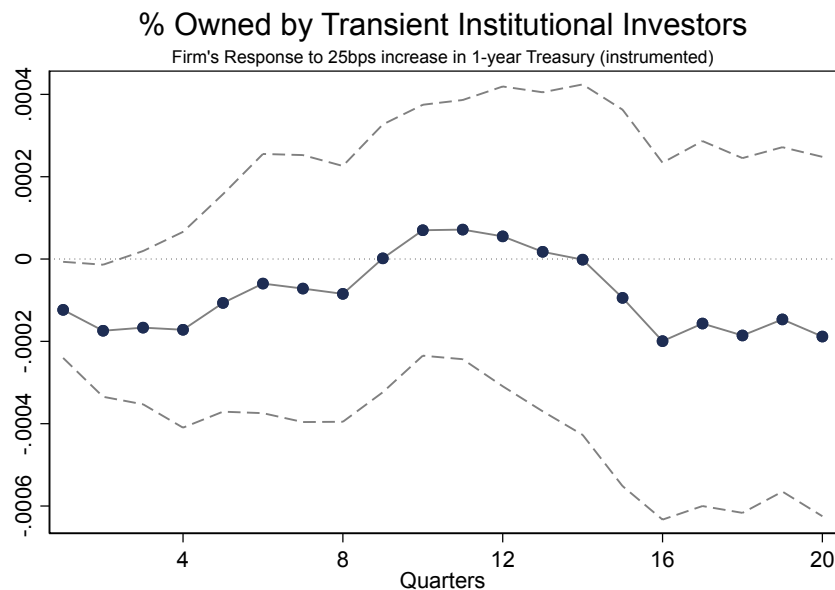
This figure shows the impulse response of the price-sensitive share of the corporate equity market (Panel A) and the corporate and foreign bond market (Panel B) to a 25 bps contractionary monetary policy shock, estimated from Equation (19) at horizons $h = 1, \dots, 20$ quarters. The dependent variable is the change in the aggregate price-sensitive share of the corresponding market from quarter $t - 1$ to quarter $t + h$, with sectors classified as in Section 4.3. The 1-year Treasury rate is instrumented by cumulative high-frequency monetary policy shocks from Jarociński and Karadi (2020). Controls include lagged log CPI, log industrial production, excess bond premium, and log employment ratio. Dashed lines are 90% confidence intervals based on robust Driscoll-Kraay standard errors.

Figure 3. The Dynamic Response of % Shares Owned by Transient Institutional Investors to Monetary Policy: Equity vs. Debt-focused Constrained Firms

(A) Panel A: Equity-Focused Constrained Firms



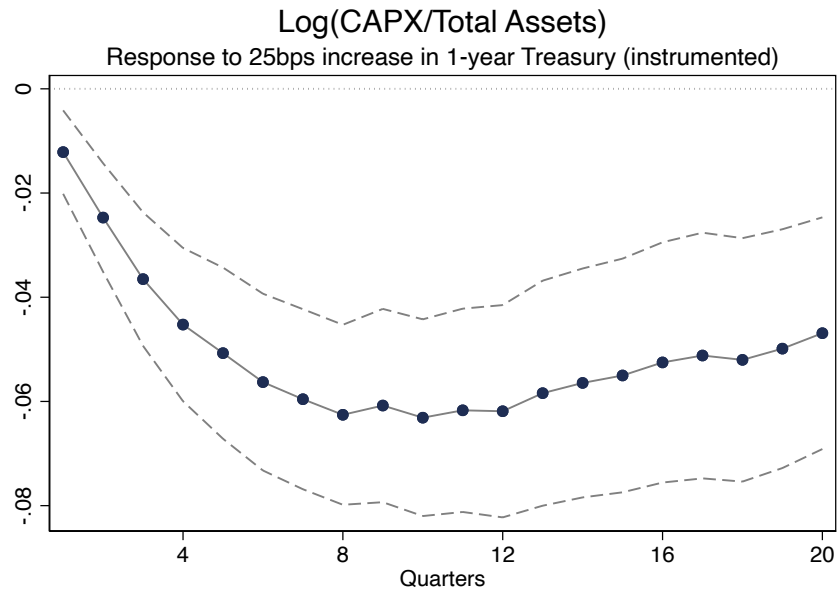
(B) Panel B: Debt-Focused Constrained Firms



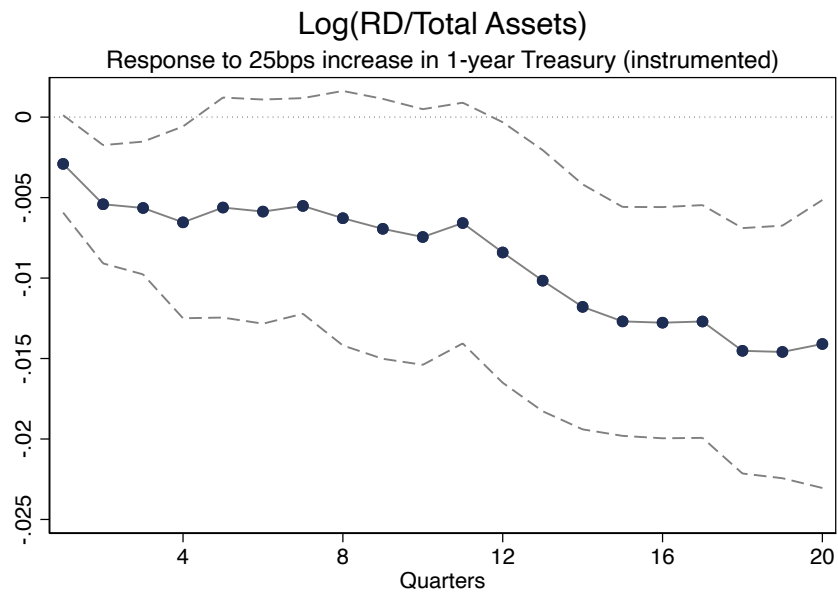
This figure shows the Impulse Response Function (IRF) for the response of % shares owned by transient institutional investors to a 25 bps increase in 1-year Treasury (instrumented), estimated using Equation (23). The outcome variable is the % of shares owned by transient institutional investors. The 1-year Treasury rate is instrumented by cumulative high-frequency monetary policy shocks measured as monetary policy shocks from the decomposition by Jarociński and Karadi (2020). Each point represents the point estimate of the coefficient of the instrumented 1-year Treasury rate interacted with the financing constraint measure $FCE_{ij,t-1}$ (Panel A) and $FCD_{ij,t-1}$ (Panel B). The dashed line represents 90% confidence intervals using heteroscedasticity and autocorrelation robust Driscoll-Kraay standard errors.

Figure 4. Dynamic Response of Investment to Monetary Policy

(A) CAPX

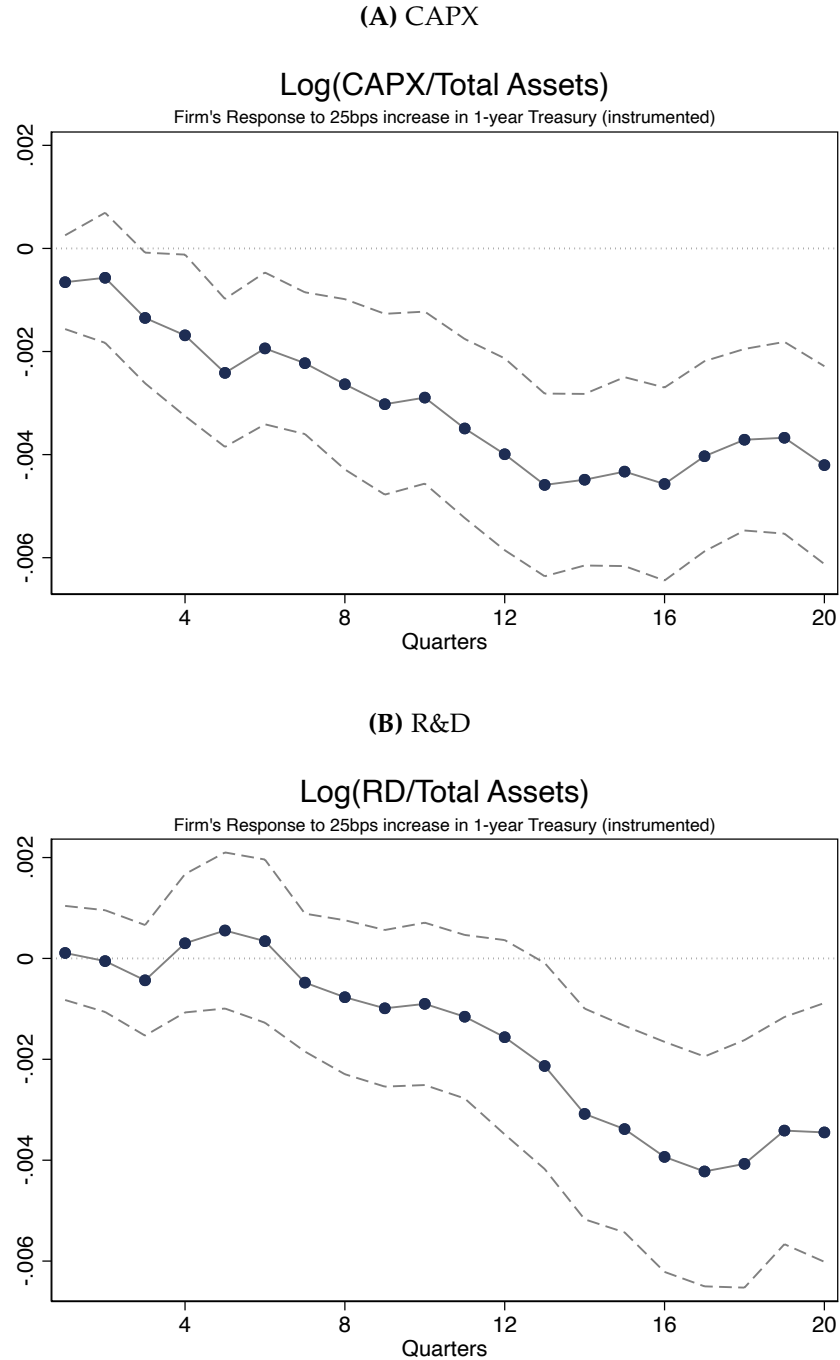


(B) R&D



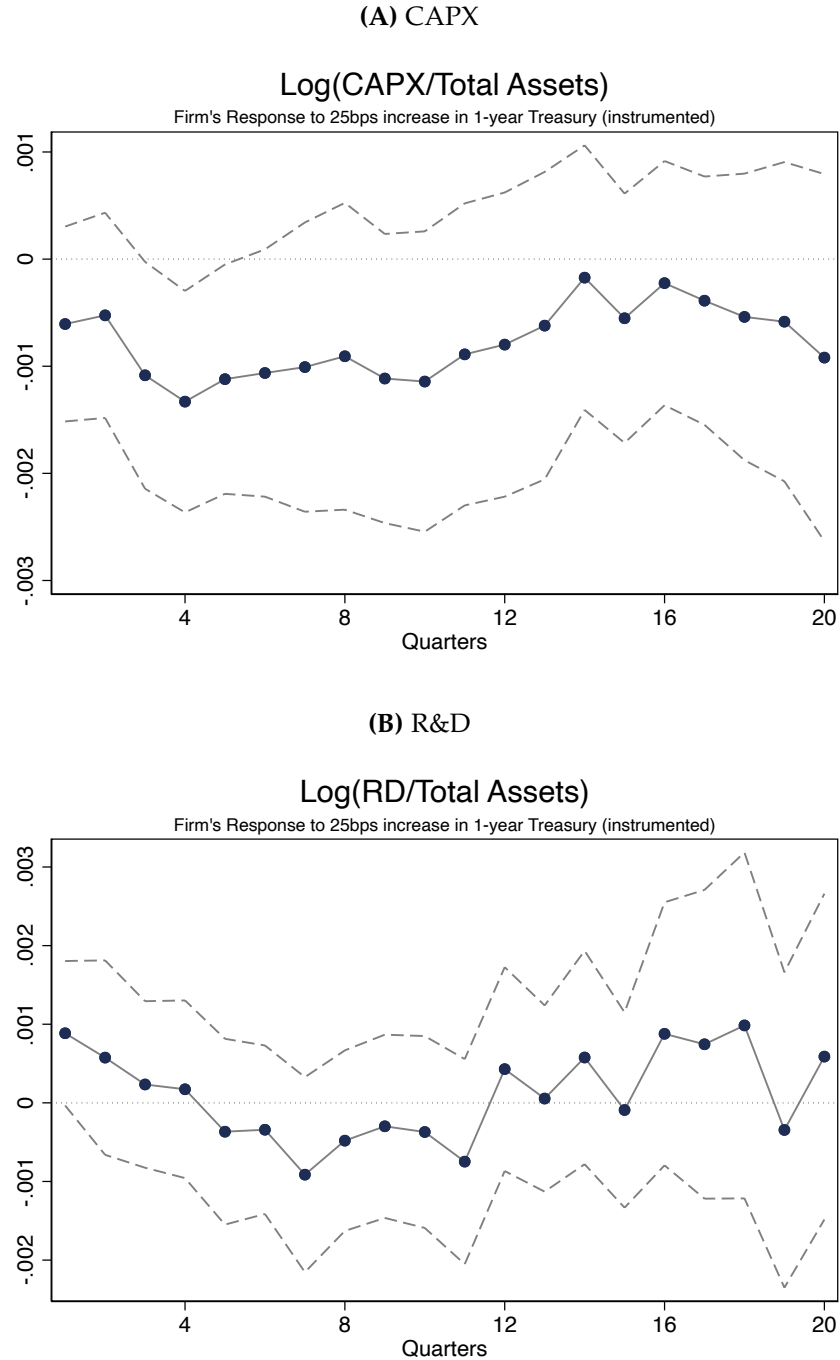
This figure shows the Impulse Response Function (IRF) for the response of CAPX, and R&D to a 25 bps increase in 1-year Treasury rate (instrumented). The 1-year Treasury rate is instrumented by cumulative high-frequency monetary policy shocks measured as monetary policy shocks from the decomposition by [Jarociński and Karadi \(2020\)](#). Each point represents the point estimate of the coefficient of the instrumented 1-year Treasury rate. The dashed line represents 90% confidence intervals using heteroscedasticity and autocorrelation robust Driscoll-Kraay standard errors.

Figure 5. Equity-Focused Constraints and the Dynamic Response of Investment to Monetary Policy



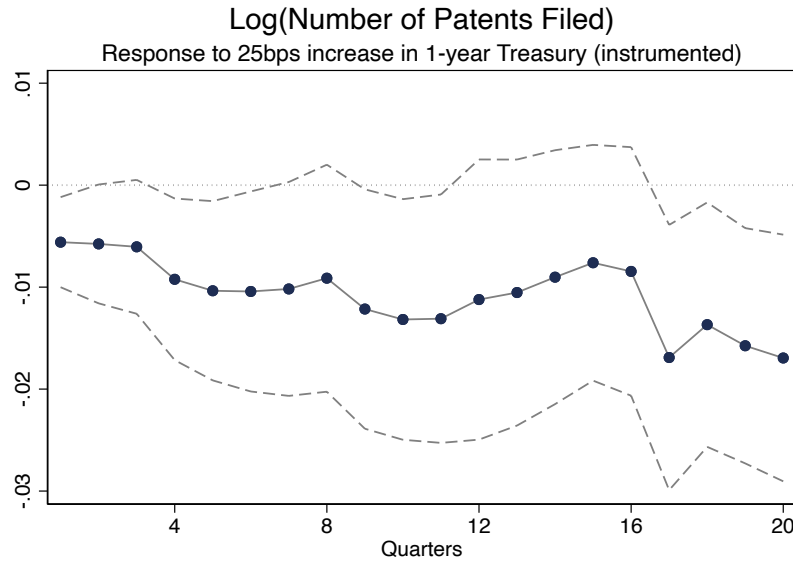
This figure shows the Impulse Response Function (IRF) for the response of CAPX, and R&D to a 25 bps increase in 1-year Treasury (instrumented), estimated using Equation (23). The 1-year Treasury rate is instrumented by cumulative high-frequency monetary policy shocks measured as monetary policy shocks from the decomposition by [Jarociński and Karadi \(2020\)](#). Each point represents the point estimate of the coefficient of the instrumented 1-year Treasury rate interacted with the financing constraint measure (β_2^h in Equation (23)). The dashed line represents 90% confidence intervals using heteroscedasticity and autocorrelation robust Driscoll-Kraay standard errors.

Figure 6. Debt-Focused Constraints and the Dynamic Response of Investment to Monetary Policy



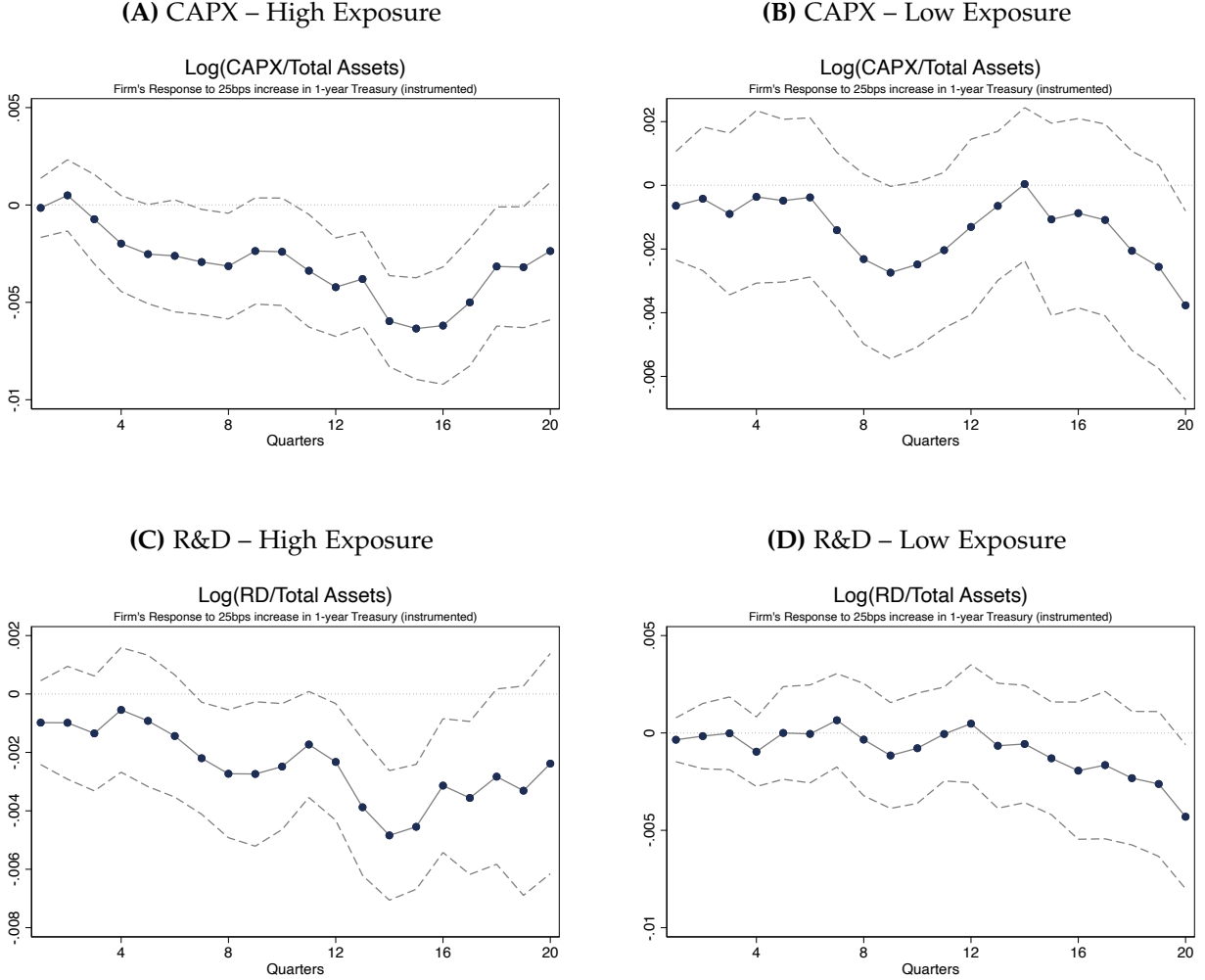
This figure shows the Impulse Response Function (IRF) for the response of CAPX, and R&D to a 25 bps increase in 1-year Treasury (instrumented), estimated using Equation (23). The 1-year Treasury rate is instrumented by cumulative high-frequency monetary policy shocks measured as monetary policy shocks from the decomposition by [Jarociński and Karadi \(2020\)](#). Each point represents the point estimate of the coefficient of the instrumented 1-year Treasury rate interacted with the financing constraint measure $FCD_{ij,t-1}$. The dashed line represents 90% confidence intervals using heteroscedasticity and autocorrelation robust Driscoll-Kraay standard errors.

Figure 7. The Dynamic Response of Innovation to Monetary Policy



This figure shows the Impulse Response Function (IRF) for the response of Innovation to a 25 bps higher 1-year Treasury rate (instrumented). The outcome variable is the log of the number of patents filed. The 1-year Treasury rate is instrumented by cumulative high-frequency monetary policy shocks measured as monetary policy shocks from the decomposition by [Jarociński and Karadi \(2020\)](#). Each point represents the point estimate of the coefficient of the instrumented 1-year Treasury rate. The dashed line represents 90% confidence intervals using heteroscedasticity and autocorrelation robust Driscoll-Kraay standard errors.

Figure 8. Equity-Focused Constraints and the Dynamic Response of Investment to Monetary Policy: The Role of Ex-Ante Transient Ownership



This figure shows the Impulse Response Function (IRF) for the response of CAPX and R&D to a 25 bps increase in the 1-year Treasury rate (instrumented), estimated using Equation (??) separately in two subsamples defined by ex-ante transient institutional ownership. *High Exposure* firms are those with lagged transient institutional ownership above the cross-sectional median; *Low Exposure* firms are those at or below the median. The 1-year Treasury rate is instrumented by cumulative high-frequency monetary policy shocks from the decomposition by Jarociński and Karadi (2020). Each point represents the point estimate of the coefficient on the instrumented 1-year Treasury rate interacted with the equity-focused constraint measure ($\beta_2^{h,g}$ in Equation (??)). The dashed lines represent 90% confidence intervals using heteroscedasticity- and autocorrelation-robust Driscoll-Kraay standard errors.

Table 1. Summary Statistics: Equity-Focused Constrained Firms vs. Unconstrained Firms

	Equity-Focused Constrained Firms			Unconstrained Firms		
	Obs	Mean	Std. dev.	Obs	Mean	Std. dev.
CAPX/ Assets	65,934	0.028	0.057	30,056	0.014	0.022
R&D/ Assets	32,598	0.044	0.064	15,902	0.018	0.022
Cash Flow	61,861	-0.011	0.085	28,219	0.038	0.035
Cash holdings	65,742	0.294	0.242	30,023	0.196	0.164
Size	65,934	4.948	1.857	30,056	6.183	1.930
Q	62,688	2.607	3.442	27,855	2.074	1.498
Duration	22,557	93.38	105.20	22,739	49.24	40.54
Age	65,934	9.474	8.441	30,056	20.77	12.49
Dividend	65,934	0.062	0.242	30,056	0.047	0.213
FCE	64,697	0.560	0.498	29,696	-0.681	0.290
FCD	64,697	-0.453	0.322	29,696	-0.442	0.395
Book Leverage	63,887	0.149	0.269	29,000	0.146	0.178
Long-term Leverage	65,475	0.118	0.219	29,743	0.122	0.167
Long-term Debt/ Assets	65,475	0.118	0.232	29,743	0.123	0.177
Short-term Debt/ Assets	64,053	0.040	0.162	29,146	0.028	0.060
Maturity	42,230	0.645	0.355	22,314	0.714	0.319
RFC	39,141	0.053	0.179	21,435	0.035	0.127
% Owned by Transient	38,494	0.271	0.181	20,863	0.213	0.149

This table provides summary statistics for basic firm characteristics (see Subsection C). We sort firms into terciles each year based on the lagged financial constraints, following the standard practice in the literature. Equity-focused constrained firms are firms in the top tercile of the *FCE* and bottom tercile of *FCD* distribution. Unconstrained firms are firms in the bottom tercile of both measures. The sample covers the years 1991 to 2019. Source: COMPUSTAT and LSEG.

Table 2. Summary Statistics: Debt-Focused Constrained Firms vs. Unconstrained Firms

	Debt-Focused Constrained Firms			Unconstrained Firms		
	Obs	Mean	Std. dev.	Obs	Mean	Std. dev.
CAPX/Assets	58,472	0.015	0.022	30,056	0.014	0.022
R&D/Assets	20,948	0.007	0.013	15,902	0.018	0.022
Cash Flow	55,341	0.032	0.029	28,219	0.038	0.035
Cash holdings	58,099	0.056	0.080	30,023	0.196	0.164
Size	58,472	5.917	1.518	30,056	6.183	1.930
Q	54,304	1.414	0.794	27,855	2.074	1.498
Duration	38,153	52.53	52.56	22,739	49.24	40.54
Age	58,472	17.40	11.84	30,056	20.77	12.49
Dividend	58,472	0.056	0.231	30,056	0.047	0.213
FCE	57,639	-0.689	0.294	29,696	-0.681	0.290
FCD	57,639	0.857	0.457	29,696	-0.442	0.395
Book Leverage	56,860	0.306	0.205	29,000	0.146	0.178
Long-term Leverage	58,243	0.254	0.202	29,743	0.122	0.167
Long-term Debt/Assets	58,243	0.255	0.212	29,743	0.123	0.177
Short-term Debt/Assets	56,974	0.059	0.102	29,146	0.028	0.060
Maturity	54,806	0.773	0.295	22,314	0.714	0.319
RFC	54,318	0.024	0.103	21,435	0.035	0.127
% Owned by Transient	36,142	0.207	0.160	20,863	0.213	0.149

This table provides summary statistics for basic firm characteristics (see Subsection C). We sort firms into terciles each year based on the lagged financial constraints, following the standard practice in the literature. Debt-focused constrained firms are firms in the top tercile of the *FCD* and bottom tercile of *FCE* distribution. Unconstrained firms are firms in the bottom tercile of both measures. The sample covers the years 1991 to 2019. Source: COMPUSTAT and LSEG.

Table 3. Stock Price Response to Monetary Policy Shocks

Window:	(0,0) (1)	(+1,+1) (2)	(0,+1) (3)	(0,+2) (4)	(0,+5) (5)
<i>mps</i>	-0.514*** (0.009)	-0.272*** (0.009)	-0.787*** (0.013)	-0.770*** (0.014)	-1.05*** (0.018)
Controls	Yes	Yes	Yes	Yes	Yes
<i>Fixed-effects</i> industry-year	Yes	Yes	Yes	Yes	Yes
<i>Fit statistics</i>					
Observations	905,306	853,799	905,017	904,738	903,908
R ²	0.019	0.015	0.024	0.023	0.029

This table reports the coefficient estimates of the cross-sectional regression by pooling all firm level returns around FOMC announcements from 1991 to 2019. The firm level returns are calculated over five different event windows and shown in Columns (1) to (5). Column (1) reports the results when we use the daily return on the day of the FOMC announcements as the dependent variable. Column (2) reports the results when we use the daily return 1 day after the FOMC announcements as the dependent variable. Column (3) to (5) report the results when we use the 1-, 2-, and 5-day cumulative returns since the FOMC announcements as the dependent variable, respectively. The *mps* variable denotes monetary policy shock from [Jarociński and Karadi \(2020\)](#). All regressions include industry-year fixed effects and control for log asset (size), book-to-market ratio, leverage, operating profitability, and cash holdings at the firm level. Robust standard errors clustered at the firm level are reported in parentheses. Significance codes: ***: 0.01, **: 0.05, *: 0.1.

Table 4. Cumulative Heterogeneous Stock Price Response to Monetary Policy Shocks

Window:	(0,+1) (1)	(0,+2) (2)	(0,+5) (3)
$mps \times equity_focused$	-0.201*** (0.055)	-0.292*** (0.063)	-0.481*** (0.081)
$mps \times debt_focused$	-0.077* (0.043)	-0.183*** (0.048)	-0.153** (0.062)
$mps \times leverage$	0.350*** (0.092)	0.210** (0.101)	0.122 (0.121)
$mps \times bm$	0.011 (0.020)	-0.031 (0.024)	-0.130*** (0.030)
$mps \times size$	-0.091*** (0.008)	-0.040*** (0.008)	-0.005 (0.010)
$mps \times profitability$	0.182*** (0.039)	0.175*** (0.042)	0.262*** (0.050)
$mps \times cashholding$	-0.449*** (0.091)	-0.344*** (0.102)	-0.175 (0.123)
<i>Fixed-effects</i> industry-year	Yes	Yes	Yes
<i>Fit statistics</i>			
Observations	843,764	843,501	842,718
R ²	0.025	0.023	0.030
χ^2	5.5018	3.1464	17.327

This table reports the coefficient estimates of the cross-sectional regression by pooling all firm level returns around FOMC announcements from 1991 to 2019. The firm level cumulative returns are calculated over three different event windows and shown in Columns (1) to (3). Columns (1) to (3) report the results when we use the 1-, 2-, and 5-day cumulative returns since the FOMC announcements as the dependent variable, respectively. The *mps* variable denotes monetary policy shock from [Jarociński and Karadi \(2020\)](#). The *equity_focused* variable denotes the indicator for firms that are in the top tercile of *FCE* and bottom tercile of *FCD*. The *debt_focused* variable denotes the indicator for firms that are in the top tercile of *FCD* and bottom tercile of *FCE*. All other group indicators are included in the regression, but omitted for brevity, except for the unconstrained group of firms. All regressions include industry and year fixed effects and control for log asset (size), book-to-market ratio, leverage, operating profitability, and cash holdings at the firm level. Robust standard errors clustered at the firm level are reported in parentheses. Significance codes: ***: 0.01, **: 0.05, *: 0.1.

Table 5. Monetary Policy and Innovation

	Log(Number of Patents Filed)	
	$h = 17$	$h = 20$
$mps \times FCE$	-0.011* (0.006)	-0.017*** (0.006)
$mps \times FCD$	-0.003 (0.007)	-0.004 (0.007)
Observations	39,634	36,079
Firm Controls	Yes	Yes
Firm FE	Yes	Yes
Fiscal Quarter FE	Yes	Yes
Industry $\times$ Time	Yes	Yes

This table reports coefficient estimates from estimating Equation (23). The dependent variable is the h -year change in the log of the number of patents filed. mps is the instrumented 1-year Treasury rate. The 1-year Treasury rate is instrumented by cumulative high-frequency monetary policy shocks measured as monetary policy shocks from the decomposition by [Jarociński and Karadi \(2020\)](#). The FCE and FCD variables denote the proxies for equity-focused and debt-focused constrained firms, respectively ([Hoberg and Maksimovic, 2015](#); [Linn and Weagley, 2023](#)). We report Driscoll-Kraay heteroscedasticity and autocorrelation robust standard errors in parenthesis. *, **, and *** indicate statistical significance at the 10%, 5%, and 1% levels, respectively.

Table 6. Effect of Monetary Policy Shocks on Financing Policies

	Δ Equity issuance/Assets (1)	Δ Equity issuance Adjusted/Assets (2)	Δ Repurchases/Assets (3)
<i>mps</i>	-0.0024*** (0.0005)	-0.0024*** (0.0005)	-0.0002*** (0.00007)
<i>mps</i> $\times$ <i>equity_focused</i>	-0.0023*** (0.0007)	-0.0026*** (0.0007)	0.0000 (0.00007)
<i>mps</i> $\times$ <i>debt_focused</i>	0.0002 (0.0004)	0.0003 (0.0004)	0.0001 (0.00009)
Observations	306,279	306,279	289,959
R ²	0.038	0.039	0.003
Firm Controls	Yes	Yes	Yes
Aggregate Controls	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes
Fiscal Quarter FE	Yes	Yes	Yes
Quarter $\times$ Sector FE	Yes	Yes	Yes

This table reports coefficient estimates from estimating Equation (24). The dependent variables are Δ Equity issuance/Assets, Δ Equity issuance Adjusted/Assets and Δ Repurchases/Assets (for details, see Section 4). *mps* is the instrumented 1-year Treasury rate. The 1-year Treasury rate is instrumented by cumulative high-frequency monetary policy shocks measured as monetary policy shocks from the decomposition by Jarociński and Karadi (2020). The *equity_focused* variable denotes the indicator for firms that are in the top tercile of *FCE* and bottom tercile of *FCD*. The *debt_focused* variable denotes the indicator for firms that are in the top tercile of *FCD* and bottom tercile of *FCE*. All other group indicators are included in the regression, but omitted for brevity except for the unconstrained group of firms. All regressions control for Q, leverage, size, cash flow, cash holdings, age, and dividend. Standard errors heteroskedasticity robust and clustered at the firm level are reported parentheses. *, **, and *** indicate statistical significance at the 10%, 5%, and 1% levels, respectively.

Table 7. Effect of Monetary Policy Shocks on Financing Policies and Cash Holdings

	Δ Public SEO issuance/Assets (1)	Δ Debt issuance/Assets (2)	Δ Cash (3)
<i>mps</i>	-0.001* (0.0006)	-0.0002 (0.0004)	-0.0029*** (0.0004)
<i>mps</i> $\times$ <i>equity_focused</i>	-0.002*** (0.0007)	-0.001** (0.0004)	0.0014** (0.0006)
<i>mps</i> $\times$ <i>debt_focused</i>	-0.000 (0.0004)	0.0003 (0.0004)	-0.0002 (0.0003)
Observations	314,614	293,471	316,593
R ²	0.013	0.014	0.0797
Firm Controls	Yes	Yes	Yes
Aggregate Controls	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes
Fiscal Quarter FE	Yes	Yes	Yes
Quarter $\times$ Sector FE	Yes	Yes	Yes

This table reports coefficient estimates from estimating Equation (24). The dependent variables are Δ Public SEO issuance/Assets and Δ Debt issuance/Assets (for details, see Section 4), and Δ Cash, the difference between cash at quarter t and cash at quarter $t - 1$, scaled by lagged total assets as in McLean (2011), where cash is measured as cash and short-term investments (COMPUSTAT's cheq). *mps* is the instrumented 1-year Treasury rate. The 1-year Treasury rate is instrumented by cumulative high-frequency monetary policy shocks measured as monetary policy shocks from the decomposition by Jarociński and Karadi (2020). The *equity_focused* variable denotes the indicator for firms that are in the top tercile of *FCE* and bottom tercile of *FCD*. The *debt_focused* variable denotes the indicator for firms that are in the top tercile of *FCD* and bottom tercile of *FCE*. All other group indicators are included in the regression, but omitted for brevity except for the unconstrained group of firms. All regressions control for Q, leverage, size, cash flow, cash holdings, age, and dividend. Standard errors heteroskedasticity robust and clustered at the firm level are reported in parentheses. *, **, and *** indicate statistical significance at the 10%, 5%, and 1% levels, respectively.

Table 8. High vs. Low Ex-Ante Transient Ownership Within FCE Firms

Variable	High Exposure		Low Exposure	
	Mean	SD	Mean	SD
Book leverage	0.153	(0.250)	0.167	(0.227)
Tobin's Q	3.161	(3.361)	2.485	(2.157)
Cash/Assets	0.373	(0.237)	0.309	(0.246)
Cash flow/Assets	0.005	(0.076)	−0.001	(0.085)
Log assets	5.531	(1.510)	5.949	(2.202)
Log age	2.172	(0.669)	2.508	(0.742)
Dividend indicator	0.067	(0.249)	0.094	(0.291)
R&D/Assets	0.044	(0.066)	0.044	(0.052)
CAPX/Assets	0.021	(0.034)	0.017	(0.036)
Log duration (Gonçalves)	4.172	(0.725)	4.076	(0.727)

High Exposure (Low Exposure) denotes firms with above-median (at-or-below-median) ex-ante transient institutional ownership within the pure equity-focused constrained group (top tercile of *FCE* and bottom tercile of *FCD*; see Section 4.1). Duration is measured following Gonçalves (2021); all other variables follow the definitions in Section C.

Internet Appendix for “The Equity Constraint Channel of Monetary Policy” by Heitor Almeida, Timothy Johnson, Sebastiao Oliveira and Yucheng Zhou

This internet appendix is organized as follows. Sections [A](#) and [B](#) provide details on the model derivation. Section [C](#) describes the firm-level variables used in our analysis. Section [D](#) shows the validity of our instrument of monetary policy measure used in the Section [5.3.2](#). Section [E](#) shows that the stock price results are robust to using different windows around FOMC meetings. Section [F](#) compares the results of stock price sensitivity with those in the literature. Section [G](#) presents additional results and robustness tests. Section [H](#) provides additional information on types of institutional investors.

A. Asset Market Equilibrium: Derivations

A.1 Equilibrium financing wedge

The PS investor’s problem [\(4\)](#) is strictly concave in $\mathbf{S}$ because $\Sigma \succ 0$, so the first-order condition [\(5\)](#) characterizes the unique optimum:

$$\mathbf{s} - \gamma \Sigma \mathbf{S} = \mathbf{0} \quad \Longleftrightarrow \quad \mathbf{S} = \frac{1}{\gamma} \Sigma^{-1} \mathbf{s}.$$

Substituting NPS demand [\(9\)](#) into market clearing [\(10\)](#) and using $\mathbf{R} \equiv \bar{\mathbf{S}} - \bar{\mathbf{D}}$,

$$\frac{1}{\gamma} \Sigma^{-1} \mathbf{s} + H \mathbf{s} = \mathbf{R}.$$

Premultiplying by $\gamma \Sigma$,

$$[I + \gamma \Sigma H] \mathbf{s} = \gamma \Sigma \mathbf{R},$$

which is (11). The matrix $I + \gamma \Sigma H$ is invertible: writing it out,

$$I + \gamma \Sigma H = \begin{pmatrix} 1 + \gamma \sigma_E^2 \eta_E & \gamma \sigma_{EB} \eta_B \\ \gamma \sigma_{EB} \eta_E & 1 + \gamma \sigma_B^2 \eta_B \end{pmatrix},$$

with determinant

$$\Delta \equiv \left(1 + \gamma \sigma_E^2 \eta_E\right) \left(1 + \gamma \sigma_B^2 \eta_B\right) - \gamma^2 \sigma_{EB}^2 \eta_E \eta_B. \quad (25)$$

Expanding the product,

$$\Delta = 1 + \gamma \sigma_E^2 \eta_E + \gamma \sigma_B^2 \eta_B + \gamma^2 \eta_E \eta_B \det(\Sigma),$$

where $\det(\Sigma) = \sigma_E^2 \sigma_B^2 - \sigma_{EB}^2 > 0$ by positive definiteness. Every term beyond the first is non-negative, so $\Delta \geq 1 > 0$ and (11) is well defined.

A.2 Level ordering of the wedges: $s_E > s_B > 0$

From the PS pricing condition (5), evaluated at the equilibrium holdings $\mathbf{S} = (S_E, \tilde{S}_B)'$,

$$s_E = \gamma \left(\sigma_E^2 S_E + \sigma_{EB} \tilde{S}_B \right), \quad (26)$$

$$s_B = \gamma \left(\sigma_{EB} S_E + \sigma_B^2 \tilde{S}_B \right). \quad (27)$$

Subtracting,

$$s_E - s_B = \gamma \left[\left(\sigma_E^2 - \sigma_{EB} \right) S_E - \left(\sigma_B^2 - \sigma_{EB} \right) \tilde{S}_B \right], \quad (28)$$

which is positive under exactly the same condition (14) that governs the ordering of the responses (see Lemma 1 below). The same condition therefore delivers both the level ordering $s_E > s_B$ (which underlies the pecking order in the firm problem) and the response ordering $\rho_E > \rho_B$. Positivity of s_B follows from (27) whenever $\sigma_{EB} > -\sigma_B^2 \tilde{S}_B / S_E$; a sufficient condition is $\sigma_{EB} \geq 0$.

Lemma 1. Suppose $\Sigma \succ 0$, $\sigma_E^2 > \sigma_B^2 > 0$, and $S_E > \tilde{S}_B > 0$. Then

$$\left(\sigma_E^2 - \sigma_{EB} \right) S_E > \left(\sigma_B^2 - \sigma_{EB} \right) \tilde{S}_B$$

for any admissible covariance σ_{EB} .

Proof. Positive definiteness of Σ implies $|\sigma_{EB}| < \sigma_E \sigma_B$. Combined with $\sigma_B < \sigma_E$, this gives $\sigma_{EB} < \sigma_E \sigma_B < \sigma_E^2$, so $\sigma_E^2 - \sigma_{EB} > 0$ always. Consider two cases.

Case 1: $\sigma_{EB} \leq \sigma_B^2$. Then $\sigma_B^2 - \sigma_{EB} \geq 0$. The risk asymmetry $\sigma_E^2 > \sigma_B^2$ implies $\sigma_E^2 - \sigma_{EB} > \sigma_B^2 - \sigma_{EB}$, and therefore

$$(\sigma_E^2 - \sigma_{EB}) S_E > (\sigma_B^2 - \sigma_{EB}) S_E \geq (\sigma_B^2 - \sigma_{EB}) \tilde{S}_B,$$

where the first inequality uses $S_E > 0$ and the second uses $S_E > \tilde{S}_B$ together with $\sigma_B^2 - \sigma_{EB} \geq 0$.

Case 2: $\sigma_{EB} > \sigma_B^2$. Then the right-hand side is strictly negative while the left-hand side is strictly positive, so the inequality holds trivially. $\square$

A.3 Responses to the monetary policy shock

Define the vector of wedge responses

$$\boldsymbol{\rho} \equiv \frac{\partial \mathbf{s}}{\partial \varepsilon^{MP}} = \begin{pmatrix} \rho_E \\ \rho_B \end{pmatrix}. \quad (29)$$

Totally differentiating the PS pricing condition $\mathbf{s} = \gamma \Sigma \mathbf{S}$ with respect to ε^{MP} , holding Σ fixed by (3),

$$\boldsymbol{\rho} = \gamma_\varepsilon \Sigma \mathbf{S} + \gamma \Sigma \frac{\partial \mathbf{S}}{\partial \varepsilon^{MP}}. \quad (30)$$

The first term is the direct effect of tighter PS risk-bearing capacity; the second is the endogenous reallocation of holdings. From market clearing, with $\mathbf{R}$ and H fixed,

$$\frac{\partial \mathbf{S}}{\partial \varepsilon^{MP}} = -H \boldsymbol{\rho}. \quad (31)$$

Substituting (31) into (30) and collecting terms yields

$$[I + \gamma \Sigma H] \boldsymbol{\rho} = \gamma_\varepsilon \Sigma \mathbf{S}, \quad (32)$$

and hence

$$\boldsymbol{\rho} = [I + \gamma \Sigma H]^{-1} \gamma_\epsilon \Sigma \mathbf{S}. \quad (33)$$

Using the explicit inverse of $I + \gamma \Sigma H$ and $\det(\Sigma) = \sigma_E^2 \sigma_B^2 - \sigma_{EB}^2$, the scalar responses are

$$\rho_E = \frac{\gamma_\epsilon}{\Delta} \left[\sigma_E^2 S_E + \sigma_{EB} \tilde{S}_B + \gamma \eta_B \det(\Sigma) S_E \right], \quad (34)$$

$$\rho_B = \frac{\gamma_\epsilon}{\Delta} \left[\sigma_{EB} S_E + \sigma_B^2 \tilde{S}_B + \gamma \eta_E \det(\Sigma) \tilde{S}_B \right]. \quad (35)$$

Setting $\eta_E = \eta_B = 0$ (so $\Delta = 1$) yields the leading-order responses for weakly elastic NPS demand,

$$\rho_E \approx \gamma_\epsilon \left(\sigma_E^2 S_E + \sigma_{EB} \tilde{S}_B \right), \quad (36)$$

$$\rho_B \approx \gamma_\epsilon \left(\sigma_{EB} S_E + \sigma_B^2 \tilde{S}_B \right). \quad (37)$$

The holdings responses (16) follow componentwise from (31).

A.4 Proof of the response ordering $\rho_E > \rho_B > 0$

To leading order in (η_E, η_B) , the difference of (34) and (35) is

$$\rho_E - \rho_B \approx \gamma_\epsilon \left[\left(\sigma_E^2 - \sigma_{EB} \right) S_E - \left(\sigma_B^2 - \sigma_{EB} \right) \tilde{S}_B \right],$$

which is strictly positive by Lemma 1 under $\sigma_E^2 > \sigma_B^2$ and $S_E > \tilde{S}_B$. Positivity of ρ_B follows from (35): the bracket is positive whenever $\sigma_{EB} > -\sigma_B^2 \tilde{S}_B / S_E$ (sufficient: $\sigma_{EB} \geq 0$), and $\gamma_\epsilon > 0$ by (13). For strictly positive but small η_a , the ordering is preserved by continuity of (34)–(35) in (η_E, η_B) . $\square$

A.5 Proof of Prediction 1

From (16), $|\partial S_E / \partial \varepsilon^{MP}| = \eta_E \rho_E$ and $|\partial \tilde{S}_B / \partial \varepsilon^{MP}| = \eta_B \rho_B$. Suppose $\eta_B \leq \eta_E$ with $\eta_E > 0$, and $\rho_E > \rho_B > 0$. If $\eta_B > 0$, then

$$\eta_E \rho_E \geq \eta_B \rho_E > \eta_B \rho_B.$$

If $\eta_B = 0$, then $\eta_E \rho_E > 0 = \eta_B \rho_B$. In either case (17) holds strictly. $\square$

B. Firm Problem: First-Order Conditions and Regime Characterization

B.1 Lagrangian and KKT conditions

Writing the firm problem (12) with internal funds used N_i as an explicit choice variable, the Lagrangian is

$$\begin{aligned} \mathcal{L} = & f_i(I_i) - (1+r)N_i - (1+r+s_B)B_i - (1+r+s_E)E_i \\ & + \xi_i(N_i + B_i + E_i - I_i) + \nu_{W,i}(W_i - N_i) + \nu_{B,i}(\mu K_i - B_i) \\ & + \lambda_{N,i}N_i + \lambda_{B,i}B_i + \lambda_{E,i}E_i, \end{aligned}$$

where ξ_i is the multiplier on the budget identity, $\nu_{W,i} \geq 0$ and $\nu_{B,i} \geq 0$ are the multipliers on the internal-funds and collateral constraints, and $\lambda_{N,i}, \lambda_{B,i}, \lambda_{E,i} \geq 0$ are the multipliers on the non-negativity constraints. The first-order conditions are

$$[I_i]: f'_i(I_i^*) = \xi_i, \tag{38}$$

$$[N_i]: \xi_i = (1+r) + \nu_{W,i} - \lambda_{N,i}, \tag{39}$$

$$[B_i]: \xi_i = (1+r+s_B) + \nu_{B,i} - \lambda_{B,i}, \tag{40}$$

$$[E_i]: \xi_i = (1+r+s_E) - \lambda_{E,i}, \tag{41}$$

together with complementary slackness: $\nu_{W,i}(W_i - N_i^*) = 0$, $\nu_{B,i}(\mu K_i - B_i^*) = 0$, and $\lambda_{x,i} x_i^* = 0$ for $x \in \{N, B, E\}$. The shadow price of funds ξ_i equals the firm's marginal cost of capital at the optimum, and (38) equates it to the marginal product of investment.

B.2 Regime characterization

Unconstrained regime ($N_i^* < W_i$, $B_i^* = E_i^* = 0$). Slackness of the internal-funds constraint gives $\nu_{W,i} = 0$, and interior $N_i^* > 0$ gives $\lambda_{N,i} = 0$; hence $\xi_i = 1 + r$ from (39) and $f'_i(I_i^*) = 1 + r$ from (38). The corners $B_i^* = E_i^* = 0$ are optimal because, from (40) and (41) with $\nu_{B,i} = 0$,

$$\lambda_{B,i} = (1 + r + s_B) - \xi_i = s_B > 0, \quad \lambda_{E,i} = (1 + r + s_E) - \xi_i = s_E > 0.$$

FCD regime ($N_i^* = W_i$, $0 < B_i^* < \mu K_i$, $E_i^* = 0$). Interior B_i^* gives $\lambda_{B,i} = 0$ and slack collateral gives $\nu_{B,i} = 0$, so $\xi_i = 1 + r + s_B$ from (40) and $f'_i(I_i^*) = 1 + r + s_B$. From (39) with $\lambda_{N,i} = 0$,

$$\nu_{W,i} = \xi_i - (1 + r) = s_B > 0,$$

confirming that the internal-funds constraint binds with shadow price s_B . The corner $E_i^* = 0$ is optimal because $\lambda_{E,i} = (1 + r + s_E) - \xi_i = s_E - s_B > 0$.

FCE regime ($N_i^* = W_i$, $B_i^* = \mu K_i$, $E_i^* > 0$). Interior E_i^* gives $\lambda_{E,i} = 0$, so $\xi_i = 1 + r + s_E$ from (41) and $f'_i(I_i^*) = 1 + r + s_E$. From (39) and (40) with $\lambda_{N,i} = \lambda_{B,i} = 0$,

$$\nu_{W,i} = s_E > 0, \quad \nu_{B,i} = \xi_i - (1 + r + s_B) = s_E - s_B > 0.$$

Both the internal-funds constraint and the collateral constraint bind, with shadow prices s_E and $s_E - s_B$, respectively.

In the FCD regime, an extra dollar of internal wealth replaces a dollar of borrowing and saves the wedge s_B ; in the FCE regime it replaces a dollar of equity issuance and saves s_E . In the FCE regime, an extra dollar of collateral capacity allows the firm to substitute debt for equity at the margin, saving $s_E - s_B$.

Two additional configurations arise at the kinks of the marginal-cost schedule. A firm with $f'_i(W_i) \in [1 + r, 1 + r + s_B]$ invests exactly $I_i^* = W_i$: it exhausts internal funds but does not access external finance. A firm with $f'_i(W_i + \mu K_i) \in [1 + r + s_B, 1 + r + s_E]$ invests exactly $I_i^* = W_i + \mu K_i$: it exhausts debt capacity but does not issue equity, and the collateral multiplier is $\nu_{B,i} = f'_i(W_i + \mu K_i) - (1 + r + s_B) \in [0, s_E - s_B]$. In both configurations, investment is locally pinned by the resource limit rather than by a marginal condition, so the response to a small monetary policy shock is zero. These firms are inframarginal to our analysis: the FCE and FCD regimes compared in Proposition 1 are the interior configurations in which the firm is actively issuing equity or debt at the margin, matching the empirical classification, which identifies firms by their reliance on equity or debt financing at the margin. The FCE group is unaffected by these configurations, since the empirical classification requires active reliance on equity issuance. To the extent that collateral-kink firms enter the empirical FCD group, their muted responses attenuate the measured FCD response, making the FCE–FCD comparison conservative.

B.3 Proof of Proposition 1

In each regime g , the first-order condition is $f'_i(I_i^*) = 1 + r + s_g$, where $s_{Unc} \equiv 0$, $s_{FCD} \equiv s_B$, and $s_{FCE} \equiv s_E$. Totally differentiating with respect to ε^{MP} , holding regime membership fixed (a local comparative static),

$$f''_i(I_i^*) \frac{dI_i^*}{d\varepsilon^{MP}} = r' + \rho_g, \quad \text{so} \quad \left| \frac{dI_i^*}{d\varepsilon^{MP}} \right|_g = \frac{|r' + \rho_g|}{|f''_i(I_i^*)|},$$

with $\rho_{Unc} = 0$, $\rho_{FCD} = \rho_B$, $\rho_{FCE} = \rho_E$. For a contractionary shock, $r' \geq 0$ and $\rho_E > \rho_B > 0$ (Appendix A), so the numerators are strictly ordered:

$$r' + \rho_E > r' + \rho_B > r' \geq 0.$$

Under the comparable-curvature benchmark $|f''_i(I_{Unc}^*)| \approx |f''_i(I_{FCD}^*)| \approx |f''_i(I_{FCE}^*)|$, the ordering of the responses follows immediately from the ordering of the numerators. $\square$

Without the benchmark, note that $s_E > s_B > 0$ implies $I_{FCE}^* < I_{FCD}^* < I_{Unc}^*$, and under standard concavity $|f''_i|$ is weakly larger at lower investment levels. The de-

nominator effect therefore attenuates the FCE and FCD responses relative to the unconstrained response. The $FCE > FCD$ leg of the ordering is preserved whenever the numerator gap $\rho_E - \rho_B$ dominates the (typically second-order) difference in curvature between the two constrained optima; the $FCD > Unc$ leg is preserved whenever ρ_B/r' exceeds the relative curvature difference between the FCD and unconstrained optima. Both conditions hold in the benchmark and for production functions whose curvature varies slowly over the relevant range of investment levels.

C. Firm-level Variables

CAPX is quarterly capital expenditures (COMPUSTAT's capxy). *R&D* is quarterly R&D (COMPUSTAT's xrdq). *Public SEO* is quarterly SDC Platinum SEO dollars raised. *Equity Issuance* is quarterly sale of common and preferred stock (COMPUSTAT's sstky). *Repurchases* is quarterly purchase of common and preferred stock (COMPUSTAT's prstkcy). *Debt issuance* is quarterly newly issued long-term debt (COMPUSTAT's dltis). *Cash flow* represents the ratio of operating income before depreciation (COMPUSTAT's oibdpq) to the lag of total assets. *Size* is given by the logarithm of total assets. *Cash holdings* are measured as the ratio of cash and short-term investments (COMPUSTAT's cheq) to total assets. *Total debt* is long-term debt (COMPUSTAT's dlttq) plus debt in Current Liabilities (COMPUSTAT's dlcq). *Book leverage* denotes the ratio of total debt to total assets. *Long-term leverage* is long-term debt maturing within one-year (COMPUSTAT's dd1q) plus long-term debt (COMPUSTAT's dlttq) divided by total assets. *Maturity* is long-term debt (COMPUSTAT's dlttq) divided by *Total debt*. *Q* is defined as the ratio of total assets plus market capitalization minus common equity minus deferred taxes and investment tax credit ($atq + prccq \times cshoq - ceqq - txditcq$) to total assets (atq). *Age* is the number of years since a firm first appears in Compustat. *Dividend* is a dummy whether $dvtq > 0$ in a given quarter. *RFC* is defined as the ratio of long term debt maturing within one year (COMPUSTAT's dd1q) to the sum of long term debt maturing within one year (COMPUSTAT's dd1q) and long-term debt (COMPUSTAT's dlttq).²³

²³The variables capxy, sstky, and prstkcy represent "year-to-date". We adjust these variables to reflect quarterly values.

D. Real Effects and Monetary Policy Measure

In this section, we show that cumulative “pure” monetary policy shocks are a strong instrument for the 1-year Treasury rate. Figure [D.1](#) plots the predicted 1-year treasury rate, and Table [D.1](#) reports the results from the first stage regression. The statistically significant coefficient estimates on the cumulative high-frequency “pure” monetary policy shocks (JK shock) and the “F stat IV” confirm we have a strong instrument for the 1-year Treasury rate.

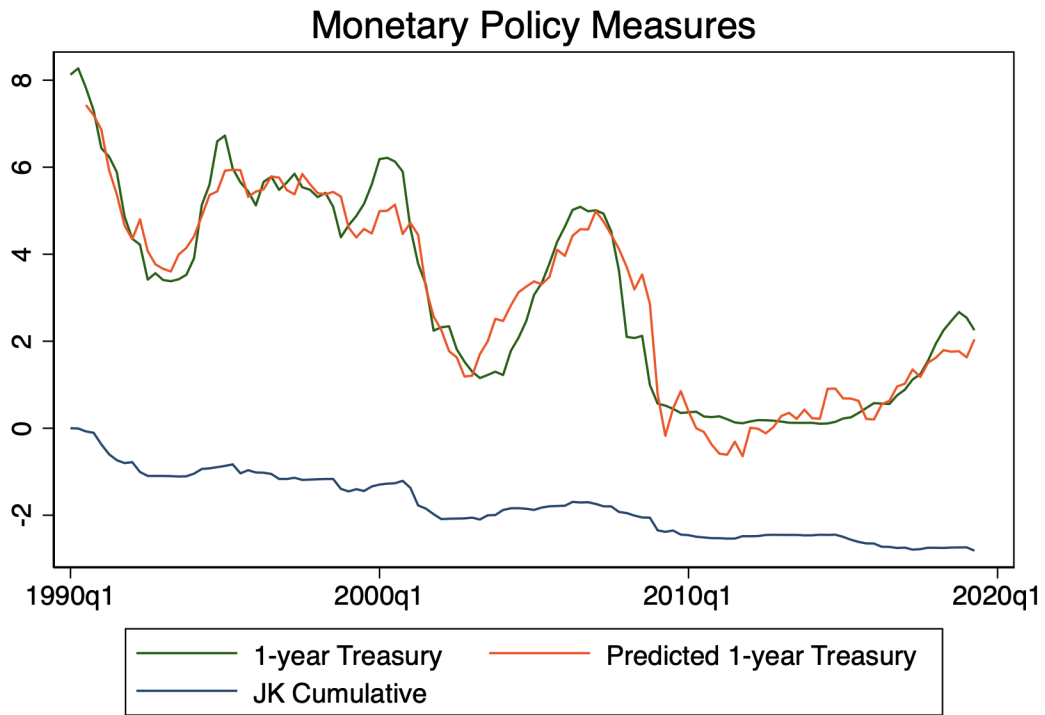
As a robustness exercise, we also instrument the 1-year Treasury rate without controlling for lagged macroeconomic variables. Figure [D.2](#) shows our results remain quantitatively the same for equity-focused constrained firms, underscoring the equity constraint channel of monetary policy. Figure [D.3](#) presents the results for debt-focused constrained firms. The amplification effect on CAPX and R&D is not statistically significant.

Appendix Table D.1. First Stage Regression

	<i>1yt</i>
JK shock	3.45*** (0.50)
Log CPI	16.0*** (3.56)
Log Industrial Production	-9.78*** (2.75)
Log Employment Ratio	54.0*** (7.77)
Excess Bond Premium	-0.50*** (0.18)
GDP Growth	0.0018* (0.0010)
Observations	116
F stat all	232
F stat IV	48.1

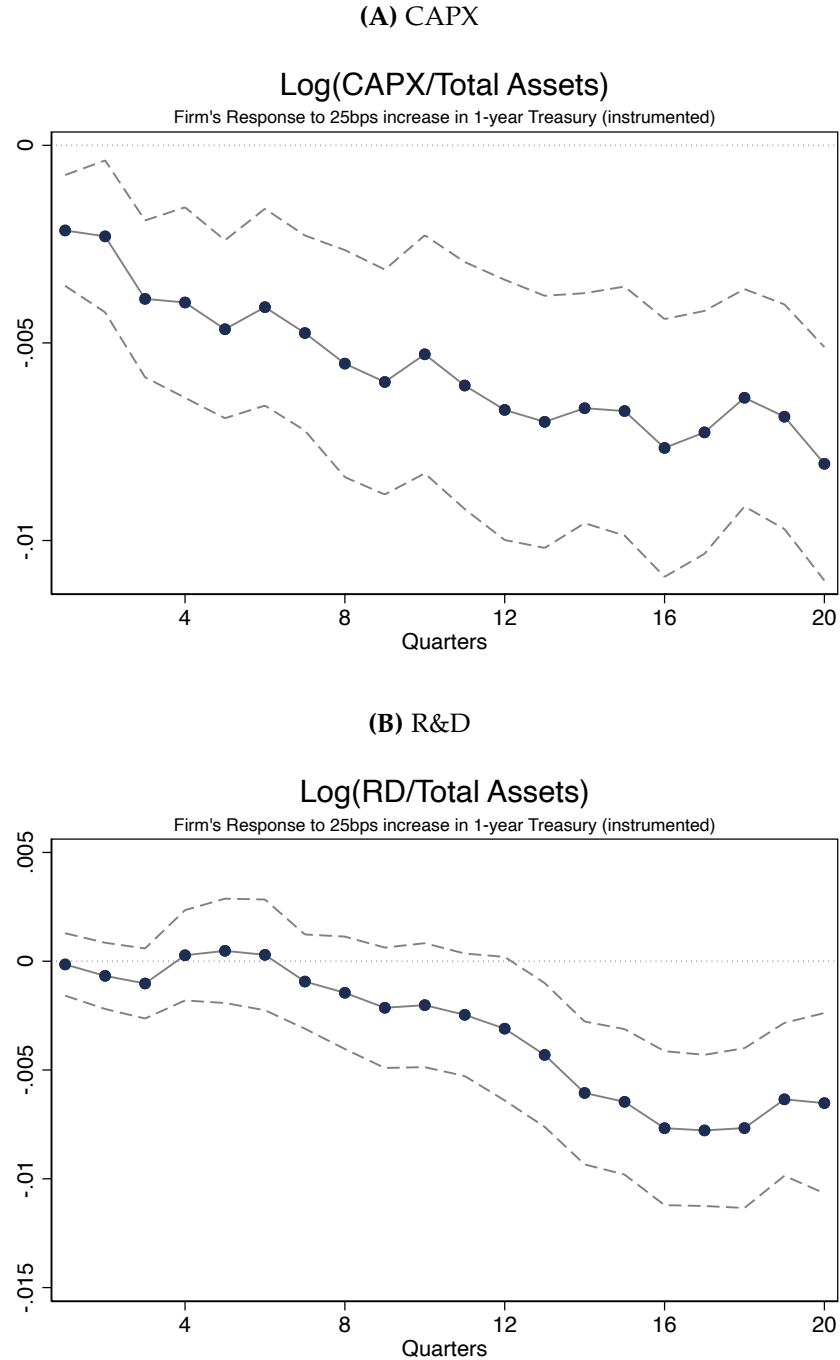
This table reports the results from the first-stage regression. The dependent variable is the 1-year Treasury rate and the instrument is the cumulative high-frequency shocks from [Jarociński and Karadi \(2020\)](#), lagged by one quarter. Newey-West standard errors are reported in parentheses. F statistics are reported for all variables and the instrument, respectively.

Appendix Figure D.1. Monetary Policy Measure (Instrumented)



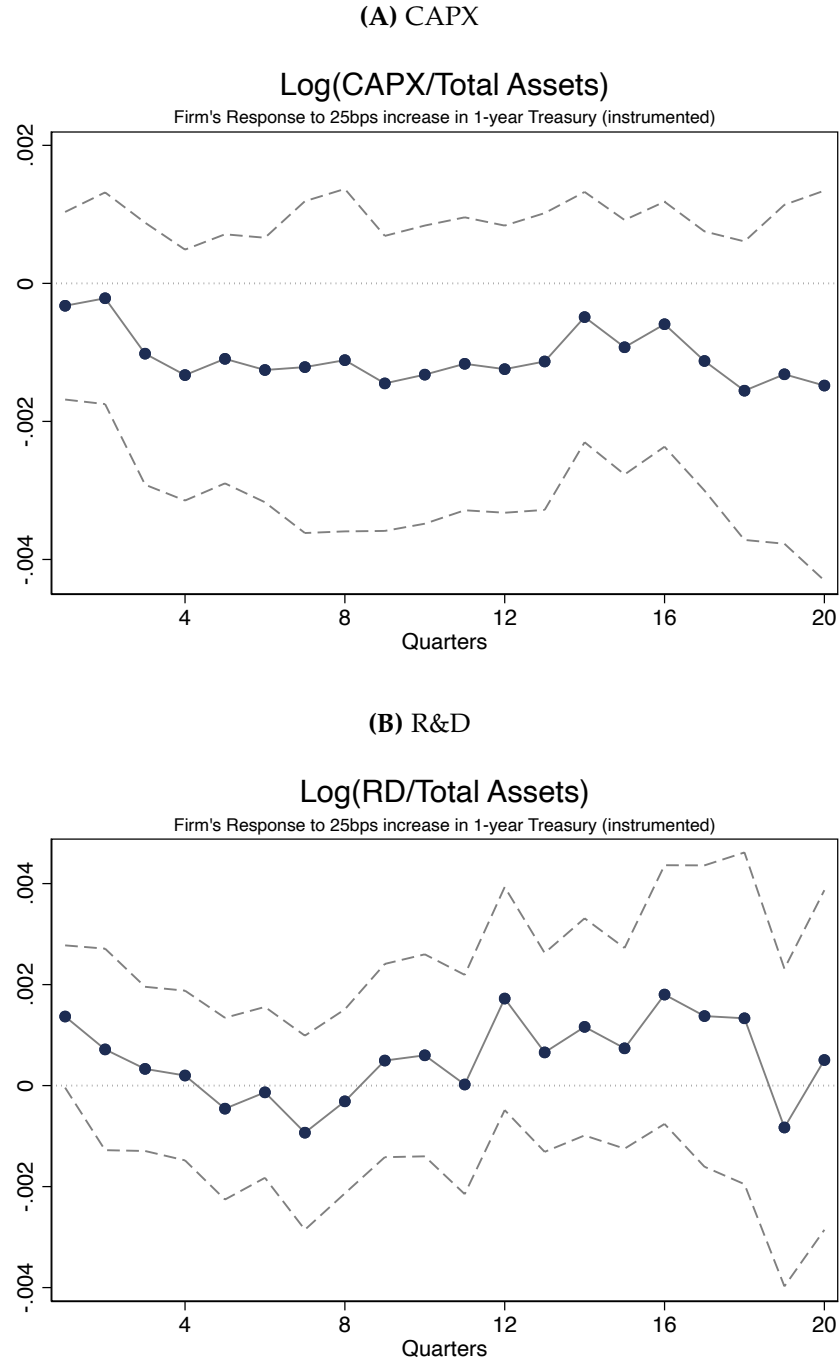
This figure plots the 1-year Treasury rate and the predicted 1-year Treasury rate predicted rate from the first-stage regression with cumulative [Jarociński and Karadi \(2020\)](#) shocks and macroeconomic control variables

Appendix Figure D.2. Equity-Focused Constraints and the Dynamic Response of Investment to Monetary Policy: Alternative Instrument



This figure shows the Impulse Response Function (IRF) for the response of CAPX, and R&D to a 25 bps increase in 1-year Treasury (instrumented), estimated using Equation (23). The 1-year Treasury rate is instrumented by cumulative high-frequency monetary policy shocks measured as monetary policy shocks from the decomposition by [Jarociński and Karadi \(2020\)](#), without controlling for macroeconomic variables. Each point represents the point estimate of the coefficient of the instrumented 1-year Treasury rate interacted with the financing constraint measure (β_2^h in Equation (23)). The dashed line represents 90% confidence intervals using heteroscedasticity and autocorrelation robust Driscoll-Kraay standard errors.

Appendix Figure D.3. Debt-Focused Constraints and the Dynamic Response of Investment to Monetary Policy: Alternative Instrument



This figure shows the Impulse Response Function (IRF) for the response of CAPX, and R&D to a 25 bps increase in 1-year Treasury (instrumented), estimated using Equation (23). The 1-year Treasury rate is instrumented by cumulative high-frequency monetary policy shocks measured as monetary policy shocks from the decomposition by [Jarociński and Karadi \(2020\)](#), without controlling for macroeconomic variables. Each point represents the point estimate of the coefficient of the instrumented 1-year Treasury rate interacted with the financing constraint measure $FCD_{ij,t-1}$. The dashed line represents 90% confidence intervals using heteroscedasticity and autocorrelation robust Driscoll-Kraay standard errors.

E. Stock Price: Additional Windows

We complement our findings of Section 5.3.1 by using different windows of stock returns in Equation (21). Table E.1 presents the results. The dependent variable in Column (1) is daily return on the day of FOMC announcements. Columns (2) and (3) report the estimates using daily returns one day after, and two days after the FOMC announcements as the dependent variable, respectively. Estimates from Table E.1 indicate that equity-focused constraints significantly amplify stock prices response. When there is a positive 25 bps monetary policy surprise, equity-focused constrained firms have an average realized return that is 0.745% lower than that of the unconstrained firms on the day of FOMC announcement, even after controlling for leverage, book-to-market ratio, size, operating profitability, and cash holdings. The response corresponds to 34.8% of the average stock price response on the day of FOMC announcement. Debt-focused constrained firms experience a 0.458% lower return than unconstrained firms do, representing 21.4% of the average response. The heterogeneous impact remains significant for daily returns 2 days after FOMC announcements.

Appendix Table E.1. Heterogeneous Stock Price Response to Monetary Policy Shocks

Window:	(0,0) (1)	(+1,+1) (2)	(+2,+2) (3)
$mps \times equity_focused$	-0.179*** (0.039)	-0.026 (0.041)	-0.119*** (0.038)
$mps \times debt_focused$	-0.110*** (0.032)	0.035 (0.033)	-0.102*** (0.030)
$mps \times leverage$	0.312*** (0.049)	0.070 (0.082)	-0.227*** (0.053)
$mps \times bm$	0.044*** (0.014)	-0.029* (0.016)	-0.069*** (0.014)
$mps \times size$	-0.082*** (0.006)	-0.008* (0.005)	0.064*** (0.005)
$mps \times profitability$	0.058*** (0.016)	0.128*** (0.038)	-0.012 (0.020)
$mps \times cashholding$	-0.381*** (0.072)	-0.049 (0.055)	0.129** (0.059)
<i>Fixed-effects</i> industry-year	Yes	Yes	Yes
<i>Fit statistics</i>			
Observations	844,031	795,949	728,257
R ²	0.020	0.016	0.015

This table reports the coefficient estimates of the cross-sectional regression by pooling all firm level returns around FOMC announcements from 1991 to 2019. The firm level returns are calculated over three different event windows and shown in Columns (1) to (3). Column (1) reports the results when we use the daily return on the day of the FOMC announcements as the dependent variable. Column (2) reports the results when we use the daily return 1 day after the FOMC announcements as the dependent variable. Column (3) reports the results when we use the daily return 2 days after the FOMC announcements as the dependent variable. The *mps* variable denotes monetary policy shock from [Jarociński and Karadi \(2020\)](#). The *equity_focused* variable denotes the indicator for firms that are in the top tercile of *FCE* and bottom tercile of *FCD*. The *debt_focused* variable denotes the indicator for firms that are in the top tercile of *FCD* and bottom tercile of *FCE*. All other group indicators are included in the regression, but omitted for brevity, except for the unconstrained group of firms. All regressions include industry and year fixed effects and control for log asset (size), book-to-market ratio, leverage, operating profitability, and cash holdings at the firm level. Robust standard errors clustered at the firm level are reported in parentheses. Significance codes: ***: 0.01, **: 0.05, *: 0.1.

F. Stock Price: Comparison with Literature

In this section, we discuss how our paper is related to two papers of the literature in more details. [Ozdagli \(2017\)](#) finds that financially constrained firms have relatively lower returns than unconstrained firms on the day of FOMC announcement when there is an expansionary monetary shock, as the stock price of these firms respond less to such a surprise. In a contemporaneous work, [Chava and Hsu \(2020\)](#) show that financially constrained firms have higher response to monetary surprises, e.g., these firms have higher relative returns when the Fed decreases the rate unexpectedly. Financial constraints are defined using [Whited and Wu \(2006\)](#) index, and the monetary policy shock is identified as high-frequency change of fed funds futures contract. On the surface, these two papers contradict with each other, and they indeed differ in several aspects, e.g., sample period and empirical specification. Moreover, [Chava and Hsu \(2020\)](#) successfully replicate the findings in [Ozdagli \(2017\)](#), but further conclude that such an effect diminishes in a few days after the FOMC announcement and eventually goes in the opposite direction, which indicates that the result of [Ozdagli \(2017\)](#) might well be a result of delayed reaction of market participants, i.e., illiquidity among stocks of financially constrained firms.

We want to first note that, we include the zero lower bound (ZLB) in order to study the stock price response in a sample period as long as possible and to gain more statistical power. Second, we use monetary policy shocks from [Jarociński and Karadi \(2020\)](#), who isolate the "information effect" of high-frequency identification of monetary policy shocks. The two papers mentioned above construct monetary policy shock as price changes of current fed funds rate futures contract, while [Jarociński and Karadi \(2020\)](#) and others use "FF4", i.e., the change in the three-month fed funds future, which is less sensitive to the "timing surprises", i.e., a short-term advancement or postponement of a widely expected policy decision.²⁴ Third, the proxy for financial constraints is another important dimension that we differ from the above two papers (see Section 4 for more discussion). Lastly, we control for the firm characteristics and their interactions with the monetary policy shocks to isolate the potential impact of these characteristics.

²⁴As a robustness check, we also use series of monetary policy shocks from [Bauer and Swanson \(2023\)](#), and the results are qualitatively similar (see Section G)

We also cluster standard errors at firm and FOMC date level as [Chava and Hsu \(2020\)](#) do. Table [F.1](#) shows that the significance of the coefficients on the interaction terms survive when we focus on the cumulative returns. The equity-focused constraint significantly amplifies the negative stock price response to monetary policy shocks. We also control for industry and FOMC date fixed effects and cluster standard errors at industry and FOMC date level. The equity constraint channel is still statistically significant, as shown in Table [F.2](#).

For comparison, we also construct financial constraint measures following [Kaplan and Zingales \(1997\)](#), [Whited and Wu \(2006\)](#), [Hadlock and Pierce \(2010\)](#).²⁵ First, these indices do not seem to align well in that the coefficients on the interaction term now are sensitive to the controls and specification. The results are not consistent across these indices as well. For example, the KZ-index seems to produce a positive but not significant heterogeneous impact, while WW-index and HP-index produce a significantly negative coefficient on the day of FOMC announcements, which does not persist in a cumulative return analysis.

²⁵Results are not included in the paper for brevity, but available upon request.

Appendix Table F.1. Cumulative Heterogeneous Stock Price Response to Monetary Policy Shocks: Two-way Cluster

Window:	(0,+1) (1)	(0,+2) (2)	(0,+5) (3)
$mps \times equity_focused$	-0.201* (0.106)	-0.292** (0.145)	-0.481*** (0.166)
$mps \times debt_focused$	-0.077 (0.058)	-0.183** (0.078)	-0.153 (0.107)
$mps \times leverage$	0.350** (0.145)	0.210 (0.197)	0.122 (0.212)
$mps \times bm$	0.011 (0.101)	-0.031 (0.129)	-0.130 (0.168)
$mps \times size$	-0.091*** (0.025)	-0.040 (0.038)	-0.005 (0.040)
$mps \times profitability$	0.182** (0.075)	0.175** (0.080)	0.262*** (0.098)
$mps \times cashholding$	-0.449** (0.198)	-0.344 (0.307)	-0.175 (0.329)
<i>Fixed-effects</i>			
sic3-year	Yes	Yes	Yes
<i>Std. error cluster</i>			
Firm-date	Yes	Yes	Yes
<i>Fit statistics</i>			
Observations	843,764	843,501	842,718
R ²	0.025	0.024	0.030

This table reports the coefficient estimates of the cross-sectional regression by pooling all firm level returns around FOMC announcements from 1991 to 2019. The firm level cumulative returns are calculated over three different event windows and shown in Columns (1) to (3). Columns (1) to (3) report the results when we use the 1-, 2-, and 5-day cumulative returns since the FOMC announcements as the dependent variable, respectively. The *mps* variable denotes monetary policy shock from [Jarociński and Karadi \(2020\)](#). The *equity_focused* variable denotes the indicator for firms that are in the top tercile of *FCE* and bottom tercile of *FCD*. The *debt_focused* variable denotes the indicator for firms that are in the top tercile of *FCD* and bottom tercile of *FCE*. All other group indicators are included in the regression, but omitted for brevity, except for the unconstrained group of firms. All regressions include industry and year fixed effects and control for log asset (size), book-to-market ratio, leverage, operating profitability, cash holding at the firm level. Robust standard errors clustered at the firm and FOMC announcement date levels are in reported parentheses. Significance codes: ***: 0.01, **: 0.05, *: 0.1.

Appendix Table F.2. Cumulative Heterogeneous Stock Price Response to Monetary Policy Shocks: Industry-Date FE

Window:	(0,+1) (1)	(0,+2) (2)	(0,+5) (3)
$mps \times equity_focused$	-0.321*** (0.049)	-0.330*** (0.074)	-0.425*** (0.092)
$mps \times debt_focused$	-0.055 (0.049)	-0.141** (0.055)	-0.134** (0.061)
$mps \times leverage$	-0.019 (0.148)	-0.058 (0.180)	-0.201 (0.236)
$mps \times bm$	-0.022 (0.085)	-0.088 (0.110)	-0.161 (0.132)
$mps \times size$	-0.052** (0.024)	-0.026 (0.042)	-0.023 (0.044)
$mps \times profitability$	0.122** (0.053)	0.137** (0.061)	0.232*** (0.062)
$mps \times cashholding$	-0.165 (0.169)	-0.179 (0.261)	-0.191 (0.303)
<i>Fixed-effects</i> sic3-date	Yes	Yes	Yes
<i>Std. error cluster</i> sic3-date	Yes	Yes	Yes
<i>Fit statistics</i> Observations	843,764	843,501	842,718
R ²	0.127	0.129	0.148

This table reports the coefficient estimates of the cross-sectional regression by pooling all firm level returns around FOMC announcements from 1991 to 2019. The firm level cumulative returns are calculated over three different event windows and shown in Columns (1) to (3). Columns (1) to (3) report the results when we use the 1-, 2-, and 5-day cumulative returns since the FOMC announcements as the dependent variable, respectively. The *mps* variable denotes monetary policy shock from [Jarociński and Karadi \(2020\)](#). The *equity_focused* variable denotes the indicator for firms that are in the top tercile of *FCE* and bottom tercile of *FCD*. The *debt_focused* variable denotes the indicator for firms that are in the top tercile of *FCD* and bottom tercile of *FCE*. All other group indicators are included in the regression, but omitted for brevity, except for the unconstrained group of firms. All regressions include industry-FOMC-announcement-date fixed effects and control for log asset (size), book-to-market ratio, leverage, operating profitability, and cash holding at the firm level. Robust standard errors clustered at the industry and FOMC announcement date levels are reported in parentheses. Significance codes: ***: 0.01, **: 0.05, *: 0.1.

G. Additional Results and Robustness

This section rules-out potential alternative mechanisms that may explain why equity-focused constraints amplify the effect of monetary policy shocks and provide robustness exercises for the response of stock prices and investment policy to monetary policy.

G.1 Equity and Debt Financing Shocks

A potential concern with the notion that changes in debt financing terms may not be a primary mechanism of monetary policy transmission is that we also observe a reduction in debt issuance by equity-focused constrained firms, albeit to a lesser extent. This might indicate that the transmission of monetary policy could still operate through debt financing rather than equity financing channel.

We address this concern as follows. Using aggregate level financing shocks for both equity (EIS) and debt (DIS) market from [Belo et al. \(2024\)](#)²⁶, we estimate the heterogeneous impact of these two financing shocks on firms' investment. We expect that investment policies of equity-focused constrained firms mainly react to EIS. That is, in contrast to EIS, DIS should weakly affect the investment of equity-focused constrained firms. If this is the case, it is very unlikely that monetary policy affects firms' investment decisions by impacting the debt issuance of equity-focused constrained firms.

We test our hypothesis formally using the following specification:

$$\begin{aligned} y_{ij,t+h} - y_{ij,t-1} = & \beta_1^h FCE_{ij,t-1} + \beta_2^h FCE_{ij,t-1} \times EIS_t + \beta_3^h FCE_{ij,t-1} \times DIS_t \\ & + \gamma_1^{h'} Z_{ij,t-1} \times EIS_t + \gamma_2^{h'} Z_{ij,t-1} \times DIS_t + \gamma_3^{h'} Z_{ij,t-1} + \alpha_i + \eta_{jt} + \epsilon_{ij,t}, \end{aligned} \quad (42)$$

where $y_{ij,t}$ is the outcome variable (CAPX/Assets and R&D/Assets in logarithm) for firm i in industry j at year t , and EIS_t and DIS_t are the financing shocks from [Belo](#)

²⁶The Equity Issuance Shocks (EIS) and Debt Issuance Shocks (DIS) are the residuals from regressions that include several aggregate variables to control for investment opportunities, and costs of equity and debt financing, thus capturing the expected normal variation in issuance activity. That way, there are two financial shocks from the time series variation in the fractions of firms issuing equity and debt in the cross-section of U.S. publicly traded firms using Compustat data. See [Belo et al. \(2024\)](#) for more details.

et al. (2024).²⁷ FCE is the proxy for equity-focused constraint. The proxy for debt-focused constraint, FCD , and its interaction with EIS_t and DIS_t are included in the regression but not shown for brevity. $Z_{ij,t-1}$ is a vector of firm-level controls, which includes Q , leverage, size, cash flow, cash holdings, age, and dividend. We also include firm fixed effects and industry $\times$ time fixed effects.

Table G.1 collects our results. We choose horizons of $h = 4$ and $h = 5$ years (at which the impulse response functions demonstrate the strongest response). We find a statistically significant coefficient of the interaction between FCE and EIS (β_2^h) for both CAPX and R&D. Therefore, unexpected changes in equity market conditions affect the investment decisions of equity-focused constrained firms. Importantly, we do not find a statistically significant coefficient of the interaction between FCE and DIS (β_3^h) for both CAPX and R&D, suggesting that unexpected changes in debt market conditions are not economically important for the investment decisions of these firms.

G.2 M&A Activity

To provide a comprehensive analysis of how monetary policy shapes firm investment, we examine how financial constraints amplify the transmission of monetary policy to M&A activity. To do so, we collect M&A deals from SDC Platinum and estimate our baseline specifications using two outcome variables: M&A deal value, which is the deal value from SDC Platinum divided by total assets, and Probability of an M&A, a binary indicator that equals one if the firm engages in an M&A transaction (i.e., if the deal value is greater than zero).

We start our analysis by investigating the average effects. Figure G.1 shows the response of M&A deal value (Panel A) and the probability of engaging in an M&A transaction (Panel B) to a 25 bps increase in the instrumented one-year Treasury rate. After eight quarters, a 25 bps increase in rates leads to a 4.51% decline in M&A deal value and a 4.79% decline in the probability of engaging in an M&A transaction.²⁸

²⁷ A positive EIS (DIS) represents an unexpected improvement in equity (debt) market conditions.

²⁸ These magnitudes are consistent with Fischer and Horn (2023), who document that a 25 bps contractionary shock reduces the total number of deals per month by about 4.3% and total deal value by about 7.3%.

We then find that financial constraints significantly amplify the impact of monetary policy on M&A activity. Figure G.2 shows that both equity-focused and debt-focused constraints strengthen the response of M&A outcomes to monetary tightening. After eight quarters, a two-standard-deviation increase in the FCE measure amplifies a firm’s M&A deal value response to a 25 bps higher one-year Treasury rate by 56%, while the amplification effect associated with FCD is 33.7%. After fourteen quarters, debt-focused constrained firms become more responsive: a two-standard-deviation increase in FCE amplifies the deal value response by 55.22%, and the corresponding effect of FCD rises to 69.54%. The results for the probability of engaging in an M&A transaction are qualitatively similar. Overall, these findings suggest that both equity- and debt-focused financial constraints amplify the sensitivity of M&A activity to monetary policy shocks.

G.3 The Duration Channel

In the recent asset pricing literature, there is evidence that firms with shorter duration tend to carry a premium in stock price (see Gonçalves (2021) for further discussion). Constrained firms focusing on equity financing tend to have longer duration as shown in Table 1, since these firms tend to invest heavily in R&D. Moreover, previous literature has suggested that firms with high duration do suffer more in the aftermath of negative monetary policy shocks.

To address this possibility, we control for duration in our specification. Table G.2 shows that our results are robust to controlling for duration in that the amplification of the equity constraint channel still prevails.²⁹ In a 5-day cumulative return window after the FOMC announcements, the equity-focused constrained firms experience a significantly 22 bps lower return than unconstrained firms do, controlling for duration. The coefficients on the interaction between monetary policy shocks and duration measure is significantly negative, consistent with the idea that firms with longer duration tend to be more sensitive to the monetary policy.

The investment response is also robust after controlling for duration. Figure G.4 shows that a two-standard-deviation increase in the FCE measure significantly increases a firm’s investment (CAPX) response to a 25 bps higher 1-year rate by 37.36 bps af-

²⁹We thank Gonçalves (2021) for providing the data on duration.

ter 12 quarters and R&D response by 43.22 bps after 14 quarters. Figure G.6 presents the results for debt-focused constrained firms. A two-standard-deviation increase in the *FCD* measure significantly increases a firm’s investment (R&D) response to a 25 bps higher 1-year rate by 20.58 bps. The differential effect is not statistically significant for CAPX. Our results show that the equity constraint channel remains significant and distinct from the duration and debt channels.

Hoberg and Maksimovic (2022) develop a novel 10-K text-based model of product life cycles. They define a four-stage product life cycle: product innovation, process innovation, maturity, and decline. They refer to these stages as Life1, Life2, Life3, and Life4, respectively. Life1 firms intensively discuss an “explanation of material product research and development to be performed during the period covered”. As Life1 firms are still in the product development stage, these firms are likely to have higher duration. We extend our baseline regressions by including the Life1 variable and its interaction with monetary policy shocks. Table G.3, Figure G.5 and G.7 show that our findings remain qualitatively the same.

G.4 Credit Risk

Although our baseline specification controls for the transmission of monetary policy through leverage, one may still be concerned that our channel operates through debt-related characteristics such as credit risk. To address this concern, we follow Altman et al. (2021) and define the Z-score as:

$$\begin{aligned} \text{Z-score} = & 1.2 \times \frac{\text{Current Assets} - \text{Current Liabilities}}{\text{Total Assets}} + 1.4 \times \frac{\text{Retained Earnings}}{\text{Total Assets}} \\ & + 3.3 \times \frac{\text{EBIT}}{\text{Total Assets}} + 0.6 \times \frac{\text{Market Value of Equity}}{\text{Total Liabilities}} + 1.0 \times \frac{\text{Sales}}{\text{Total Assets}}. \end{aligned}$$

We then re-estimate our baseline specification controlling for the Z-score and its interaction with the monetary policy shock. Figures G.8 and G.9 show that our main results remain virtually unchanged. We consistently find that equity-focused constraints significantly amplify the transmission of monetary policy to firm investment.

G.5 The Refinancing Constraints Channel

Debt-focused constrained firms frequently report issues related to covenant violations (Hoberg and Maksimovic, 2015). In our baseline results, we control for *FCD* while estimating the effect of monetary policy shocks on equity-focused constrained firms. One potential concern with these results is that covenant violations may not be the only mechanism through which debt-related constraints affect firms. For example, there is evidence that refinancing constraints can amplify the effects of monetary policy shocks (Jungherr et al., 2024; Oliveira et al., 2024). If equity-focused constrained firms are also likely to face refinancing risk, the refinancing constraints channel could be potentially attenuating the equity constraint channel.

In line with this idea, we follow Almeida et al. (2012) and use the ex-ante maturity structure of long-term debt to predict firms' financial position in a given year. Our measure of refinancing constraint is:

$$RFC_{ij,t} = \frac{dd1q_{ij,t}}{dd1q_{ij,t} + dl ttq_{ij,t}}, \quad (43)$$

COMPUSTAT's *dd1q* is the amount of long-term debt maturing during the first year after the annual report, e.g., the long-term debt maturing in 2008 for firms with a December 2007 fiscal year-end. COMPUSTAT's *dl ttq* represents the amount of long-term debt that matures in more than one year. Therefore, the one-year lag of the ratio of *dd1q* to *dd1q + dl ttq* is the fraction of a firm's long-term debt due in a given year as predicted in the previous year.

We then estimate Equation (23) controlling for $RFC_{ij,t-1}$ and its interaction with the instrumented 1-year Treasury rate $\hat{y}t_t$. It is worth noting that we are controlling for both refinancing constraints and *FCD*. Figure G.10 shows that the equity constraint channel remains economically and statistically significant. Figure G.11 shows the results for debt-focused constrained firms. The differential effect is not statistically significant for both types of investment, CAPX and R&D. Overall, our results show that the equity channel is distinct and quantitatively more important than the debt channel.

G.6 Information Effect and Investment Response

Hsu et al. (2023) argue that the Fed’s private information about economic conditions revealed through FOMC announcements affect firm investment and show that the sensitivity of the investment rate to a Fed information shock is greater for more cyclical firms. To rule out that our results are driven by the information effect, we estimate Equation (23) adding the information shock from Jarociński and Karadi (2020) interacted with the FCE and FCD measures as controls variables. Figures G.16 and G.17 show that our results are virtually unchanged.

G.7 Alternative Monetary Policy Shocks

One potential concern is whether our results are robust to the measure of monetary policy. We address this concern by using an alternative shock measure, the monetary policy shocks from Bauer and Swanson (2023). By orthogonalizing the high-frequency identified monetary policy shocks with respect to the macroeconomic and financial data observed before the FOMC announcements, Bauer and Swanson (2023) construct a series of shocks that eliminate any attenuation bias or “price puzzle” types of effects in output, inflation, or other variables in a structural VAR or local projections framework, providing better estimates of monetary policy’s true effects.

We start by analyzing firms’ stock price response. Tables G.4 and G.5 report the coefficient estimates of Equation (21), using the monetary policy shocks from Bauer and Swanson (2023). Equity-focused constrained firms have a realized return that is 31.6 bps lower than that of unconstrained firms in a 5-day cumulative return window, while the debt-focused constrained firms do not show a statistically significant larger response than unconstrained firms do.

We then analyze the real effects. First, using the same approach explained in Subsection 5.3.2, we instrument the treasury rate using cumulative monetary policy shocks from Bauer and Swanson (2023). We then estimate Equation (23). Figures G.12 and G.13 show the results are virtually the same. We conclude that our findings are robust to the choice of shock construction.

G.8 Cyclical and Investment Response

We guarantee that the results are not driven by differences in cyclical or other observable differences between equity-focused constrained firms and unconstrained firms, time-invariant unobservable firm characteristics, nor by economy-wide or industry-specific trends by estimating Equation (23) and adding the FCE and FCD measures interacted with GDP growth. It is worth noting that this specification includes firm characteristics interacted with the monetary policy shock, FCE and FCD measures interacted with GDP growth, firm fixed effects, and industry-time fixed effects. Figures G.14 and G.15 show that our results are remain very similar after controlling for cyclical.

G.9 2-year Treasury Rate

We measure the stance of monetary policy using the instrumented 2-year U.S. Treasury rate, following the procedure outlined in Section 5.3.2. Figures G.18 and G.19 shows our findings are virtually the same.

G.10 Zero Lower Bound

Brennan et al. (2024) show that different series of high-frequency monetary shocks can have a low correlation coefficient, and that the shock series become even more distinct when the federal funds rate is at its effective lower bound (ELB) due to data. One concern is that our results might be driven by the use of a specific shock or by the zero lower bound period. As discussed above, we first address this concern by showing that our results are robust to the monetary policy shocks from Bauer and Swanson (2023) and to the use of the instrumented 2-year U.S. Treasury rate. We next estimate the results excluding the zero lower bound period (January 2009 to December 2015). Tables G.6, G.7 and Figures G.20, G.21 display the findings. Equity-focused constraints significantly amplify the effect of monetary policy shocks on stock returns, CAPX, and R&D. Overall, our results suggest that the “equity constraint channel” remains robust even when the zero lower bound period is excluded from our sample.

G.11 Symmetry

Recent studies suggest asymmetric transmission mechanisms of monetary policy ([Perez-Orive et al., 2024](#)), emphasizing directional changes depending on whether monetary shocks are expansionary or contractionary. We separately estimate the stock price response for expansionary and contractionary shocks. Table [G.8](#) shows that coefficients of interest remain negative, suggesting that the amplification of equity financing constraint is symmetric, though the effect is more concentrated in contractionary shocks, consistent with the literature. The effect of leverage and size, on the other hand, depends on the sign of shocks, consistent with [Perez-Orive et al. \(2024\)](#), who find asymmetric effects when the constraint is measured as distance to default.

G.12 Intangibility

There is evidence that intangible firms are less sensitive to monetary policy shocks ([Caggese and Pérez-Orive, 2022](#); [Döttling and Ratnovski, 2023](#)). Although our focus is on equity-focused constrained firms rather than intangible firms, the former tend to be more R&D-intensive, as suggested by Table [1](#). This indicates a potential correlation between our proxy for equity-focused constraints and intangibility, which could introduce biases into our estimations.

We address the concern discussed above as follows. First, we adopt the methodology of [Peters and Taylor \(2017\)](#) to measure intangible capital, incorporating both externally purchased and internally created intangible capital. Externally purchased intangible capital is defined as intangible assets reported on the balance sheet (Compustat item *intan*). Internally created intangible capital is calculated as the sum of knowledge capital and organizational capital. Knowledge capital is measured by capitalizing R&D expenses, while organizational capital is measured by capitalizing selling, general, and administrative (SG&A) expenses weighted by a factor of 0.3. We directly use this measure from [Peters and Taylor \(2017\)](#), available through Wharton Research Data Services (WRDS).

Next, we compute the intangible capital ratio, defined as intangible capital divided by the sum of intangible capital and physical capital (Compustat item *ppent*). To mitigate potential biases, we control for this variable and its interaction with monetary policy shocks in Equation [\(23\)](#). Tables [G.9](#) and [G.10](#), and

Figures G.22 and G.23 show that our results remain robust to this additional control, underscoring the unique role of equity-focused financing constraints in the transmission of monetary policy to the corporate sector.

G.13 Growth/Value and Stock Price Response

We split the sample by book-to-market ratio to make sure that the results are not entirely driven by the growth firms. We estimate Equation (21) within each subsample and report the coefficients in Tables G.11 and G.12 for value firms and growth firms, respectively. We find that the strong amplification of equity-focused constraint on stock return response to monetary policy shocks is prevalent in both subsamples.

G.14 Component of Stock Price Response to the Market Fluctuations

In this section, we re-estimate the heterogeneous stock price response after subtracting the component of response to the market fluctuations. The dependent variable is calculated as $r_{i,t} - \hat{\beta}_i \times r_{M,t}$, where $r_{i,t}$ is raw daily returns, $\hat{\beta}_i$ is estimated CAPM β for each stock, and $r_{M,t}$ is the daily return on days of FOMC announcements. CAPM β is estimated using daily returns over the whole sample period. Table G.13 presents the results of the estimation. We first observe that the amplification of both constraints remain statistically significant. Moreover, the magnitudes of equity financing constraints decrease relative to those in Table 4. When the component of response to market fluctuations is subtracted, both constraints amplify the monetary policy shocks at roughly the same magnitudes.

This is consistent with the argument that equity-focused constrained firms are more exposed to monetary policy shocks, because monetary policy also affects the cost of equity. In other words, a large share of the monetary policy transmission to stock market that we aim to capture in Equation (21) can be associated with the stock sensitivity to market-wide fluctuations induced by monetary policy shocks. Once this channel is off, we see that equity and debt financing constraints play a quantitatively similar role. This evidence further supports the proposed equity constraint channel.

G.15 Pre-trends

To examine whether differences between equity-focused and debt-focused constrained firms were already present before the monetary policy surprises, we conduct a pre-trend exercise by estimating how investment changed in the four quarters prior to the shock. In particular, we construct dependent variables to assess pre-trends in firm CAPX and R&D activity prior to monetary policy shocks. Specifically, for each firm i and quarter t , we define $\text{inv}_{h,it}^{\text{pre}} = \log(\text{Inv}_{i,t-h}) - \log(\text{Inv}_{i,t-5})$, $\text{rd}_{h,it}^{\text{pre}} = \log(\text{R\&D}_{i,t-h}) - \log(\text{R\&D}_{i,t-5})$ for $h \in \{1, 2, 3, 4\}$. These variables capture cumulative log changes in CAPX and R&D over the four quarters preceding the shock, relative to a common pre-shock baseline at $t-5$. Table G.14 shows no evidence of pre-trends, which provides further credibility to our identification strategy.

G.16 Equity Reliance

To address the concern that our results are driven by differential exposure to equity financing costs rather than financial constraints per se, we compare firms with similar equity reliance but differing levels of financial constraints. To do so, we first construct a measure of equity reliance by scaling equity issuance by total assets and then computing a five-year average. The five-year window mitigates the influence of outliers and ensures that the measure captures persistent reliance on equity financing. We then re-estimate our baseline Equation (23), controlling for this variable and its interaction with the monetary policy shock. In doing so, we test whether, among firms with similar equity reliance, those that are more equity-constrained exhibit a different sensitivity to monetary policy.

Figure G.24 shows that our results remain unchanged. In particular, the interaction between equity-focused financial constraints and the monetary policy shock remains economically and statistically significant, even after controlling for equity reliance and its interaction with the shock. This indicates that our findings are not driven by differential exposure to equity financing costs per se. Holding equity reliance constant, firms that are more equity-constrained continue to exhibit a stronger sensitivity to monetary policy shocks. These results reinforce the interpretation that the amplification operates through financial constraints rather than mechanical differences in firms' reliance on equity markets.

G.17 Cost of Adjustment

One potential concern is that our results are driven by differences in capital adjustment costs. Consider an increase in the policy rate that raises firms' costs of external financing (through both equity and debt). If debt-focused constrained firms face high capital adjustment costs, they tend to resist reducing debt in order to maintain their capital stock. If equity-focused constrained firms face lower adjustment costs, they are more willing to reduce equity issuance. To address this concern, we control for adjustment costs using the inflexibility measure from [Gu et al. \(2018\)](#).

We first construct an inflexibility measure following the methodology in [Gu et al. \(2018\)](#). The variable $INFLEX_{i,t}$ is defined as:

$$INFLEX_{i,t} = \frac{\max_{i,0,t} \frac{OPC}{Sales} - \min_{i,0,t} \frac{OPC}{Sales}}{std_{i,0,t} \Delta \log \frac{Sales}{Assets}}$$

where $\max_{i,0,t} \frac{OPC}{Sales} - \min_{i,0,t} \frac{OPC}{Sales}$ is the range of firm's operating cost (compustat item `xsga + cogs`) over sales (compustat item `sale`) over the period of year 0 to year t and $std_{i,0,t} \Delta \log \frac{Sales}{Assets}$ is the standard deviation of the quarterly growth rate of sales over total assets (Compustat item `at`). Year 0 is firm's beginning year in compustat. Intuitively, a firm with less flexible operations (higher adjustment costs) will wait longer before altering its scale to adjust to changes in profitability.

We then re-estimate our baseline Equation (23) controlling for $INFLEX_{i,t}$ and its interaction with the monetary policy shock. Figures [G.25](#) and [G.26](#) show that our main results remain very similar, indicating that differences in capital adjustment costs do not drive our findings.

Appendix Table G.1. Effect of Financing Shocks on firms' Investment

	$h = 4$		$h = 5$	
	ΔI_t	ΔRD	ΔI_t	ΔRD
$EIS \times FCE$	0.011*** (0.004)	0.006* (0.003)	0.007** (0.003)	0.005* (0.002)
$DIS \times FCE$	0.001 (0.001)	0.001 (0.001)	-0.0001 (0.001)	0.001 (0.001)
R^2	0.4385	0.4580	0.4589	0.4967
Observations	50,581	18,911	44,707	16,657
Firm Controls	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes
Industry $\times$ Time	Yes	Yes	Yes	Yes

This table reports coefficient estimates from estimating Equation (42). The dependent variable is the h-year change in the log CAPX over assets and R&D over assets, respectively (for details, see Section 4). EIS and DIS are the financing shocks from [Belo et al. \(2024\)](#). The Equity Issuance Shocks (EIS) and Debt Issuance Shocks (DIS) are the residuals from regressions that include several aggregate variables to control for investment opportunities, and costs of equity and debt financing, thus capturing the expected normal variation in issuance activity (see [Belo et al. \(2024\)](#) for details). The *FCE* variable denotes the proxy for equity-focused constrained firms ([Hoberg and Maksimovic, 2015](#); [Linn and Weagley, 2023](#)). We report heteroskedasticity robust and two-way clustered at the firm and year level standard errors in parenthesis. *, **, and *** indicate statistical significance at the 10%, 5%, and 1% levels, respectively.

Appendix Table G.2. Heterogeneous Stock Price Response Controlling for Duration (Gonçalves, 2021)

Window:	(0,0) (1)	(0,+1) (2)	(0,+2) (3)	(0,+5) (4)
$mps \times equity_constraint$	-0.104** (0.047)	-0.114* (0.063)	-0.109 (0.072)	-0.220** (0.098)
$mps \times debt_constraint$	0.004 (0.032)	0.030 (0.042)	-0.116** (0.048)	-0.134** (0.062)
$mps \times duration$	-0.001*** (0.0002)	-0.001*** (0.0004)	-0.001*** (0.0004)	-0.002*** (0.0004)
<i>Fixed-effects</i>				
sic3-year	Yes	Yes	Yes	Yes
<i>Fit statistics</i>				
Observations	576,494	576,326	576,158	575,649
R ²	0.020	0.026	0.025	0.032

This table reports the coefficient estimates of the cross-sectional regression by pooling all firm level returns around FOMC announcements from 1997 to 2019. The firm-level cumulative returns are calculated over four different event windows and shown in Columns (1) to (4). Column (1) reports the results when we use the daily return on the day of the FOMC announcements as the dependent variable. Columns (2) to (4) report the results when we use the 1-, 2-, and 5-day cumulative returns since the FOMC announcements as the dependent variable, respectively. The *mps* variable denotes monetary policy shock from Jarociński and Karadi (2020). The *equity-focused* variable denotes the indicator for firms that are in the top tercile of *FCE* and bottom tercile of *FCD*. The *debt-focused* variable denotes the indicator for firms that are in the top tercile of *FCD* and bottom tercile of *FCE*. All other group indicators are included in the regression, but omitted for brevity, except for the unconstrained group of firms. All regressions include industry and year fixed effects and control for log asset (size), book-to-market ratio, operating profitability, and duration at the firm level. Robust standard errors clustered at the firm level are reported in parentheses. Significance codes: ***: 0.01, **: 0.05, *: 0.1.

Appendix Table G.3. Heterogeneous Stock Price Response Controlling for Duration (Hoberg and Maksimovic, 2022)

Window:	(0,0) (1)	(0,+1) (2)	(0,+2) (3)	(0,+5) (4)
$mps \times equity_constraint$	-0.277*** (0.042)	-0.328*** (0.060)	-0.403*** (0.070)	-0.630*** (0.092)
$mps \times debt_constraint$	-0.034 (0.033)	0.048 (0.046)	-0.111** (0.052)	-0.162** (0.067)
$mps \times duration$	-0.357*** (0.073)	-0.265** (0.116)	-0.355*** (0.130)	-0.585*** (0.164)
<i>Fixed-effects</i>				
sic3-year	Yes	Yes	Yes	Yes
<i>Fit statistics</i>				
Observations	542,331	542,143	541,959	541,417
R ²	0.025	0.032	0.029	0.038

This table reports the coefficient estimates of the cross-sectional regression by pooling all firm level returns around FOMC announcements from 1997 to 2019. The firm-level cumulative returns are calculated over four different event windows and shown in Columns (1) to (4). Column (1) reports the results when we use the daily return on the day of the FOMC announcements as the dependent variable. Columns (2) to (4) report the results when we use the 1-, 2-, and 5-day cumulative returns since the FOMC announcements as the dependent variable, respectively. The *mps* variable denotes monetary policy shock from Jarociński and Karadi (2020). The *equity-focused* variable denotes the indicator for firms that are in the top tercile of *FCE* and bottom tercile of *FCD*. The *debt-focused* variable denotes the indicator for firms that are in the top tercile of *FCD* and bottom tercile of *FCE*. All other group indicators are included in the regression, but omitted for brevity, except for the unconstrained group of firms. All regressions include industry and year fixed effects and control for log asset (size), book-to-market ratio, operating profitability, and duration at the firm level. Robust standard errors clustered at the firm level are reported in parentheses. Significance codes: ***: 0.01, **: 0.05, *: 0.1.

Appendix Table G.4. Heterogeneous Stock Price Response to [Bauer and Swanson \(2023\)](#) Shock

Window:	(0,0) (1)	(+1,+1) (2)	(+2,+2) (3)
$mps \times equity_focused$	-0.161*** (0.029)	-0.060* (0.032)	-0.113*** (0.037)
$mps \times debt_focused$	-0.035 (0.023)	-0.002 (0.026)	-0.097*** (0.029)
$mps \times leverage$	0.317*** (0.039)	0.199*** (0.046)	-0.164*** (0.046)
$mps \times bm$	0.059*** (0.011)	0.053*** (0.013)	-0.030** (0.013)
$mps \times size$	-0.047*** (0.004)	-0.037*** (0.004)	0.039*** (0.004)
$mps \times profitability$	0.028** (0.013)	0.083*** (0.023)	-0.013 (0.018)
<i>Fixed-effects</i>			
sic3-year	Yes	Yes	Yes
<i>Fit statistics</i>			
Observations	939,535	875,227	789,621
R ²	0.015	0.016	0.014

This table reports the coefficient estimates of the cross-sectional regression by pooling all firm level returns around FOMC announcements from 1989 to 2019. The firm level returns are calculated over three different event windows and shown in Columns (1) to (3). Column (1) reports the results when we use the daily return on the day of the FOMC announcements as the dependent variable. Column (2) reports the results when we use the daily return 1 day after the FOMC announcements as the dependent variable. Column (3) reports the results when we use the daily return 2 days after the FOMC announcements as the dependent variable. The *mps* variable denotes monetary policy shock from [Bauer and Swanson \(2023\)](#). The *equity_focused* variable denotes the indicator for firms that are in the top tercile of *FCE* and bottom tercile of *FCD*. The *debt_focused* variable denotes the indicator for firms that are in the top tercile of *FCD* and bottom tercile of *FCE*. All other group indicators are included in the regression, but omitted for brevity, except for the unconstrained group of firms. All regressions include industry and year fixed effects and control for log asset (size), book-to-market ratio, leverage, and operating profitability at the firm level. Robust standard errors clustered at the firm level are reported in parentheses. Significance codes: ***: 0.01, **: 0.05, *: 0.1.

Appendix Table G.5. Cumulative Heterogeneous Stock Price Response to [Bauer and Swanson \(2023\)](#) Shock

Window:	(0,+1) (1)	(0,+2) (2)	(0,+5) (3)
<i>Variables</i>			
$mps \times equity_focused$	-0.2158*** (0.0400)	-0.3104*** (0.0475)	-0.3158*** (0.0650)
$mps \times debt_focused$	-0.0285 (0.0331)	-0.1335*** (0.0383)	-0.0665 (0.0523)
$mps \times leverage$	0.4839*** (0.0545)	0.4160*** (0.0630)	0.5164*** (0.0849)
$mps \times bm$	0.1005*** (0.0157)	0.0915*** (0.0183)	0.0215 (0.0241)
$mps \times size$	-0.0843*** (0.0051)	-0.0596*** (0.0056)	-0.0410*** (0.0075)
$mps \times op$	0.1107*** (0.0243)	0.1019*** (0.0272)	0.1889*** (0.0352)
<i>Fixed-effects</i>			
sic3-year	Yes	Yes	Yes
<i>Fit statistics</i>			
Observations	939,235	938,944	938,066
R ²	0.02066	0.01976	0.02600

This table reports the coefficient estimates of the cross-sectional regression by pooling all firm level returns around FOMC announcements from 1989 to 2019. The firm level cumulative returns are calculated over three different event windows and shown in Columns (1) to (3). Columns (1) to (3) report the results when we use the 1-, 2-, and 5-day cumulative returns since the FOMC announcements as the dependent variable, respectively. The *mps* variable denotes monetary policy shock from [Bauer and Swanson \(2023\)](#). The *equity_focused* variable denotes the indicator for firms that are in the top tercile of *FCE* and bottom tercile of *FCD*. The *debt_focused* variable denotes the indicator for firms that are in the top tercile of *FCD* and bottom tercile of *FCE*. All other group indicators are included in the regression, but omitted for brevity, except for the unconstrained group of firms. All regressions include industry and year fixed effects and control for log asset (size), book-to-market ratio, leverage, and operating profitability at the firm level. Robust standard errors clustered at the firm level are reported in parentheses. Significance codes: ***: 0.01, **: 0.05, *: 0.1.

Appendix Table G.6. Heterogeneous Stock Price Response: Excluding the ZLB

Window:	(0,0) (1)	(+1,+1) (2)	(+2,+2) (3)
$mps \times equity_focused$	-0.196*** (0.040)	-0.049 (0.040)	-0.113*** (0.040)
$mps \times debt_focused$	-0.119*** (0.033)	0.010 (0.033)	-0.094*** (0.032)
$mps \times leverage$	0.352*** (0.051)	0.155*** (0.052)	-0.166*** (0.056)
$mps \times bm$	0.052*** (0.013)	-0.034** (0.016)	-0.059*** (0.014)
$mps \times size$	-0.091*** (0.005)	-0.014*** (0.005)	0.062*** (0.005)
$mps \times profitability$	0.065*** (0.017)	0.110*** (0.032)	-0.011 (0.021)
$mps \times cashholding$	-0.408*** (0.075)	-0.057 (0.058)	0.147** (0.062)
<i>Fixed-effects</i>			
sic3-year	Yes	Yes	Yes
<i>Fit statistics</i>			
Observations	706,300	658,251	595,320
R ²	0.021	0.015	0.014

This table reports the coefficient estimates of the cross-sectional regression by pooling all firm level returns around FOMC announcements from 1991 to 2019, excluding ZLB (January 2009-December 2015). The firm level returns are calculated over three different event windows and shown in Columns (1) to (3). Column (1) reports the results when we use the daily return on the day of the FOMC announcements as the dependent variable. Column (2) reports the results when we use the daily return 1 day after the FOMC announcements as the dependent variable. Column (3) reports the results when we use the daily return 2 days after the FOMC announcements as the dependent variable. The *mps* variable denotes monetary policy shock from [Jarociński and Karadi \(2020\)](#). The *equity_focused* variable denotes the indicator for firms that are in the top tercile of *FCE* and bottom tercile of *FCD*. The *debt_focused* variable denotes the indicator for firms that are in the top tercile of *FCD* and bottom tercile of *FCE*. All other group indicators are included in the regression, but omitted for brevity, except for the unconstrained group of firms. All regressions include industry and year fixed effects and control for log asset (size), book-to-market ratio, leverage, operating profitability, and cash holding at the firm level. Robust standard errors clustered at the firm level are reported in parentheses. Significance codes: ***: 0.01, **: 0.05, *: 0.1.

Appendix Table G.7. Cumulative Heterogeneous Stock Price Response: Excluding the ZLB

Window:	(0,+1) (1)	(0,+2) (2)	(0,+5) (3)
$mps \times equity_focused$	-0.241*** (0.055)	-0.319*** (0.063)	-0.538*** (0.083)
$mps \times debt_focused$	-0.111** (0.046)	-0.215*** (0.051)	-0.227*** (0.066)
$mps \times leverage$	0.469*** (0.069)	0.386*** (0.082)	0.294*** (0.109)
$mps \times bm$	0.013 (0.020)	-0.019 (0.024)	-0.094*** (0.030)
$mps \times size$	-0.106*** (0.007)	-0.058*** (0.007)	-0.026*** (0.010)
$mps \times profitability$	0.172*** (0.034)	0.168*** (0.038)	0.250*** (0.047)
$mps \times cashholding$	-0.484*** (0.096)	-0.381*** (0.106)	-0.287** (0.129)
<i>Fixed-effects</i>			
sic3-year	Yes	Yes	Yes
<i>Fit statistics</i>			
Observations	706,061	705,829	705,166
R ²	0.028	0.026	0.030

This table reports the coefficient estimates of the cross-sectional regression by pooling all firm level returns around FOMC announcements from 1991 to 2019, excluding ZLB (January 2009-December 2015). The firm level cumulative returns are calculated over three different event windows and shown in Columns (1) to (3). Columns (1) to (3) report the results when we use the 1-, 2-, and 5-day cumulative returns since the FOMC announcements as the dependent variable, respectively. The *mps* variable denotes monetary policy shock from [Jarociński and Karadi \(2020\)](#). The *equity_focused* variable denotes the indicator for firms that are in the top tercile of *FCE* and bottom tercile of *FCD*. The *debt_focused* variable denotes the indicator for firms that are in the top tercile of *FCD* and bottom tercile of *FCE*. All other group indicators are included in the regression, but omitted for brevity, except for the unconstrained group of firms. All regressions include industry and year fixed effects and control for log asset (size), book-to-market ratio, leverage, operating profitability, and cash holdings at the firm level. Robust standard errors clustered at the firm level are reported in parentheses. Significance codes: ***: 0.01, **: 0.05, *: 0.1.

Appendix Table G.8. Symmetric Cumulative Heterogeneous Stock Price Response

	Contractionary shocks			Expansionary shocks		
Window:	(0,+1)	(0,+2)	(0,+5)	(0,+1)	(0,+2)	(0,+5)
	(1)	(2)	(3)	(4)	(5)	(6)
<i>mps</i> × <i>equity_focused</i>	-0.207*** (0.057)	-0.398*** (0.072)	-0.533*** (0.097)	-0.170** (0.077)	-0.093 (0.087)	-0.110 (0.112)
<i>mps</i> × <i>debt_focused</i>	-0.064 (0.044)	-0.079 (0.058)	0.038 (0.080)	-0.011 (0.064)	-0.057 (0.072)	0.009 (0.092)
<i>mps</i> × <i>leverage</i>	-0.201*** (0.078)	-0.318*** (0.096)	-0.334*** (0.129)	0.374*** (0.123)	0.341** (0.137)	0.166 (0.164)
<i>mps</i> × <i>bm</i>	0.120*** (0.028)	0.188*** (0.033)	0.267*** (0.042)	-0.002 (0.027)	-0.052* (0.032)	-0.140*** (0.039)
<i>mps</i> × <i>size</i>	0.059*** (0.006)	0.092*** (0.008)	0.086*** (0.011)	-0.105*** (0.010)	-0.047*** (0.011)	-0.002 (0.014)
<i>mps</i> × <i>profitability</i>	0.009 (0.026)	-0.036 (0.033)	0.078* (0.042)	0.182*** (0.055)	0.205*** (0.059)	0.288*** (0.070)
<i>mps</i> × <i>cashholding</i>	0.225*** (0.084)	0.366*** (0.103)	0.002 (0.134)	-0.510*** (0.133)	-0.346** (0.149)	-0.048 (0.175)
<i>Fixed-effects</i> sic3-year	Yes	Yes	Yes	Yes	Yes	Yes
<i>Fit statistics</i>						
Observations	395,569	395,435	395,010	448,195	448,066	447,708
R ²	0.024	0.026	0.035	0.040	0.037	0.047

This table reports the coefficient estimates of the cross-sectional regression by pooling all firm level returns around FOMC announcements from 1991 to 2019, separately for contractionary and expansionary shocks. The firm level cumulative returns are calculated over three different event windows and shown in Columns (1) to (6). Columns (1) to (6) report the results when we use the 1-, 2-, and 5-day cumulative returns since the FOMC announcements as the dependent variable, respectively. The *mps* variable denotes monetary policy shock from [Jarociński and Karadi \(2020\)](#). The *equity_focused* variable denotes the indicator for firms that are in the top tercile of *FCE* and bottom tercile of *FCD*. The *debt_focused* variable denotes the indicator for firms that are in the top tercile of *FCD* and bottom tercile of *FCE*. All other group indicators are included in the regression, but omitted for brevity, except for the unconstrained group of firms. All regressions include industry and year fixed effects and control for log asset (size), book-to-market ratio, leverage, operating profitability, and cash holdings at the firm level. Robust standard errors clustered at the firm level are reported in parentheses. Significance codes: ***: 0.01, **: 0.05, *: 0.1.

Appendix Table G.9. Heterogeneous Stock Price Response Controlling for Intangibility

Window:	(0,0) (1)	(+1,+1) (2)	(+2,+2) (3)
$mps \times equity_focused$	-0.183*** (0.040)	-0.018 (0.037)	-0.126*** (0.038)
$mps \times debt_focused$	-0.120*** (0.033)	0.013 (0.031)	-0.101*** (0.030)
$mps \times intangible$	-0.214*** (0.033)	-0.058** (0.028)	-0.002 (0.031)
<i>Fixed-effects</i>			
sic3-year	Yes	Yes	Yes
<i>Fit statistics</i>			
Observations	813,363	766,761	701,424
R ²	0.020	0.016	0.016

This table reports the coefficient estimates of the cross-sectional regression by pooling all firm level returns around FOMC announcements from 1991 to 2019. The firm level returns are calculated over three different event windows and shown in Columns (1) to (3). Column (1) reports the results when we use the daily return on the day of the FOMC announcements as the dependent variable. Column (2) reports the results when we use the daily return 1 day after the FOMC announcements as the dependent variable. Column (3) reports the results when we use the daily return 2 days after the FOMC announcements as the dependent variable. The *mps* variable denotes monetary policy shock from [Jarociński and Karadi \(2020\)](#). The *equity_focused* variable denotes the indicator for firms that are in the top tercile of *FCE* and bottom tercile of *FCD*. The *debt_focused* variable denotes the indicator for firms that are in the top tercile of *FCD* and bottom tercile of *FCE*. All other group indicators are included in the regression, but omitted for brevity, except for the unconstrained group of firms. All regressions include industry and year fixed effects and control for log asset (size), book-to-market ratio, leverage, operating profitability, cash holdings, and intangible ratio at the firm level. Robust standard errors clustered at the firm level are reported in parentheses. Significance codes: ***: 0.01, **: 0.05, *: 0.1.

Appendix Table G.10. Cumulative Heterogeneous Stock Price Response Controlling for Intangibility

Window:	(0,+1) (1)	(0,+2) (2)	(0,+5) (3)
$mps \times equity_focused$	-0.197*** (0.053)	-0.288*** (0.060)	-0.492*** (0.079)
$mps \times debt_focused$	-0.109** (0.044)	-0.214*** (0.049)	-0.205*** (0.064)
$mps \times intangible$	-0.260*** (0.043)	-0.208*** (0.049)	-0.315*** (0.064)
<i>Fixed-effects</i> industry-year	Yes	Yes	Yes
<i>Fit statistics</i>			
Observations	813,264	813,173	812,891
R ²	0.026	0.024	0.031

This table reports the coefficient estimates of the cross-sectional regression by pooling all firm level returns around FOMC announcements from 1991 to 2019. The firm level cumulative returns are calculated over three different event windows and shown in Columns (1) to (3). Columns (1) to (3) report the results when we use the 1-, 2-, and 5-day cumulative returns since the FOMC announcements as the dependent variable, respectively. The *mps* variable denotes monetary policy shock from [Jarociński and Karadi \(2020\)](#). The *equity_focused* variable denotes the indicator for firms that are in the top tercile of *FCE* and bottom tercile of *FCD*. The *debt_focused* variable denotes the indicator for firms that are in the top tercile of *FCD* and bottom tercile of *FCE*. All other group indicators are included in the regression, but omitted for brevity, except for the unconstrained group of firms. All regressions include industry and year fixed effects and control for log asset (size), book-to-market ratio, leverage, operating profitability, cash holdings, and intangible ratio at the firm level. Robust standard errors clustered at the firm level are reported in parentheses. Significance codes: ***: 0.01, **: 0.05, *: 0.1.

Appendix Table G.11. Cumulative Heterogeneous Stock Price Response: Value Firms

Window:	(0,+1) (1)	(0,+2) (2)	(0,+5) (3)
<i>Variables</i>			
$mps \times equity_focused$	-0.1977** (0.0840)	-0.3398*** (0.1033)	-0.5040*** (0.1323)
$mps \times debt_focused$	-0.0315 (0.0565)	-0.1579** (0.0667)	-0.1093 (0.0873)
$mps \times leverage$	0.3202*** (0.0929)	0.0264 (0.1128)	-0.3880*** (0.1462)
$mps \times bm$	-0.0069 (0.0253)	-0.0353 (0.0299)	-0.1460*** (0.0383)
$mps \times size$	-0.1223*** (0.0092)	-0.0673*** (0.0108)	-0.0323** (0.0140)
$mps \times op$	0.5482*** (0.0927)	0.5504*** (0.0982)	0.8960*** (0.1312)
<i>Fixed-effects</i>			
sic3-year	Yes	Yes	Yes
<i>Fit statistics</i>			
Observations	423,247	423,092	422,671
R ²	0.03064	0.02888	0.03711

This table reports the coefficient estimates of the cross-sectional regression by pooling firm level returns of value firms (firms whose book-to-market ratio is above median) around FOMC announcements from 1991 to 2019. The firm level cumulative returns are calculated over three different event windows and shown in Columns (1) to (3). Columns (1) to (3) report the results when we use the 1-, 2-, and 5-day cumulative returns since the FOMC announcements as the dependent variable, respectively. The *mps* variable denotes monetary policy shock from [Jarociński and Karadi \(2020\)](#). The *equity_focused* variable denotes the indicator for firms that are in the top tercile of *FCE* and bottom tercile of *FCD*. The *debt_focused* variable denotes the indicator for firms that are in the top tercile of *FCD* and bottom tercile of *FCE*. All other group indicators are included in the regression, but omitted for brevity, except for the unconstrained group of firms. All regressions include industry and year fixed effects and control for log asset (size), book-to-market ratio, leverage, and operating profitability at the firm level. Robust standard errors clustered at the firm level are reported in parentheses. Significance codes: ***: 0.01, **: 0.05, *: 0.1.

Appendix Table G.12. Cumulative Heterogeneous Stock Price Response: Growth Firms

Window:	(0,+1) (1)	(0,+2) (2)	(0,+5) (3)
<i>Variables</i>			
$mps \times equity_focused$	-0.2109*** (0.0698)	-0.2451*** (0.0761)	-0.3742*** (0.0981)
$mps \times debt_focused$	-0.0601 (0.0628)	-0.1686** (0.0682)	-0.2317*** (0.0899)
$mps \times leverage$	0.6705*** (0.1538)	0.5779*** (0.1595)	0.6066*** (0.1799)
$mps \times bm$	0.9515*** (0.1495)	0.8376*** (0.1603)	0.7715*** (0.1864)
$mps \times size$	-0.0630*** (0.0121)	-0.0136 (0.0123)	0.0215 (0.0146)
$mps \times op$	0.1242*** (0.0437)	0.1198** (0.0469)	0.1781*** (0.0541)
<i>Fixed-effects</i>			
sic3-year	Yes	Yes	Yes
<i>Fit statistics</i>			
Observations	420,517	420,409	420,047
R ²	0.02985	0.02826	0.03514

This table reports the coefficient estimates of the cross-sectional regression by pooling firm level returns of growth firms (firms whose book-to-market ratio is below median) around FOMC announcements from 1991 to 2019. The firm level cumulative returns are calculated over three different event windows and shown in Columns (1) to (3). Columns (1) to (3) report the results when we use the 1-, 2-, and 5-day cumulative returns since the FOMC announcements as the dependent variable, respectively. The *mps* variable denotes monetary policy shock from [Jarociński and Karadi \(2020\)](#). The *equity_focused* variable denotes the indicator for firms that are in the top tercile of *FCE* and bottom tercile of *FCD*. The *debt_focused* variable denotes the indicator for firms that are in the top tercile of *FCD* and bottom tercile of *FCE*. All other group indicators are included in the regression, but omitted for brevity, except for the unconstrained group of firms. All regressions include industry and year fixed effects and control for log asset (size), book-to-market ratio, leverage, and operating profitability at the firm level. Robust standard errors clustered at the firm level are reported in parentheses. Significance codes: ***: 0.01, **: 0.05, *: 0.1.

Appendix Table G.13. Cumulative Heterogeneous Stock Price Response: Subtract Response to Market Fluctuations

Window:	(0,+1) (1)	(0,+2) (2)	(0,+5) (3)
<i>Variables</i>			
$mps \times equity_focused$	-0.106** (0.052)	-0.158*** (0.061)	-0.184** (0.077)
$mps \times debt_focused$	-0.084* (0.043)	-0.156*** (0.049)	-0.114* (0.061)
$mps \times leverage$	0.111 (0.090)	0.059 (0.100)	-0.023 (0.117)
$mps \times bm$	-0.072*** (0.020)	-0.130*** (0.024)	-0.233*** (0.029)
$mps \times size$	0.003 (0.007)	0.024*** (0.008)	0.056*** (0.010)
$mps \times profitability$	0.147*** (0.039)	0.161*** (0.042)	0.262*** (0.049)
$mps \times cashholding$	-0.185** (0.088)	-0.176* (0.099)	-0.128 (0.119)
<i>Fixed-effects</i>			
sic3-year	Yes	Yes	Yes
<i>Fit statistics</i>			
Observations	843,764	843,501	842,718
R ²	0.011	0.013	0.016

This table reports the coefficient estimates of the cross-sectional regression by pooling all firm level returns around FOMC announcements from 1991 to 2019. The dependent variable is calculated as $r_{i,t} - \hat{\beta}_i \times r_{M,t}$, where $r_{i,t}$ is raw daily returns, $\hat{\beta}_i$ is estimated CAPM β for each stock, and $r_{M,t}$ is the daily return on days of FOMC announcements. The firm level cumulative returns are calculated over three different event windows and shown in Columns (1) to (3). Columns (1) to (3) report the results when we use the 1-, 2-, and 5-day cumulative returns since the FOMC announcements as the dependent variable, respectively. The *mps* variable denotes monetary policy shock from [Jarociński and Karadi \(2020\)](#). The *equity_focused* variable denotes the indicator for firms that are in the top tercile of *FCE* and bottom tercile of *FCD*. The *debt_focused* variable denotes the indicator for firms that are in the top tercile of *FCD* and bottom tercile of *FCE*. All other group indicators are included in the regression, but omitted for brevity, except for the unconstrained group of firms. All regressions include industry and year fixed effects and control for log asset (size), book-to-market ratio, leverage, operating profitability, cash holdings, and intangible ratio at the firm level. Robust standard errors clustered at the firm level are reported in parentheses. Significance codes: ***: 0.01, **: 0.05, *: 0.1.

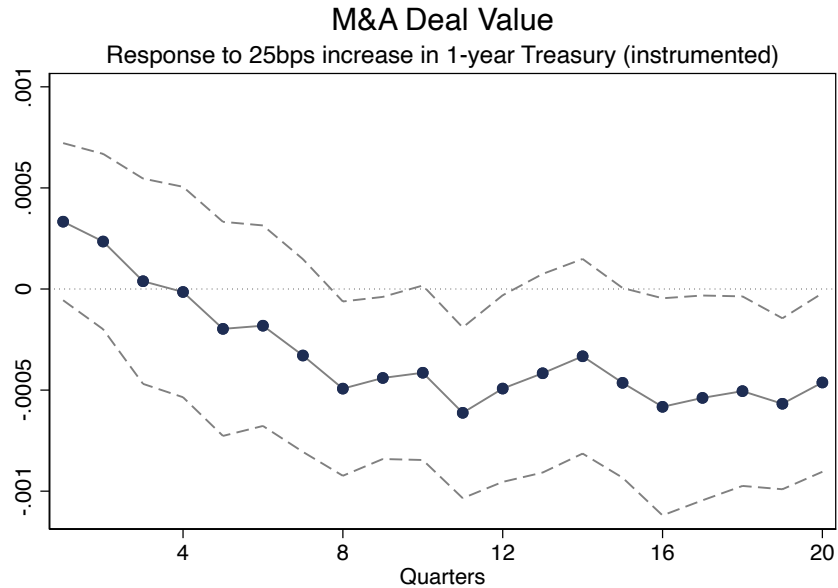
Appendix Table G.14. Real Effects of Monetary Policy Shocks: Pre-trends

	CAPX/Assets				R&D/Assets			
	Q-4	Q-3	Q-2	Q-1	Q-4	Q-3	Q-2	Q-1
$MPS \times FCE$	0.0008 (0.0015)	0.0008 (0.0025)	0.0029 (0.0031)	0.0034 (0.0032)	-0.0014 (0.0013)	-0.0027 (0.0021)	-0.0013 (0.0022)	-0.0006 (0.0025)
$MPS \times FCD$	-0.0007 (0.0015)	-0.0003 (0.0021)	0.0003 (0.0022)	-0.0030 (0.0021)	0.0008 (0.0017)	0.0031 (0.0028)	0.0025 (0.0028)	0.0003 (0.0029)
Observations	254,279	253,629	253,952	254,884	63,555	63,260	63,140	71,399
Firm Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Fiscal Quarter FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Industry $\times$ Time FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

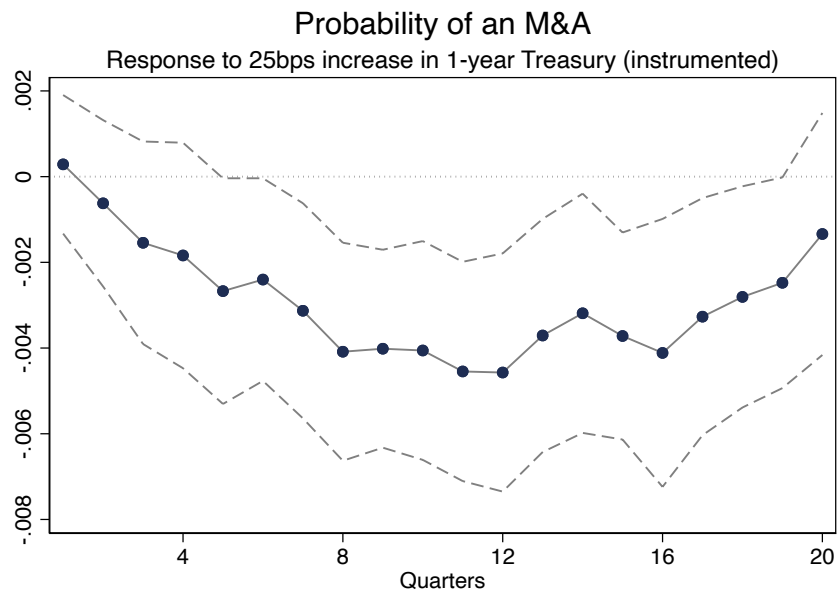
This table reports coefficient estimates from Equation (23). The dependent variables are constructed to assess pre-trends in firm CAPX and R&D activity prior to monetary policy shocks. Specifically, for each firm i and quarter t , we define $inv_{h,it}^{\text{pre}} = \log(\text{Inv}_{i,t-h}) - \log(\text{Inv}_{i,t-5})$, $rd_{h,it}^{\text{pre}} = \log(\text{R\&D}_{i,t-h}) - \log(\text{R\&D}_{i,t-5})$ for $h \in \{1, 2, 3, 4\}$. These variables capture cumulative log changes in CAPX and R&D over the four quarters preceding the shock, relative to a common pre-shock baseline at $t-5$. The 1-year Treasury rate is instrumented by cumulative high-frequency monetary policy shocks measured as monetary policy shocks from the decomposition by [Jarociński and Karadi \(2020\)](#). The *FCE* and *FCD* variables denote the proxies for equity-focused and debt-focused constrained firms ([Hoberg and Maksimovic, 2015](#); [Linn and Weagley, 2023](#)). All regressions control for Q, leverage, size, cash flow, cash holdings, age, and dividend. Heteroscedasticity and autocorrelation robust Driscoll-Kraay standard errors are reported parentheses. *, **, and *** indicate statistical significance at the 10%, 5%, and 1% levels, respectively.

Appendix Figure G.1. The Dynamic Response of M&A to Monetary Policy

(A) Panel A: Deal Value

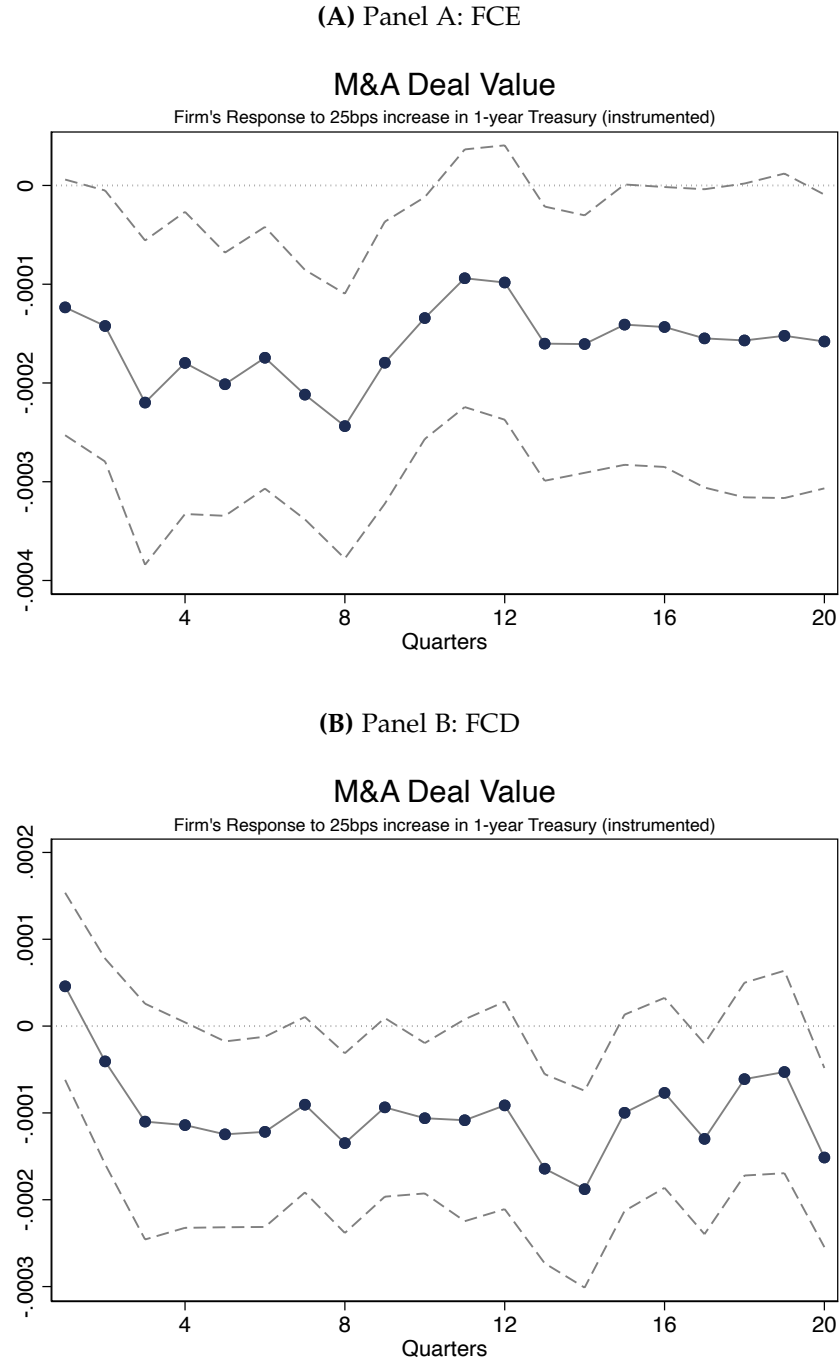


(B) Panel B: Probability of an M&A



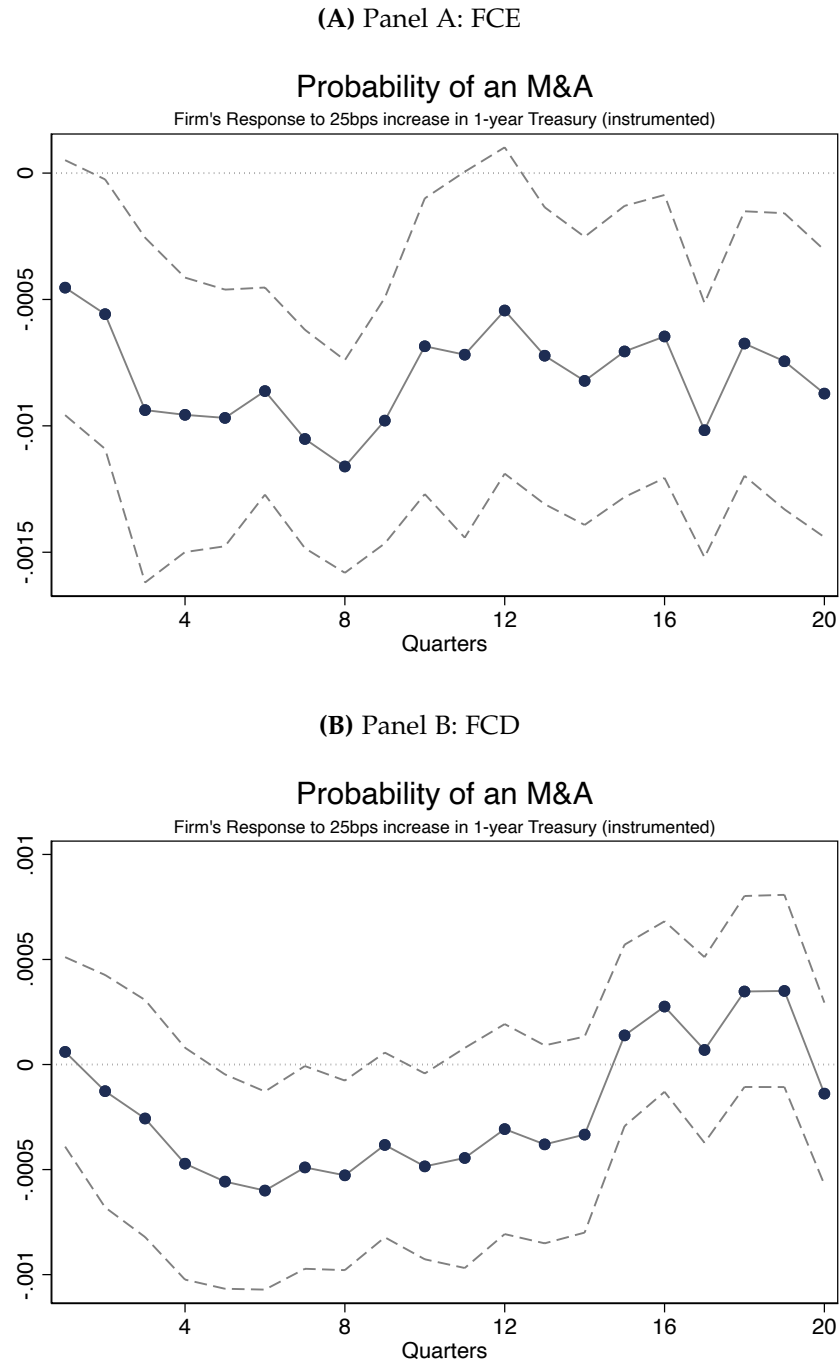
This figure shows the Impulse Response Function (IRF) for the response of M&A activity to a 25 bps increase in 1-year Treasury rate (instrumented). The outcome variables are M&A Deal Value, defined as the M&A deal value divided by total assets, and probability of an M&A, defined by a binary indicator that equals one if the firm engages in an M&A transaction (i.e., if the deal value is greater than zero). The 1-year Treasury rate is instrumented by cumulative high-frequency monetary policy shocks measured as monetary policy shocks from the decomposition by [Jarociński and Karadi \(2020\)](#). Each point represents the point estimate of the coefficient of the instrumented 1-year Treasury rate. The dashed line represents 90% confidence intervals using heteroscedasticity and autocorrelation robust Driscoll-Kraay standard errors.

Appendix Figure G.2. The Dynamic Response of M&A to Monetary Policy: Equity vs. Debt-focused Constrained Firms



This figure shows the effect of financing constraints on firm's response to a 25 bps increase in 1-year Treasury (instrumented), estimated using Equation (23). The outcome variable is the M&A deal value divided by total assets. The 1-year Treasury rate is instrumented by cumulative high-frequency monetary policy shocks measured as monetary policy shocks from the decomposition by Jarociński and Karadi (2020). Each point represents the point estimate of the coefficient of the instrumented 1-year Treasury rate interacted with the financing constraint measure $FCE_{ij,t-1}$ (Panel A) and $FCD_{ij,t-1}$ (Panel B). The dashed line represents 90% confidence intervals using heteroscedasticity and autocorrelation robust Driscoll-Kraay standard errors.

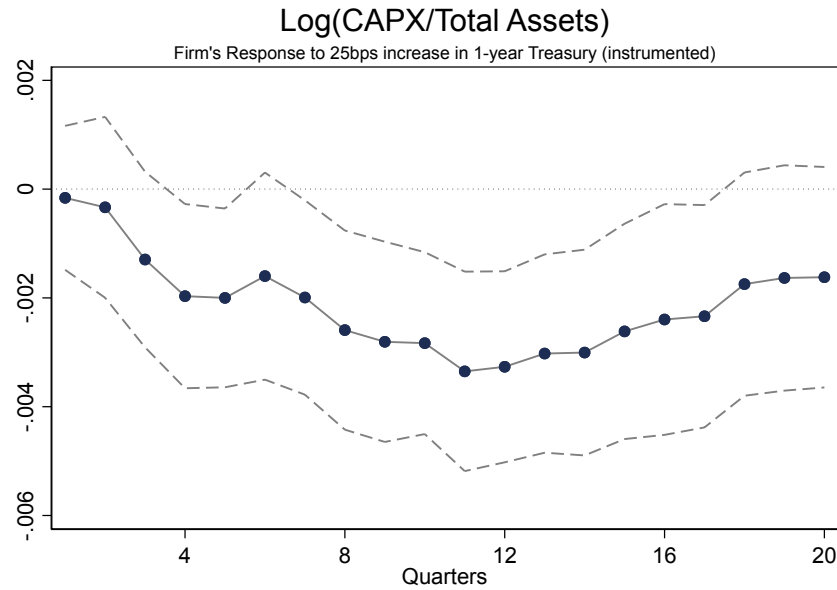
Appendix Figure G.3. The Dynamic Response of M&A to Monetary Policy: Equity vs. Debt-focused Constrained Firms



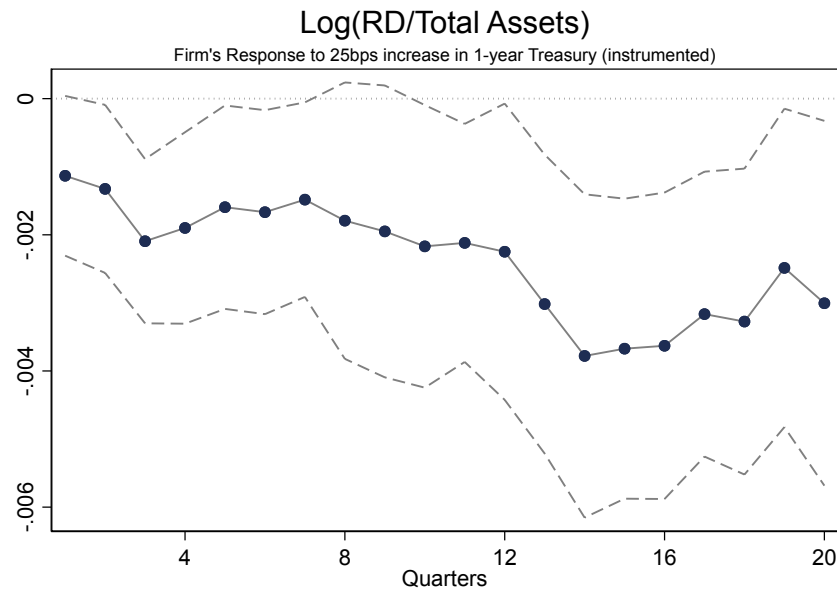
This figure shows the effect of financing constraints on firm's response to a 25 bps increase in 1-year Treasury (instrumented), estimated using Equation (23). The outcome variable is a binary indicator that equals one if the firm engages in an M&A transaction (i.e., if the deal value is greater than zero). The 1-year Treasury rate is instrumented by cumulative high-frequency monetary policy shocks measured as monetary policy shocks from the decomposition by Jarociński and Karadi (2020). Each point represents the point estimate of the coefficient of the instrumented 1-year Treasury rate interacted with the financing constraint measure $FCE_{ij,t-1}$ (Panel A) and $FCD_{ij,t-1}$ (Panel B). The dashed line represents 90% confidence intervals using heteroscedasticity and autocorrelation robust Driscoll-Kraay standard errors.

Appendix Figure G.4. Equity-Focused Constraints and the Dynamic Response of Investment to Monetary Policy: Controlling for Duration (Gonçalves, 2021)

(A) CAPX



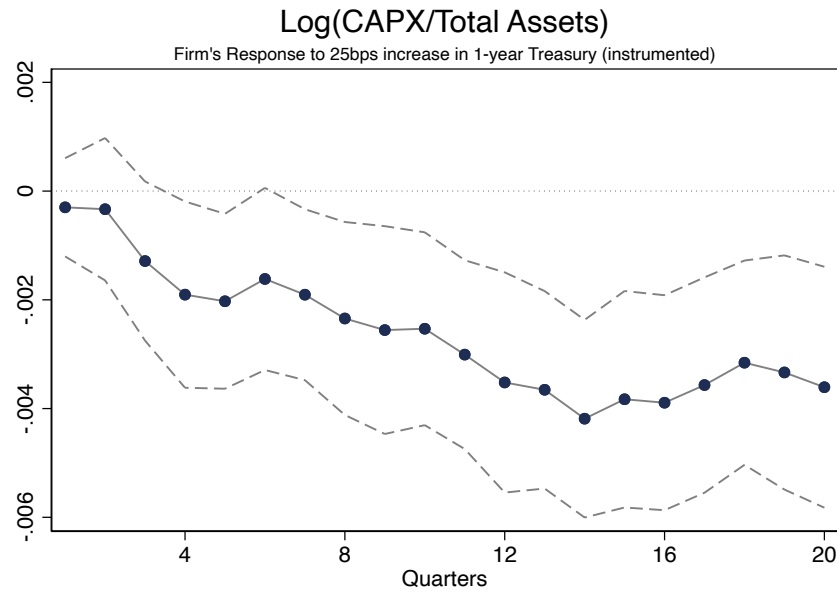
(B) R&D



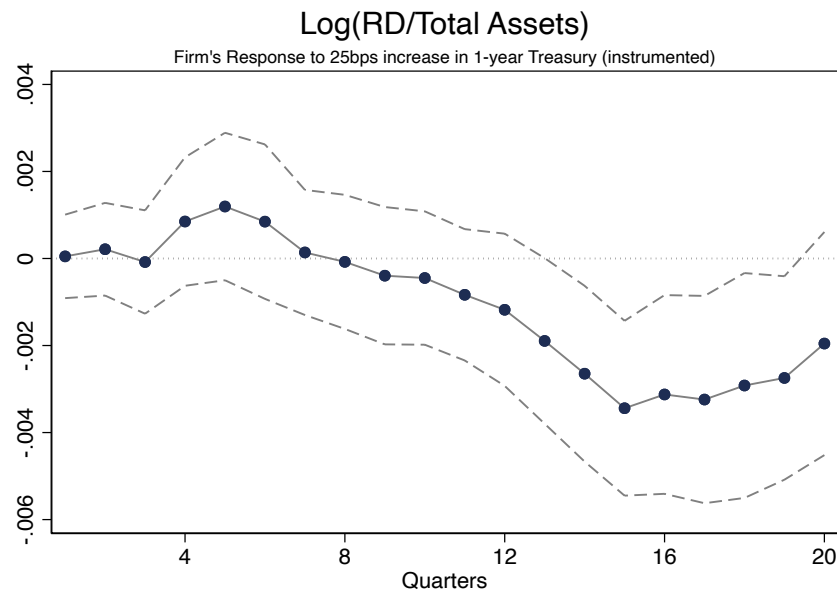
This figure shows the Impulse Response Function (IRF) for the response of CAPX, and R&D to a 25 bps increase in 1-year Treasury (instrumented), estimated using Equation (23) and controlling for Duration (Gonçalves, 2021). The 1-year Treasury rate is instrumented by cumulative high-frequency monetary policy shocks measured as monetary policy shocks from the decomposition by Jarociński and Karadi (2020). Each point represents the point estimate of the coefficient of the instrumented 1-year Treasury rate interacted with the financing constraint measure (β_2^h in Equation (23)). The dashed line represents 90% confidence intervals using heteroscedasticity and autocorrelation robust Driscoll-Kraay standard errors.

Appendix Figure G.5. Equity-Focused Constraints and the Dynamic Response of Investment to Monetary Policy: Controlling for Duration (Hoberg and Maksimovic, 2022)

(A) CAPX



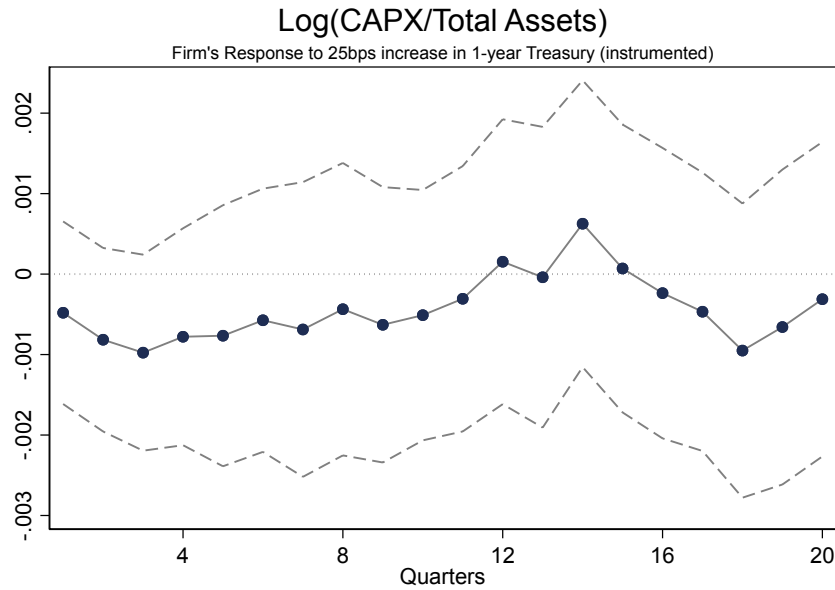
(B) R&D



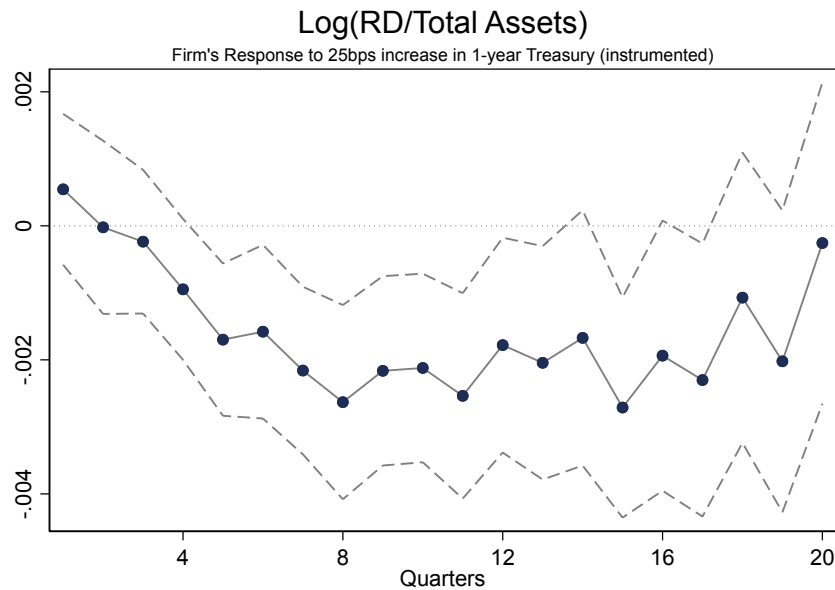
This figure shows the Impulse Response Function (IRF) for the response of CAPX, and R&D to a 25 bps increase in 1-year Treasury (instrumented), estimated using Equation (23) and controlling for Duration (Hoberg and Maksimovic, 2022). The 1-year Treasury rate is instrumented by cumulative high-frequency monetary policy shocks measured as monetary policy shocks from the decomposition by Jarociński and Karadi (2020). Each point represents the point estimate of the coefficient of the instrumented 1-year Treasury rate interacted with the financing constraint measure (β_2^h in Equation (23)). The dashed line represents 90% confidence intervals using heteroscedasticity and autocorrelation robust Driscoll-Kraay standard errors.

Appendix Figure G.6. Debt-Focused Constraints and the Dynamic Response of Investment to Monetary Policy: Controlling for Duration (Gonçalves, 2021)

(A) CAPX



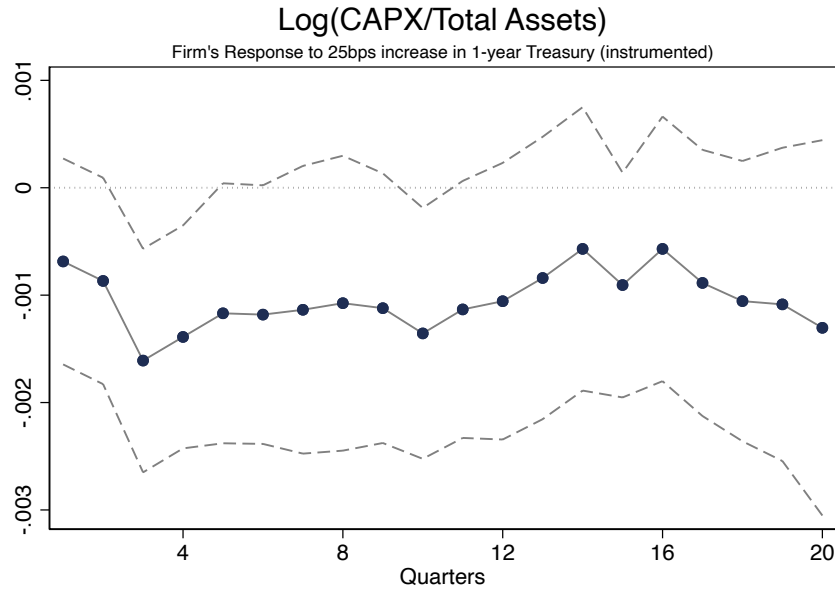
(B) R&D



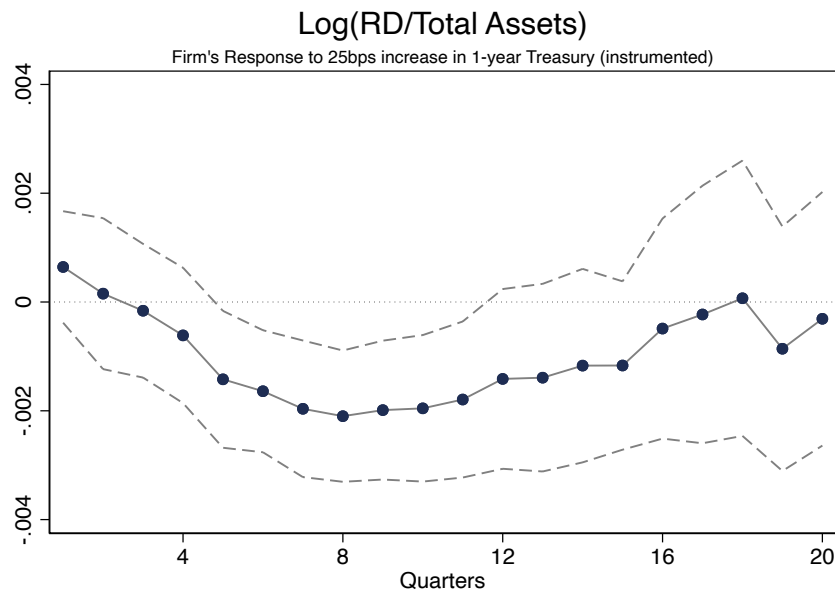
This figure shows the Impulse Response Function (IRF) for the response of CAPX, and R&D to a 25 bps increase in 1-year Treasury (instrumented), estimated using Equation (23) and controlling for Duration (Gonçalves, 2021). The 1-year Treasury rate is instrumented by cumulative high-frequency monetary policy shocks measured as monetary policy shocks from the decomposition by Jarociński and Karadi (2020). Each point represents the point estimate of the coefficient of the instrumented 1-year Treasury rate interacted with the financing constraint measure $FCD_{ij,t-1}$. The dashed line represents 90% confidence intervals using heteroscedasticity and autocorrelation robust Driscoll-Kraay standard errors.

Appendix Figure G.7. Debt-Focused Constraints and the Dynamic Response of Investment to Monetary Policy: Controlling for Duration (Hoberg and Maksimovic, 2022)

(A) CAPX

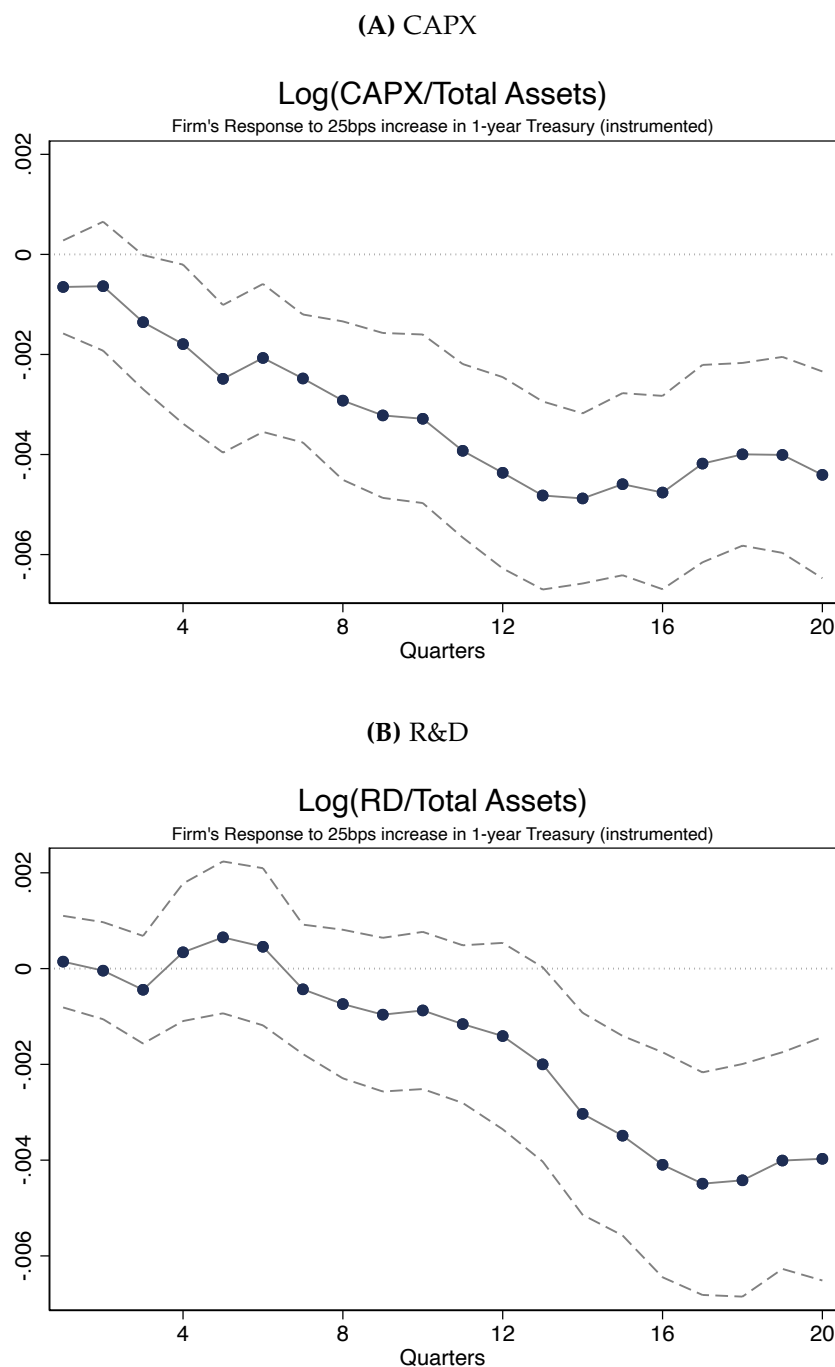


(B) R&D



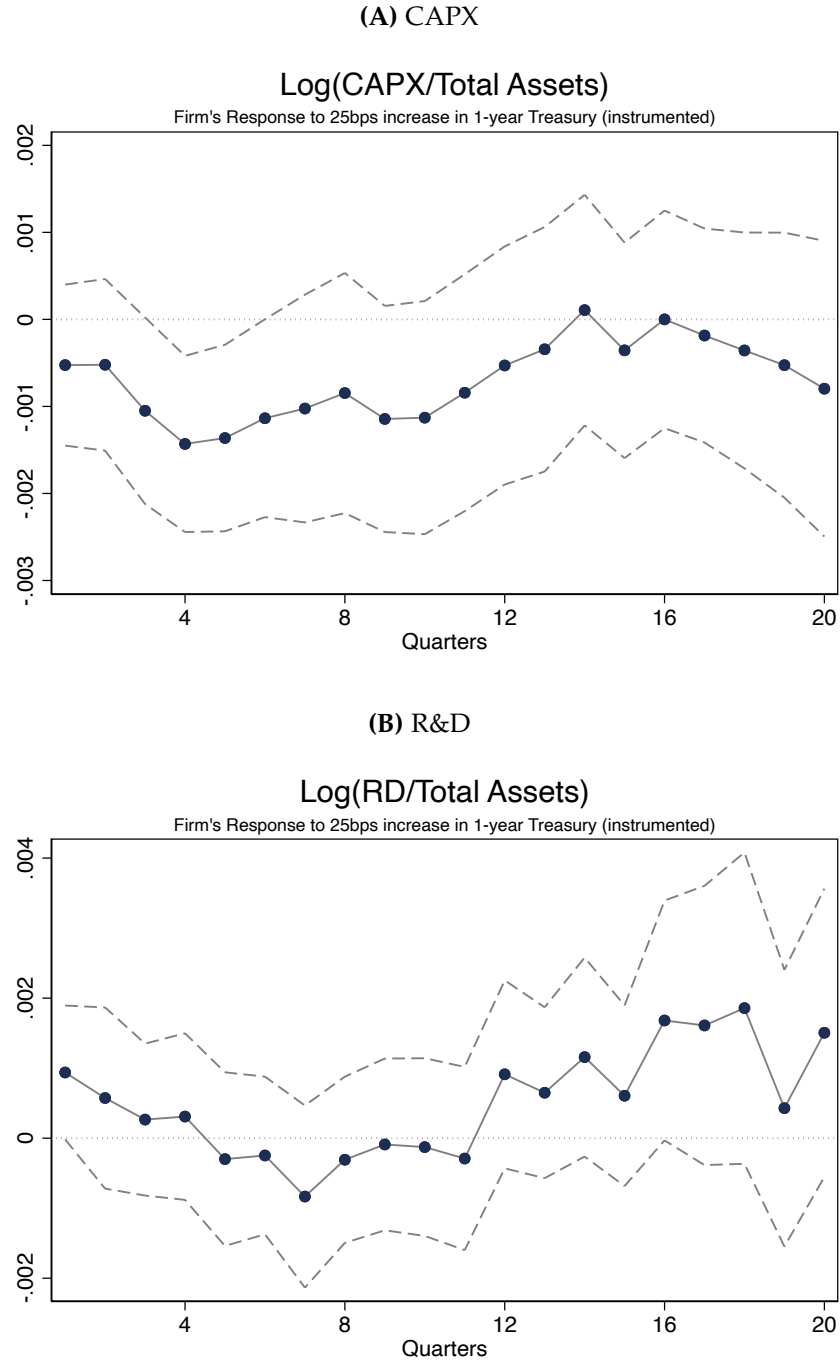
This figure shows the Impulse Response Function (IRF) for the response of CAPX, and R&D to a 25 bps increase in 1-year Treasury (instrumented), estimated using Equation (23) and controlling for Duration (Hoberg and Maksimovic, 2022). The 1-year Treasury rate is instrumented by cumulative high-frequency monetary policy shocks measured as monetary policy shocks from the decomposition by Jarociński and Karadi (2020). Each point represents the point estimate of the coefficient of the instrumented 1-year Treasury rate interacted with the financing constraint measure $FCD_{ij,t-1}$. The dashed line represents 90% confidence intervals using heteroscedasticity and autocorrelation robust Driscoll-Kraay standard errors.

Appendix Figure G.8. Equity-Focused Constraints and the Dynamic Response of Investment to Monetary Policy: Controlling for Z-score (Altman et al., 2021)



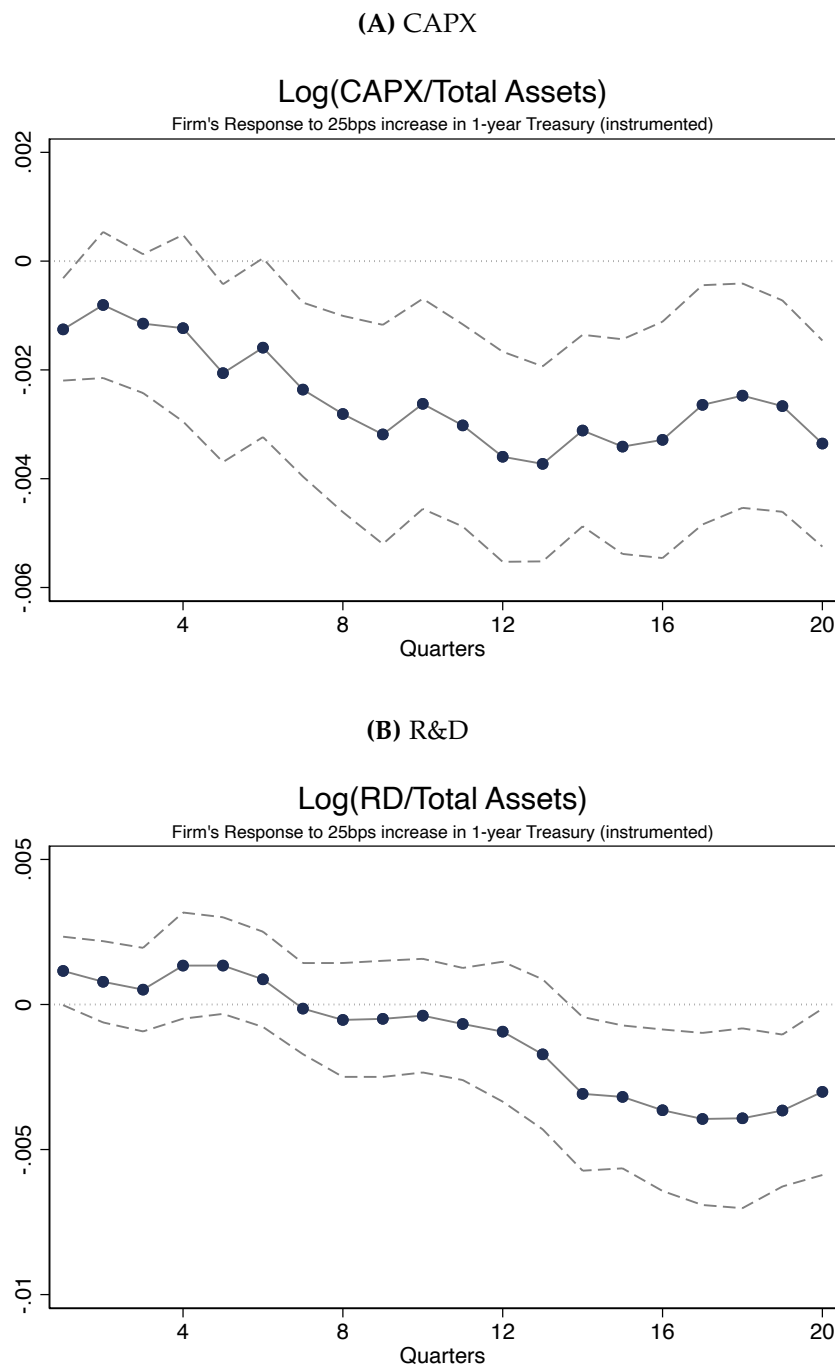
This figure shows the Impulse Response Function (IRF) for the response of CAPX, and R&D to a 25 bps increase in 1-year Treasury (instrumented), estimated using Equation (23) and controlling for Z-score (Altman et al., 2021). The 1-year Treasury rate is instrumented by cumulative high-frequency monetary policy shocks measured as monetary policy shocks from the decomposition by Jarociński and Karadi (2020). Each point represents the point estimate of the coefficient of the instrumented 1-year Treasury rate interacted with the financing constraint measure (β_2^h in Equation (23)). The dashed line represents 90% confidence intervals using heteroscedasticity and autocorrelation robust Driscoll-Kraay standard errors.

Appendix Figure G.9. Debt-Focused Constraints and the Dynamic Response of Investment to Monetary Policy: Controlling for Z-score (Altman et al., 2021)



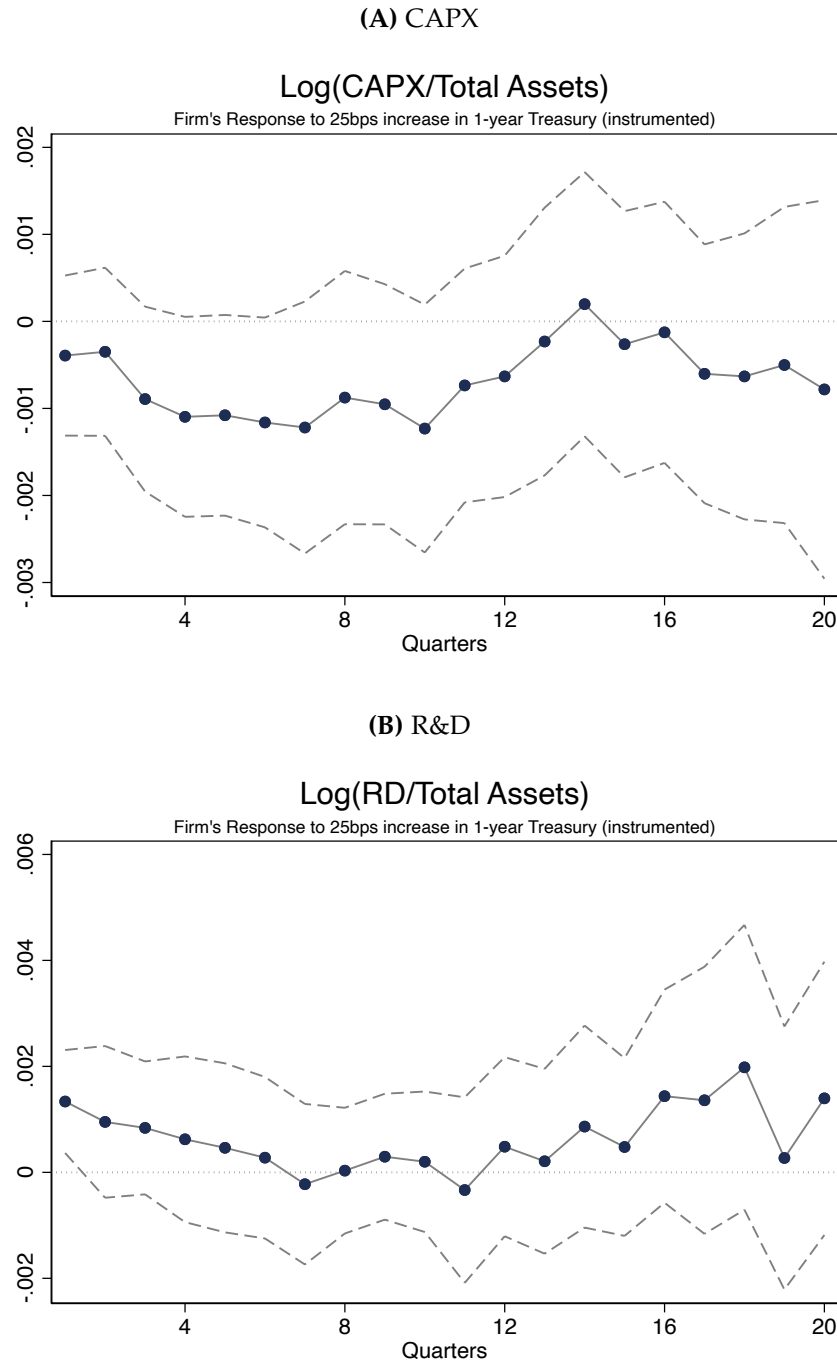
This figure shows the Impulse Response Function (IRF) for the response of CAPX, and R&D to a 25 bps increase in 1-year Treasury (instrumented), estimated using Equation (23) and controlling for Z-score (Altman et al., 2021). The 1-year Treasury rate is instrumented by cumulative high-frequency monetary policy shocks measured as monetary policy shocks from the decomposition by Jarociński and Karadi (2020). Each point represents the point estimate of the coefficient of the instrumented 1-year Treasury rate interacted with the financing constraint measure $FCD_{ij,t-1}$. The dashed line represents 90% confidence intervals using heteroscedasticity and autocorrelation robust Driscoll-Kraay standard errors.

Appendix Figure G.10. Equity-Focused Constraints and the Dynamic Response of Investment to Monetary Policy: Controlling for Refinancing Constraints



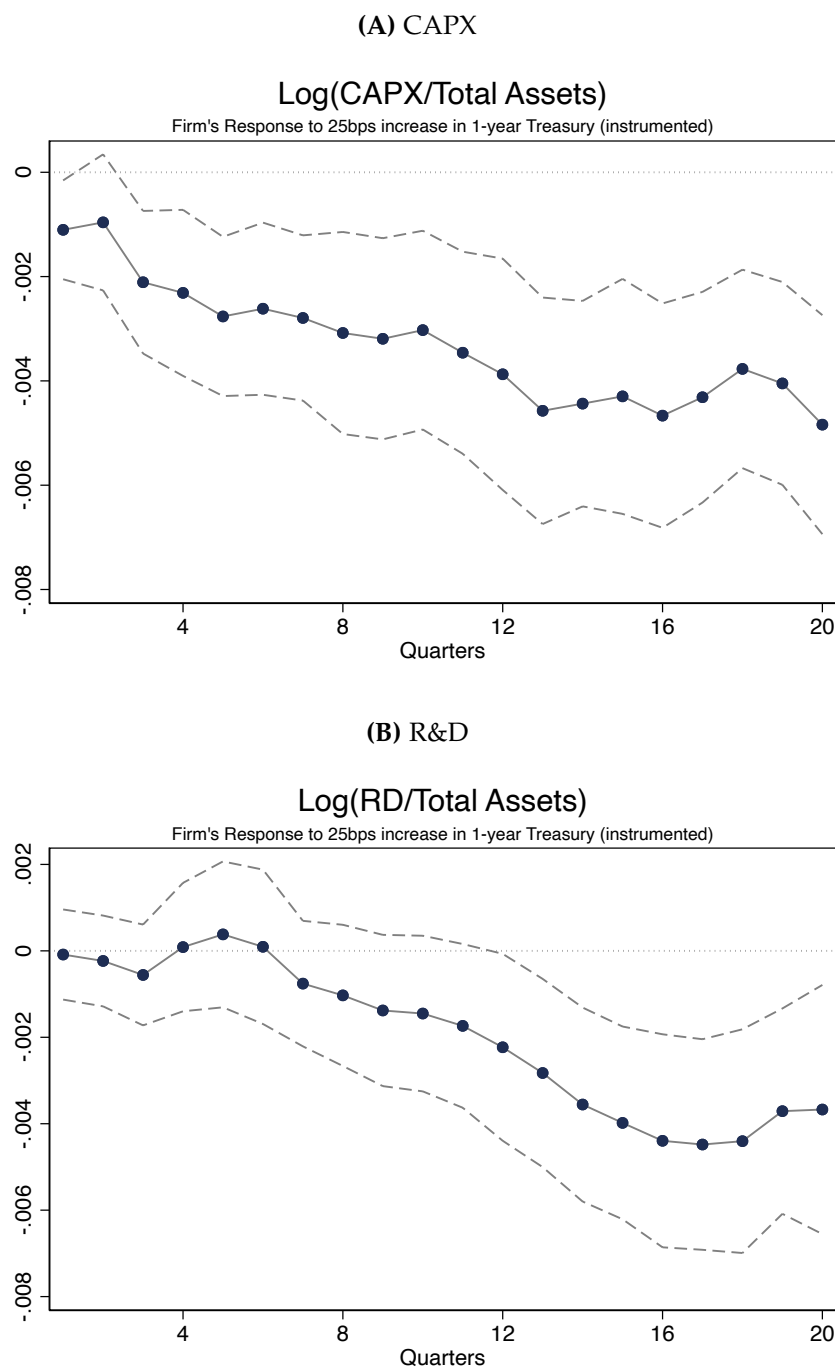
This figure shows the Impulse Response Function (IRF) for the response of CAPX, and R&D to a 25 bps increase in 1-year Treasury (instrumented), estimated using Equation (23) and controlling for refinancing constraints (see Subsection G.5). The 1-year Treasury rate is instrumented by cumulative high-frequency monetary policy shocks measured as monetary policy shocks from the decomposition by Jarociński and Karadi (2020). Each point represents the point estimate of the coefficient of the instrumented 1-year Treasury rate interacted with the financing constraint measure (β_2^h in Equation (23)). The dashed line represents 90% confidence intervals using heteroscedasticity and autocorrelation robust Driscoll-Kraay standard errors.

Appendix Figure G.11. Debt-Focused Constraints and the Dynamic Response of Investment to Monetary Policy: Controlling for Refinancing Constraints



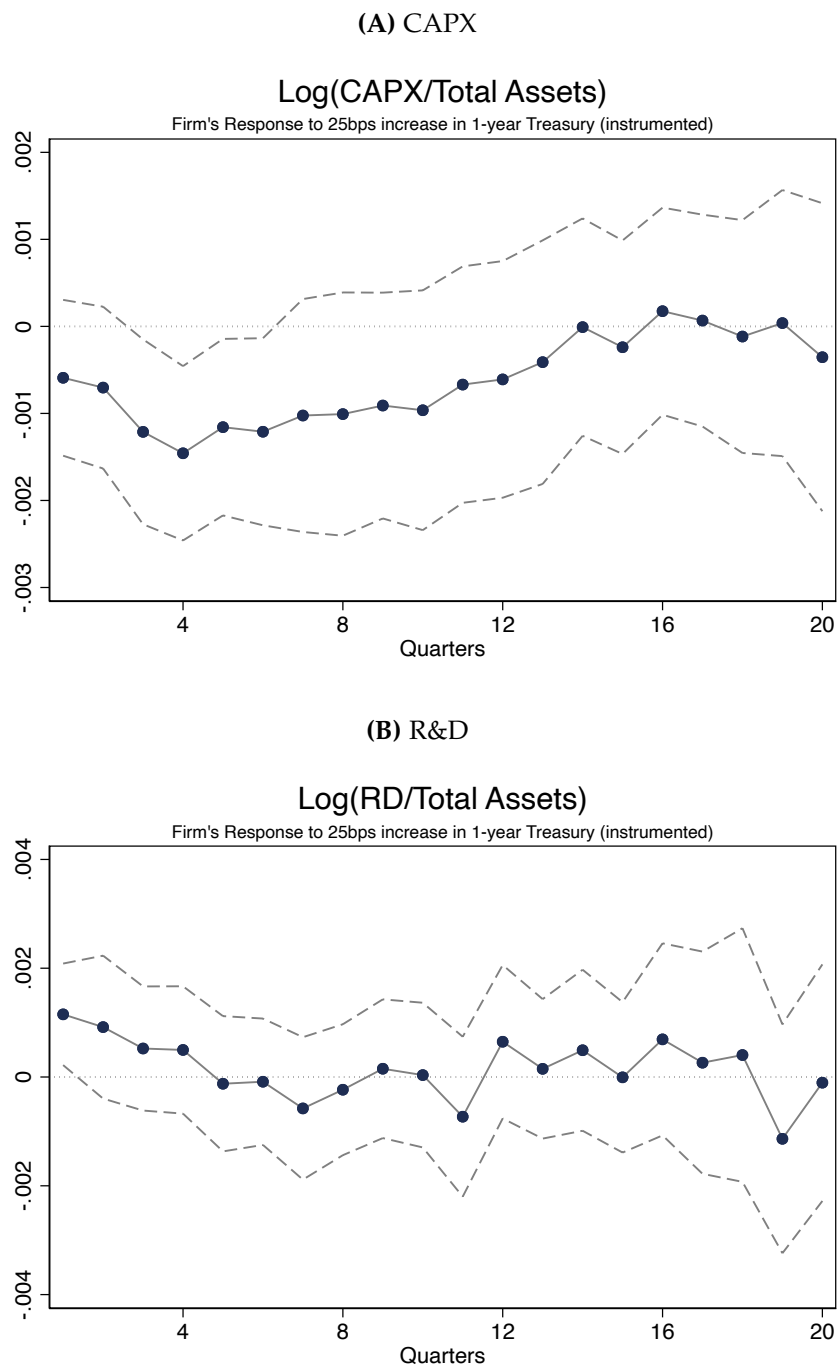
This figure shows the Impulse Response Function (IRF) for the response of CAPX, and R&D to a 25 bps increase in 1-year Treasury (instrumented), estimated using Equation (23) and controlling for refinancing constraints (see Subsection G.5). The 1-year Treasury rate is instrumented by cumulative high-frequency monetary policy shocks measured as monetary policy shocks from the decomposition by Jarociński and Karadi (2020). Each point represents the point estimate of the coefficient of the instrumented 1-year Treasury rate interacted with the financing constraint measure $FCD_{ij,t-1}$. The dashed line represents 90% confidence intervals using heteroscedasticity and autocorrelation robust Driscoll-Kraay standard errors.

Appendix Figure G.12. Equity-Focused Constraints and the Dynamic Response of Investment to Monetary Policy Shocks from [Bauer and Swanson \(2023\)](#)



This figure shows the Impulse Response Function (IRF) for the response of CAPX, and R&D to a 25 bps increase in 1-year Treasury (instrumented), estimated using Equation (23). The 1-year Treasury rate is instrumented by cumulative high-frequency monetary policy shocks measured as monetary policy shocks from [Bauer and Swanson \(2023\)](#). Each point represents the point estimate of the coefficient of the instrumented 1-year Treasury rate interacted with the financing constraint measure (β_2^h in Equation (23)). The dashed line represents 90% confidence intervals using heteroscedasticity and autocorrelation robust Driscoll-Kraay standard errors.

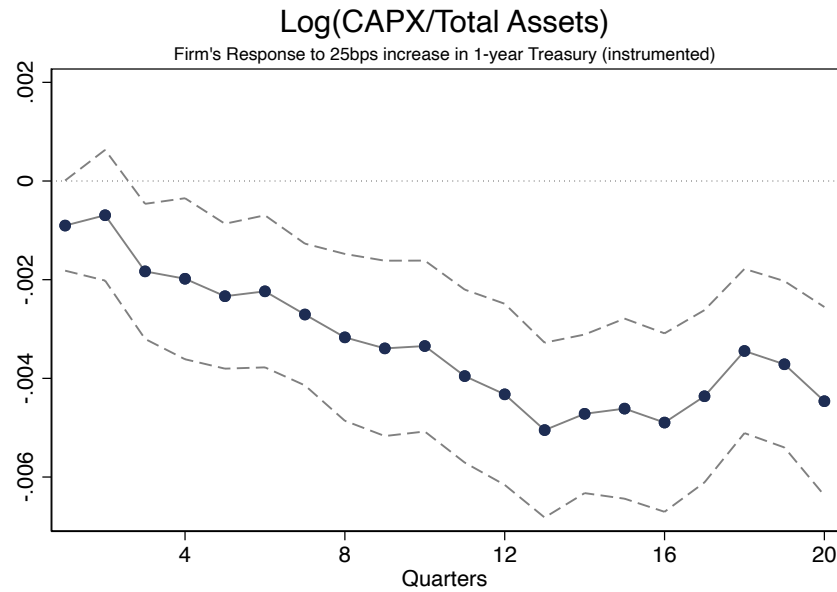
Appendix Figure G.13. Debt-Focused Constraints and the Dynamic Response of Investment to Monetary Policy Shocks from [Bauer and Swanson \(2023\)](#)



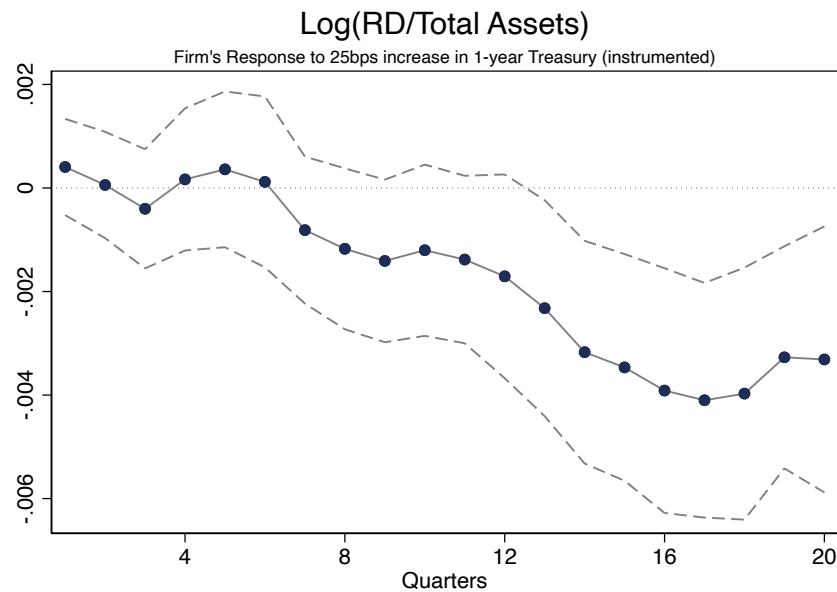
This figure shows the Impulse Response Function (IRF) for the response of CAPX, and R&D to a 25 bps increase in 1-year Treasury (instrumented), estimated using Equation (23). The 1-year Treasury rate is instrumented by cumulative high-frequency monetary policy shocks measured as monetary policy shocks from [Bauer and Swanson \(2023\)](#). Each point represents the point estimate of the coefficient of the instrumented 1-year Treasury rate interacted with the financing constraint measure $FCD_{ij,t-1}$. The dashed line represents 90% confidence intervals using heteroscedasticity and autocorrelation robust Driscoll-Kraay standard errors.

Appendix Figure G.14. Equity-Focused Constraints and the Dynamic Response of Investment to Monetary Policy: Controlling for Cyclical

(A) CAPX

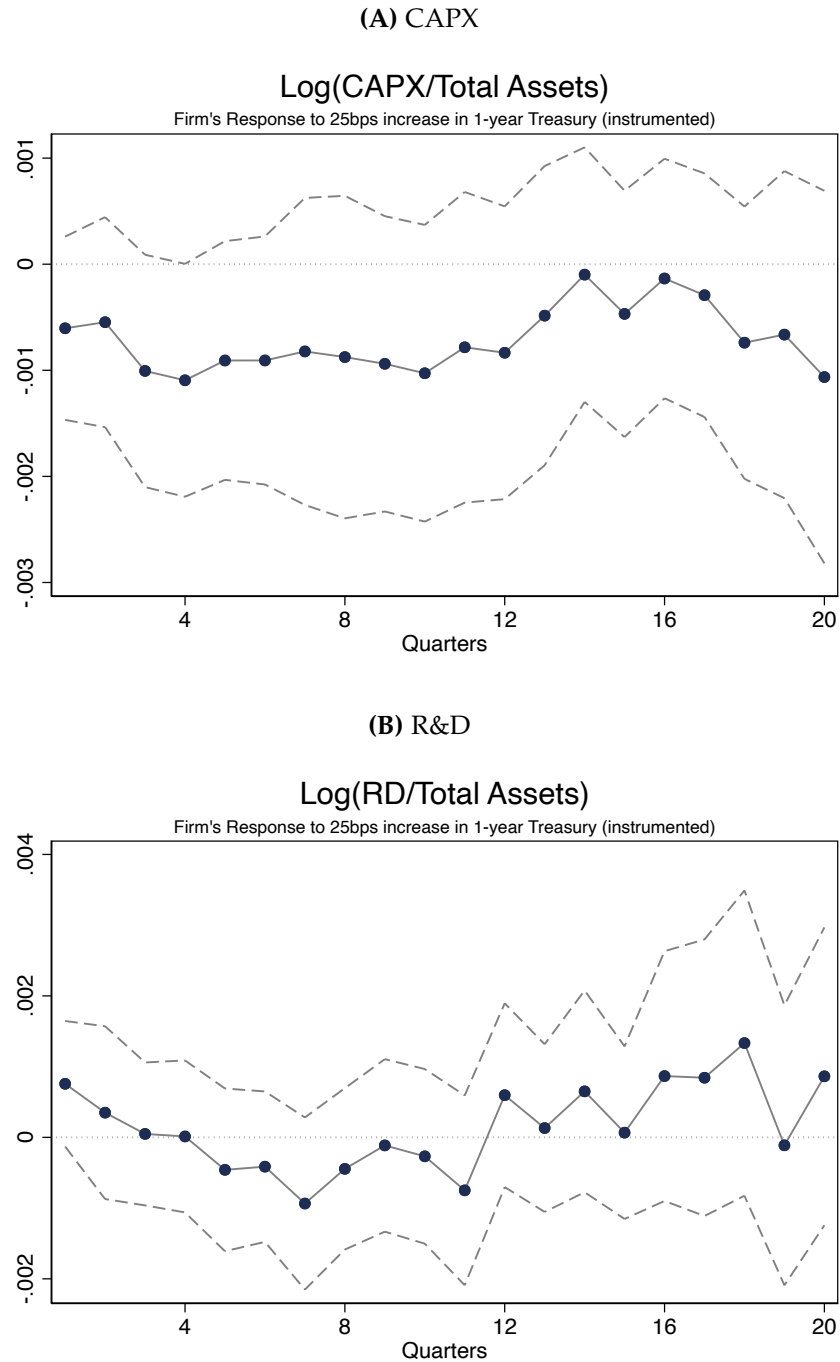


(B) R&D



This figure shows the Impulse Response Function (IRF) for the response of CAPX, and R&D to a 25 bps increase in 1-year Treasury (instrumented), estimated using Equation (23) and controlling for cyclical. The 1-year Treasury rate is instrumented by cumulative high-frequency monetary policy shocks measured as monetary policy shocks from the decomposition by Jarociński and Karadi (2020). Each point represents the point estimate of the coefficient of the instrumented 1-year Treasury rate interacted with the financing constraint measure (β_2^h in Equation (23)). The dashed line represents 90% confidence intervals using heteroscedasticity and autocorrelation robust Driscoll-Kraay standard errors.

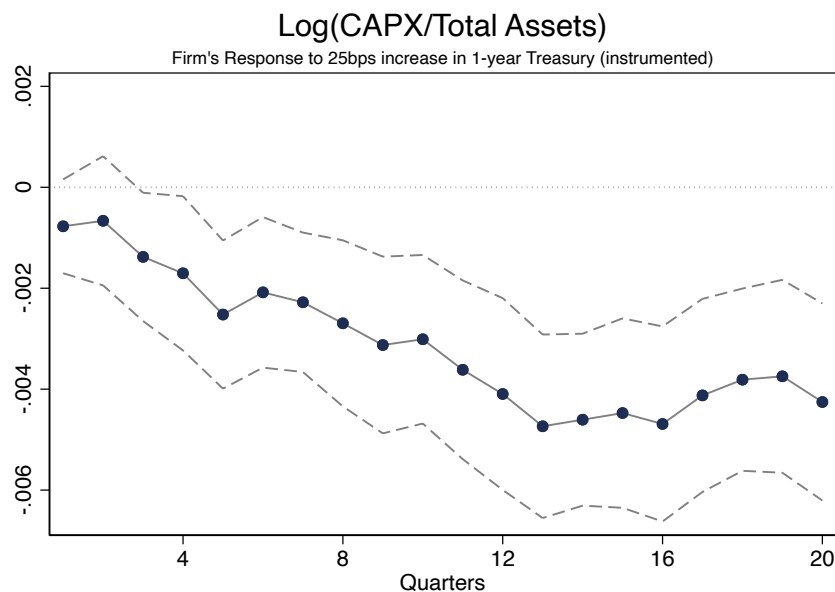
Appendix Figure G.15. Debt-Focused Constraints and the Dynamic Response of Investment to Monetary Policy: Controlling for Cyclical



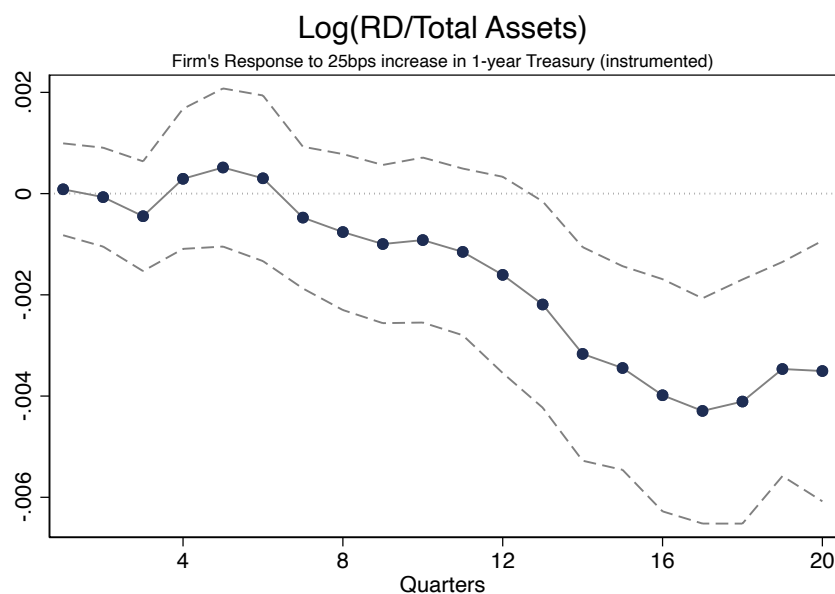
This figure shows the Impulse Response Function (IRF) for the response of CAPX, and R&D to a 25 bps increase in 1-year Treasury (instrumented), estimated using Equation (23) and controlling for cyclical. The 1-year Treasury rate is instrumented by cumulative high-frequency monetary policy shocks measured as monetary policy shocks from the decomposition by [Jarociński and Karadi \(2020\)](#). Each point represents the point estimate of the coefficient of the instrumented 1-year Treasury rate interacted with the financing constraint measure $FCD_{ij,t-1}$. The dashed line represents 90% confidence intervals using heteroscedasticity and autocorrelation robust Driscoll-Kraay standard errors.

Appendix Figure G.16. Equity-Focused Constraints and the Dynamic Response of Investment to Monetary Policy: Controlling for the Information Effect

(A) CAPX



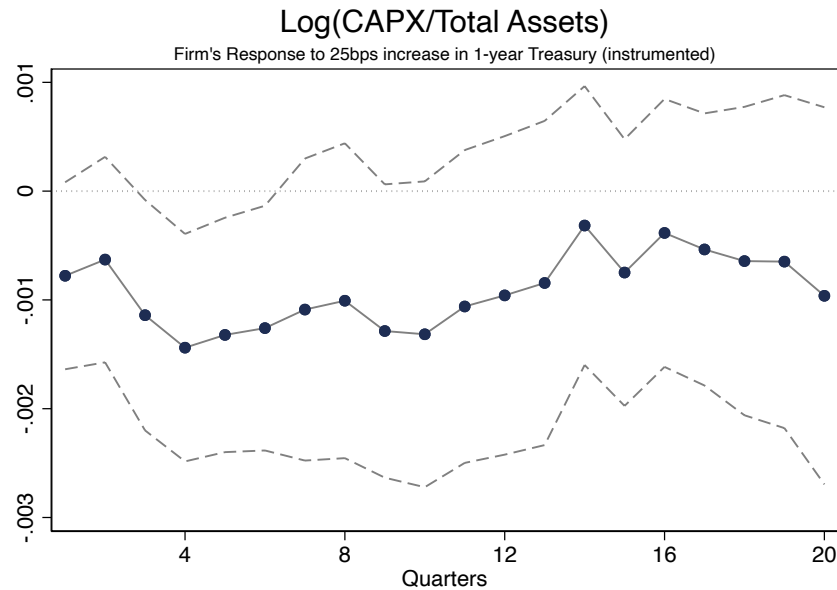
(B) R&D



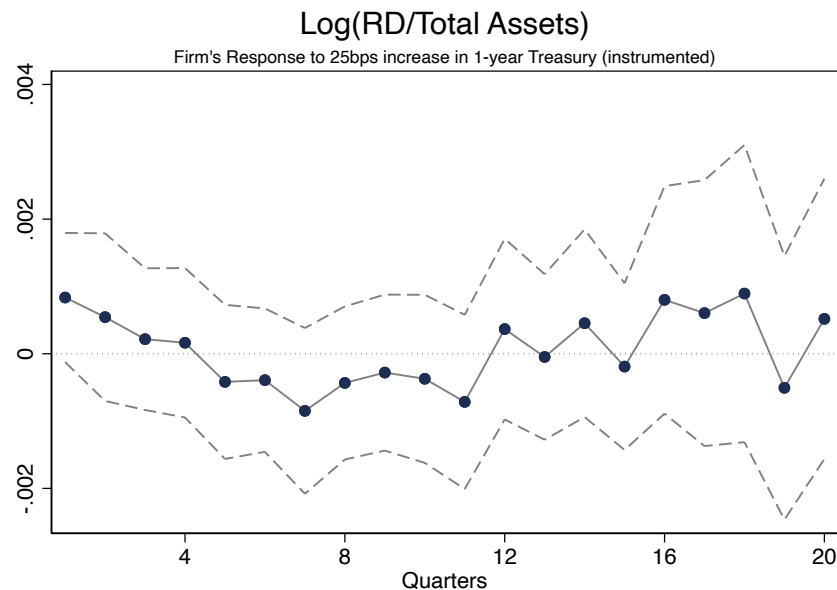
This figure shows the Impulse Response Function (IRF) for the response of CAPX, and R&D to a 25 bps increase in 1-year Treasury (instrumented), estimated using Equation (23) and controlling for the information effect. The 1-year Treasury rate is instrumented by cumulative high-frequency monetary policy shocks measured as monetary policy shocks from the decomposition by Jarociński and Karadi (2020). Each point represents the point estimate of the coefficient of the instrumented 1-year Treasury rate interacted with the financing constraint measure (β_2^h in Equation (23)). The dashed line represents 90% confidence intervals using heteroscedasticity and autocorrelation robust Driscoll-Kraay standard errors.

Appendix Figure G.17. Debt-Focused Constraints and the Dynamic Response of Investment to Monetary Policy: Controlling for the Information Effect

(A) CAPX

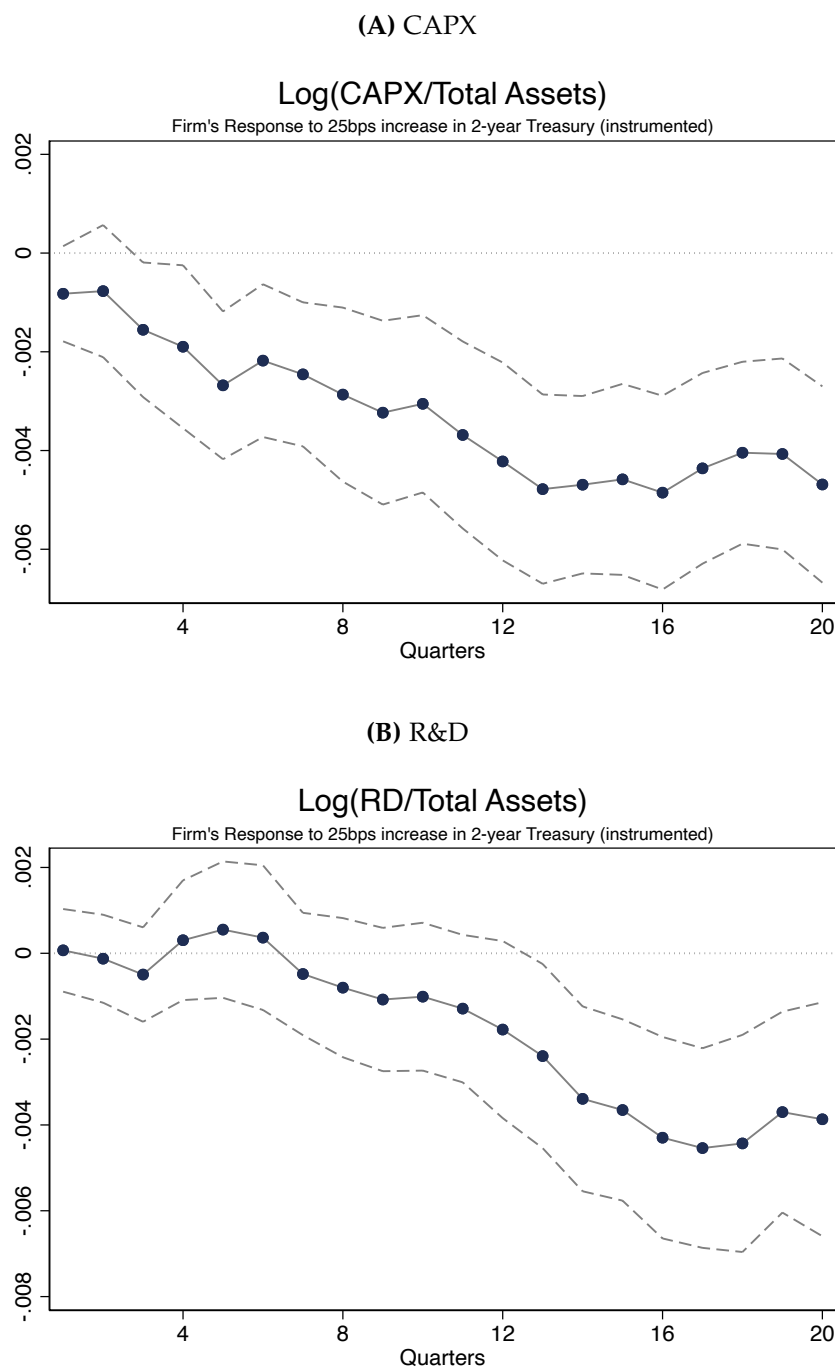


(B) R&D



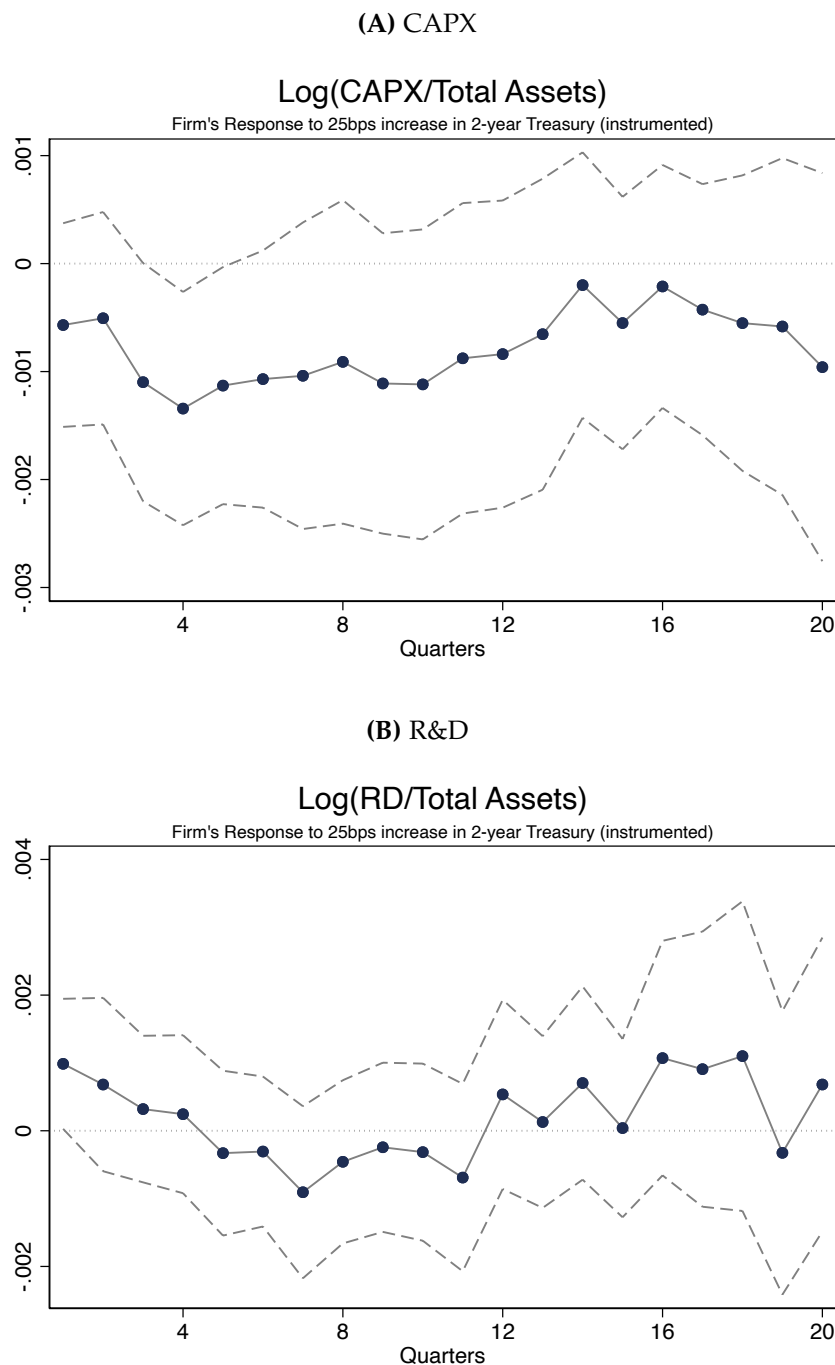
This figure shows the Impulse Response Function (IRF) for the response of CAPX, and R&D to a 25 bps increase in 1-year Treasury (instrumented), estimated using Equation (23) and controlling for the information effect. The 1-year Treasury rate is instrumented by cumulative high-frequency monetary policy shocks measured as monetary policy shocks from the decomposition by [Jarociński and Karadi \(2020\)](#). Each point represents the point estimate of the coefficient of the instrumented 1-year Treasury rate interacted with the financing constraint measure $FCD_{ij,t-1}$. The dashed line represents 90% confidence intervals using heteroscedasticity and autocorrelation robust Driscoll-Kraay standard errors.

Appendix Figure G.18. Equity-Focused Constraints and the Dynamic Response of Investment to Monetary Policy Shocks: 2-year Treasury Rate



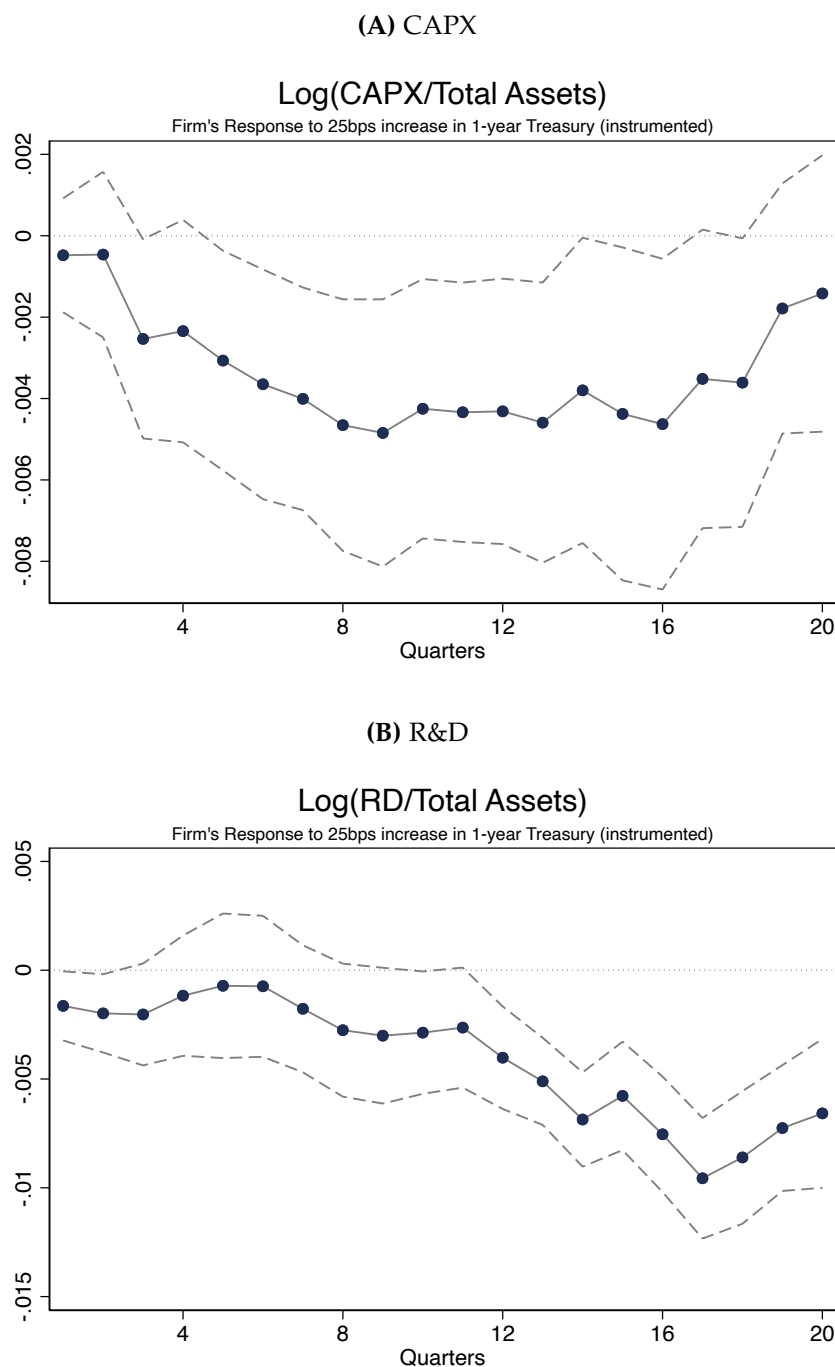
This figure shows the Impulse Response Function (IRF) for the response of CAPX, and R&D to a 25 bps increase in 2-year Treasury (instrumented), estimated using Equation (23). The 2-year Treasury rate is instrumented by cumulative high-frequency monetary policy shocks measured as monetary policy shocks from [Jarociński and Karadi \(2020\)](#). Each point represents the point estimate of the coefficient of the instrumented 2-year Treasury rate interacted with the financing constraint measure (β_2^h in Equation (23)). The dashed line represents 90% confidence intervals using heteroscedasticity and autocorrelation robust Driscoll-Kraay standard errors.

Appendix Figure G.19. Debt-Focused Constraints and the Dynamic Response of Investment to Monetary Policy Shocks: 2-year Treasury Rate



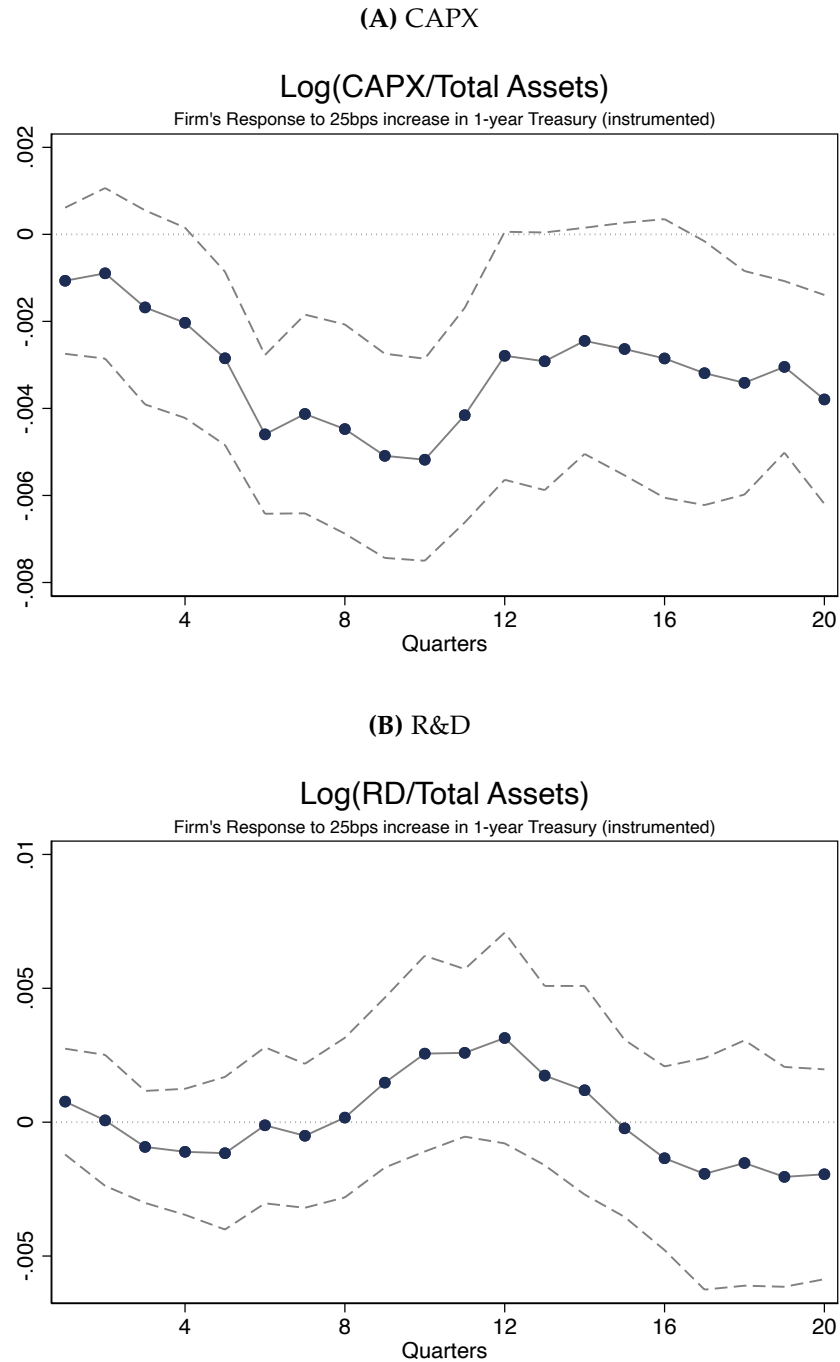
This figure shows the Impulse Response Function (IRF) for the response of CAPX, and R&D to a 25 bps increase in 2-year Treasury (instrumented), estimated using Equation (23). The 2-year Treasury rate is instrumented by cumulative high-frequency monetary policy shocks measured as monetary policy shocks from [Jarociński and Karadi \(2020\)](#). Each point represents the point estimate of the coefficient of the instrumented 2-year Treasury rate interacted with the financing constraint measure $FCD_{ij,t-1}$. The dashed line represents 90% confidence intervals using heteroscedasticity and autocorrelation robust Driscoll-Kraay standard errors.

Appendix Figure G.20. Equity-Focused Constraints and the Dynamic Response of Investment to Monetary Policy: Excluding the ZLB



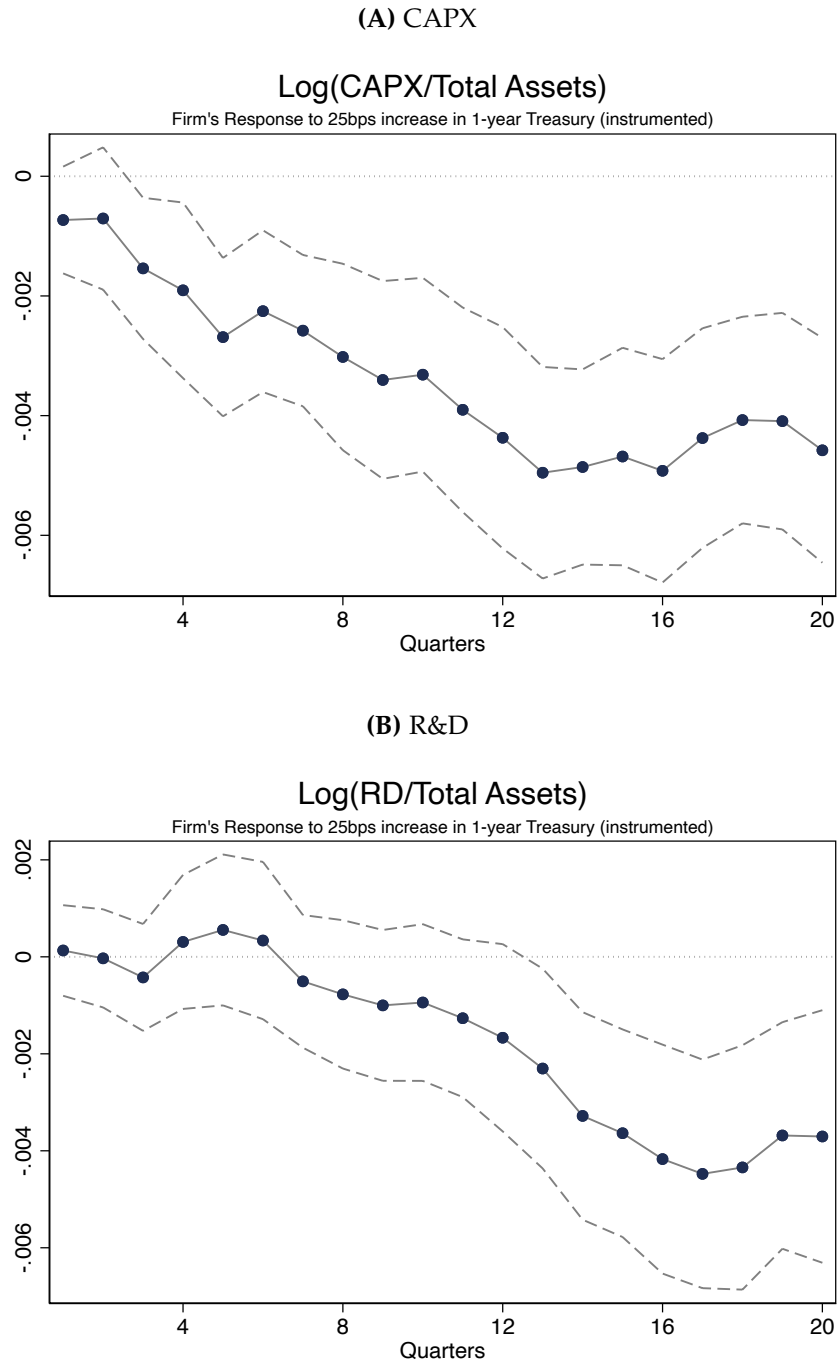
This figure shows the Impulse Response Function (IRF) for the response of CAPX, and R&D to a 25 bps increase in 1-year Treasury (instrumented), estimated using Equation (23) and excluding the zero lower bound (ZLB) period (January 2009 to December 2015). The 1-year Treasury rate is instrumented by cumulative high-frequency monetary policy shocks measured as monetary policy shocks from the decomposition by [Jarociński and Karadi \(2020\)](#). Each point represents the point estimate of the coefficient of the instrumented 1-year Treasury rate interacted with the financing constraint measure (β_2^h in Equation (23)). The dashed line represents 90% confidence intervals using heteroscedasticity and autocorrelation robust Driscoll-Kraay standard errors.

Appendix Figure G.21. Debt-Focused Constraints and the Dynamic Response of Investment to Monetary Policy: Excluding the ZLB



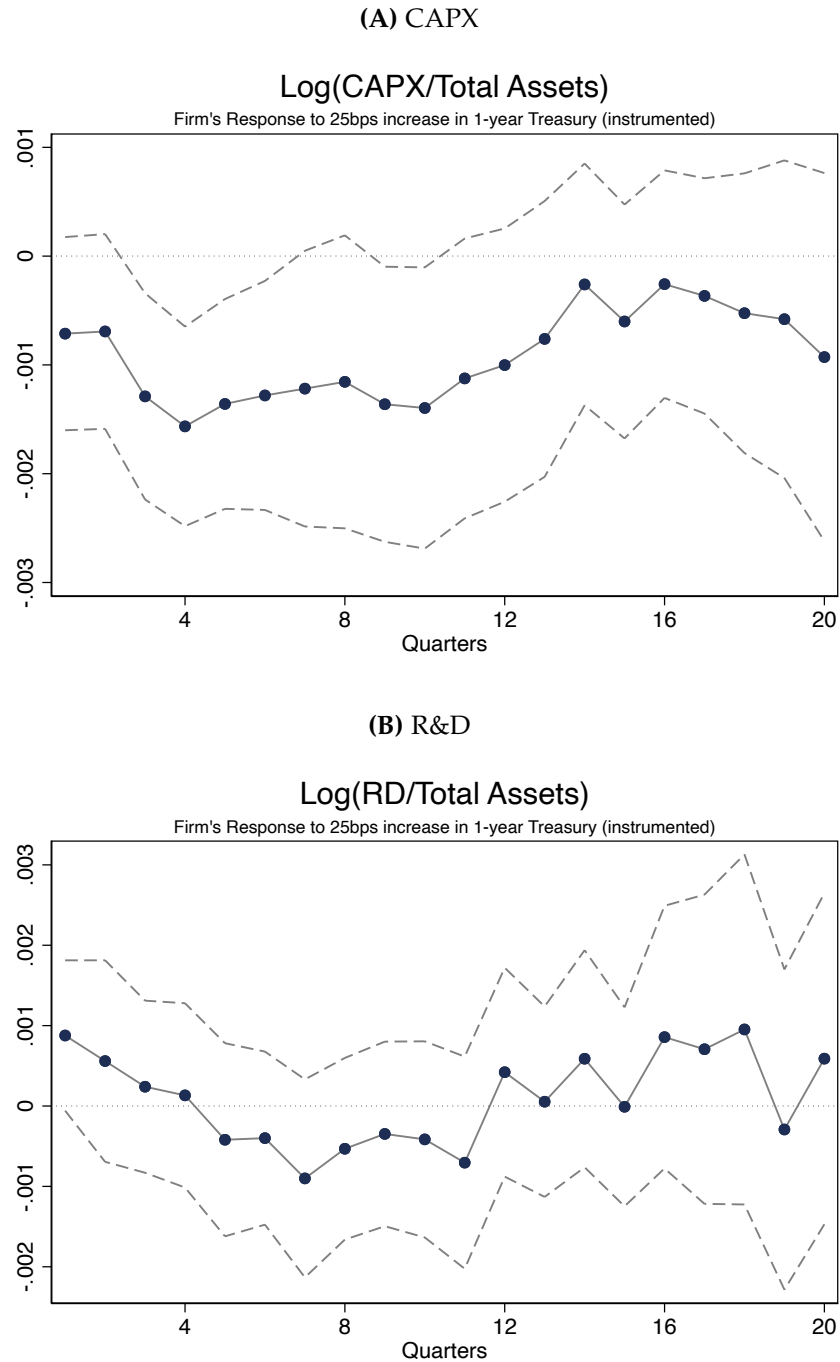
This figure shows the Impulse Response Function (IRF) for the response of CAPX, and R&D to a 25 bps increase in 1-year Treasury (instrumented), estimated using Equation (23) and excluding the zero lower bound (ZLB) period (January 2009 to December 2015). The 1-year Treasury rate is instrumented by cumulative high-frequency monetary policy shocks measured as monetary policy shocks from the decomposition by [Jarociński and Karadi \(2020\)](#). Each point represents the point estimate of the coefficient of the instrumented 1-year Treasury rate interacted with the financing constraint measure $FCD_{ij,t-1}$. The dashed line represents 90% confidence intervals using heteroscedasticity and autocorrelation robust Driscoll-Kraay standard errors.

Appendix Figure G.22. Equity-Focused Constraints and the Dynamic Response of Investment to Monetary Policy: Controlling for Intangibility



This figure shows the Impulse Response Function (IRF) for the response of CAPX, and R&D to a 25 bps increase in 1-year Treasury (instrumented), estimated using Equation (23) and controlling for intangibility. The 1-year Treasury rate is instrumented by cumulative high-frequency monetary policy shocks measured as monetary policy shocks from the decomposition by [Jarociński and Karadi \(2020\)](#). Each point represents the point estimate of the coefficient of the instrumented 1-year Treasury rate interacted with the financing constraint measure (β_2^h in Equation (23)). The dashed line represents 90% confidence intervals using heteroscedasticity and autocorrelation robust Driscoll-Kraay standard errors.

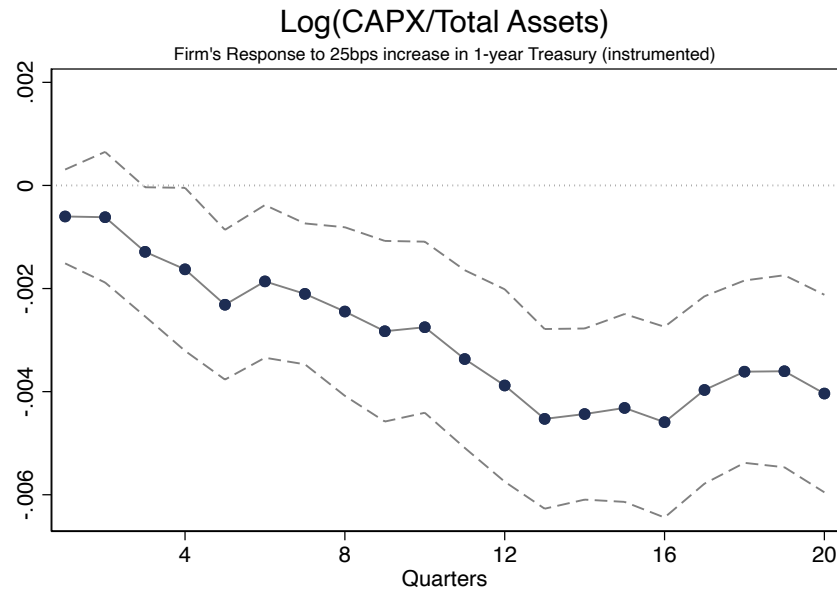
Appendix Figure G.23. Debt-Focused Constraints and the Dynamic Response of Investment to Monetary Policy: Controlling for Intangibility



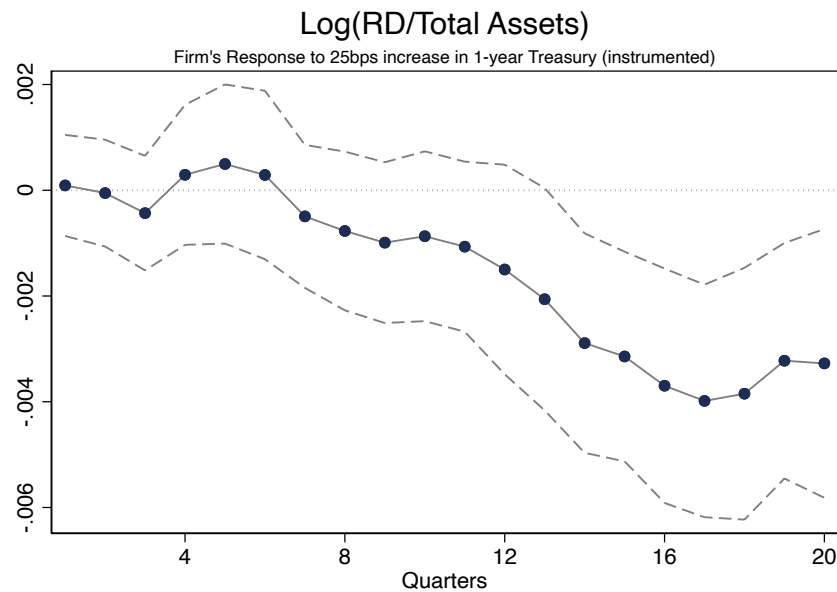
This figure shows the Impulse Response Function (IRF) for the response of CAPX, and R&D to a 25 bps increase in 1-year Treasury (instrumented), estimated using Equation (23) and controlling for intangibility. The 1-year Treasury rate is instrumented by cumulative high-frequency monetary policy shocks measured as monetary policy shocks from the decomposition by [Jarociński and Karadi \(2020\)](#). Each point represents the point estimate of the coefficient of the instrumented 1-year Treasury rate interacted with the financing constraint measure $FCD_{ij,t-1}$. The dashed line represents 90% confidence intervals using heteroscedasticity and autocorrelation robust Driscoll-Kraay standard errors.

Appendix Figure G.24. Equity-Focused Constraints and the Dynamic Response of Investment to Monetary Policy: Controlling for Equity Reliance

(A) CAPX

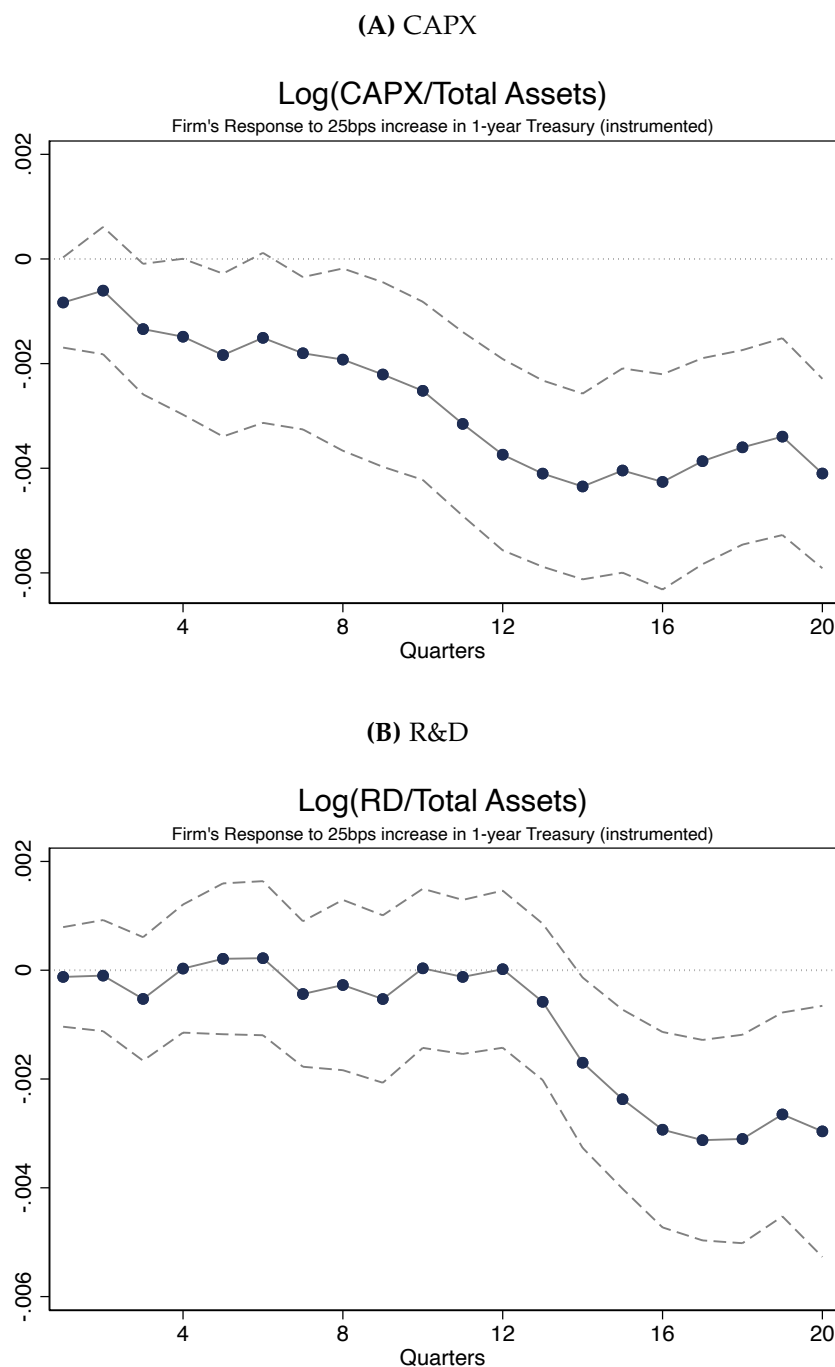


(B) R&D



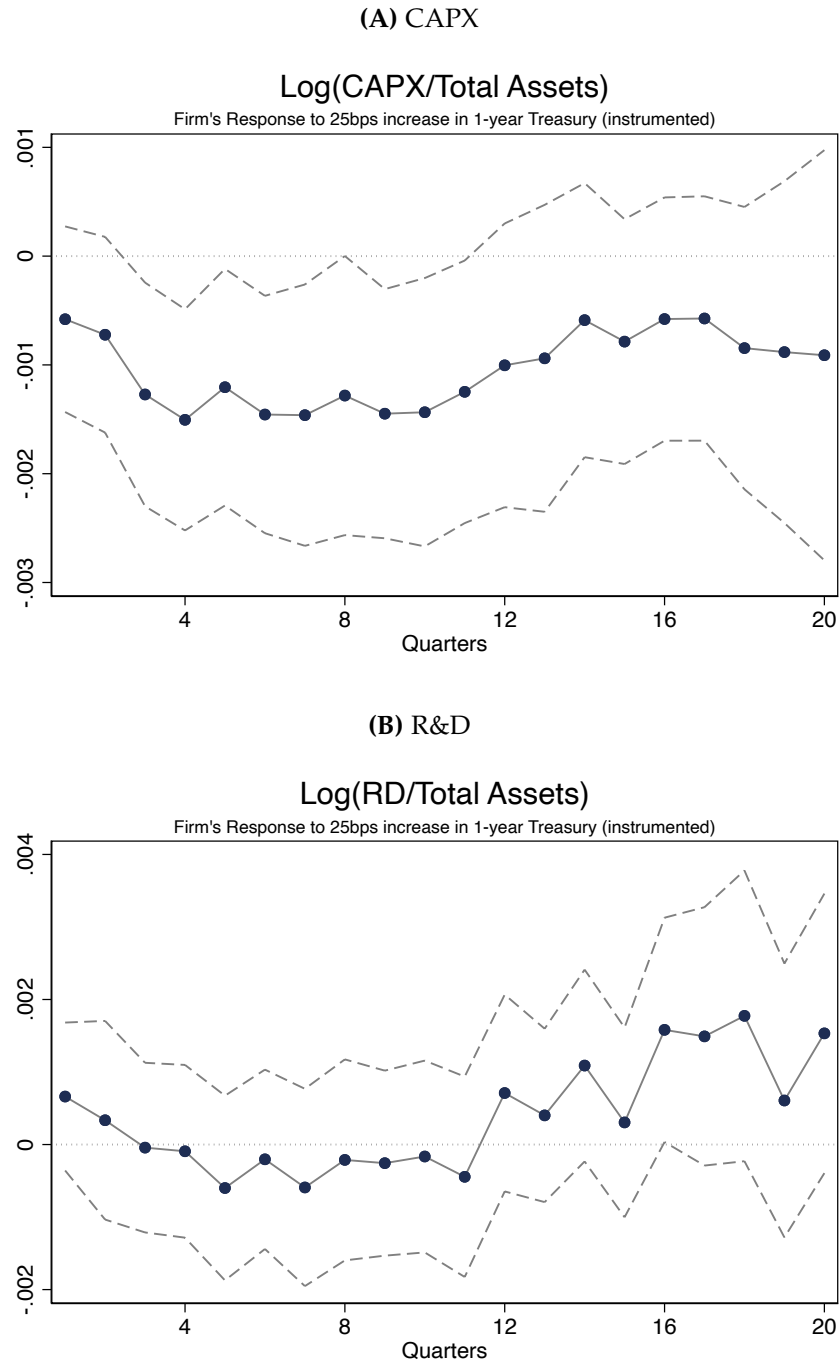
This figure shows the Impulse Response Function (IRF) for the response of CAPX, and R&D to a 25 bps increase in 1-year Treasury (instrumented), estimated using Equation (23) and controlling for equity reliance. The 1-year Treasury rate is instrumented by cumulative high-frequency monetary policy shocks measured as monetary policy shocks from the decomposition by [Jarociński and Karadi \(2020\)](#). Each point represents the point estimate of the coefficient of the instrumented 1-year Treasury rate interacted with the financing constraint measure (β_2^h in Equation (23)). The dashed line represents 90% confidence intervals using heteroscedasticity and autocorrelation robust Driscoll-Kraay standard errors.

Appendix Figure G.25. Equity-Focused Constraints and the Dynamic Response of Investment to Monetary Policy: Controlling for Inflexibility



This figure shows the Impulse Response Function (IRF) for the response of CAPX, and R&D to a 25 bps increase in 1-year Treasury (instrumented), estimated using Equation (23) and controlling for inflexibility. The 1-year Treasury rate is instrumented by cumulative high-frequency monetary policy shocks measured as monetary policy shocks from the decomposition by [Jarociński and Karadi \(2020\)](#). Each point represents the point estimate of the coefficient of the instrumented 1-year Treasury rate interacted with the financing constraint measure (β_2^h in Equation (23)). The dashed line represents 90% confidence intervals using heteroscedasticity and autocorrelation robust Driscoll-Kraay standard errors.

Appendix Figure G.26. Debt-Focused Constraints and the Dynamic Response of Investment to Monetary Policy: Controlling for Inflexibility



This figure shows the Impulse Response Function (IRF) for the response of CAPX, and R&D to a 25 bps increase in 1-year Treasury (instrumented), estimated using Equation (23) and controlling for inflexibility. The 1-year Treasury rate is instrumented by cumulative high-frequency monetary policy shocks measured as monetary policy shocks from the decomposition by [Jarociński and Karadi \(2020\)](#). Each point represents the point estimate of the coefficient of the instrumented 1-year Treasury rate interacted with the financing constraint measure $FCD_{ij,t-1}$. The dashed line represents 90% confidence intervals using heteroscedasticity and autocorrelation robust Driscoll-Kraay standard errors.

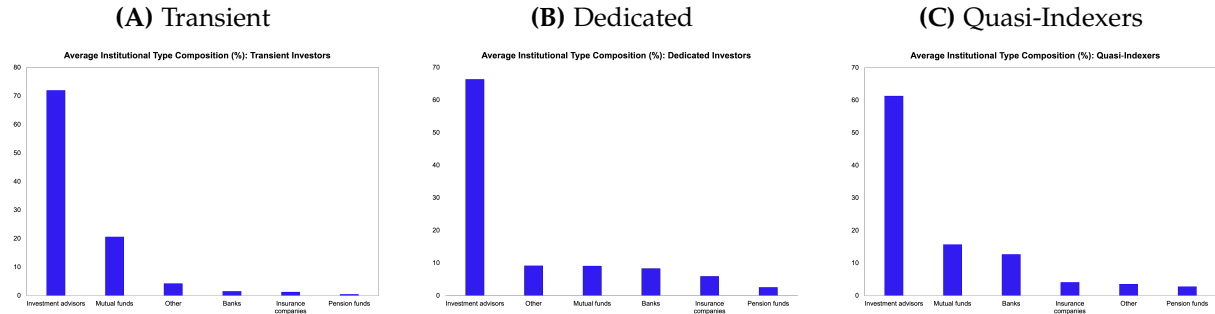
H. Institutional Investors Types

Appendix Table H.1. Institutional Investors by Average Portfolio Size

Panel A: Top 5 Transient institutional investors by average portfolio size		
Name	Type	Average Portfolio Size (\$ Billions)
Blackrock	Mutual Funds	721.13
Morgan Stanley D Witter	Mutual Funds	208.40
Janus Capital	Mutual Funds	111.67
Putnam Management	Mutual Funds	82.35
AIM Management	Mutual Funds	70.13
Panel B: Top 5 Dedicated institutional investors by average portfolio size		
Name	Type	Average Portfolio Size (\$ Billions)
Capital Research & Management	Investment Advisors	344.41
Berkshire Hathaway	Investment Advisors	78.84
Sanford Bernstein & Co	Mutual Funds	53.17
Bank of New York Asset Management	Banks	21.91
Charles Schwab Investment Advisory	Investment Advisors	18.97
Panel C: Top 5 Quasi-Indexer institutional investors by average portfolio size		
Name	Type	Average Portfolio Size (\$ Billions)
Vanguard Group	Mutual Funds	830.74
State Street Corporation	Banks	624.57
Capital World Investors	Investment advisors	323.16
T. Rowe Price Associates	Mutual Funds	287.20
Wellington Management	Mutual Funds	270.46

Panels (A), (B) and (C) of this table provide the list of transient, dedicated, and quasi-indexer institutional investors by average portfolio size. We classify institutional investor types based on the combination of portfolio turnover and holdings concentration. Investors are classified as “transient” if they have short investment horizons reflected by high portfolio turnover and highly diversified portfolio holdings. Analogously, “dedicated” investors have long investment horizons reflected by low portfolio turnover and focused portfolio holdings. The third class of investors, “quasi-indexers,” are long-horizon, low turnover investors that are highly diversified [Bushee \(1998, 2001\)](#); [Borochin and Yang \(2017\)](#). Average portfolio size is the mean total market value of an institutional investor’s equity holdings over the sample period, where the total market value in a given quarter is calculated as the sum of the market values of all individual stock positions reported by that investor in Form 13F filings.

Appendix Figure H.1. Average Institutional Type Composition



This figure shows the average fraction of managers in each investor category of [Bushee \(1998, 2001\)](#) (transient, dedicated, or quasi-indexer) accounted for by each institutional type, following the classification in [Kojien and Yogo \(2019\)](#). For each quarter, within a given Bushee category, we count the number of distinct managers of each institutional type. We then calculate the percentage share of each type in that category for the quarter and average these percentages across all quarters in the sample.

I. Construction of the Aggregate Transient Share

This appendix describes the construction of the aggregate transient share used in Section 5.2 to link the firm-level transient-ownership measure to the Flow of Funds evidence of Section 5.1.

Construction. The aggregate transient share is the market value of corporate equity held by transient institutional investors, divided by the total market value of corporate equities outstanding. Each 13F-filing institution is assigned the permanent investor classification of [Bushee \(1998, 2001\)](#) by manager-year. Each quarter, we take every transient-classified institution's end-of-quarter share positions from the LSEG (Thomson) 13F holdings data, keeping the earliest filing vintage per manager-quarter-position to avoid double counting amended filings, value each position at its end-of-quarter share price, and sum across positions and institutions. We divide the resulting aggregate by total corporate equities outstanding at market value from Flow of Funds Table F.51.1.s (formerly L.223), the same denominator underlying the price-sensitive shares of Section 5.1, so that

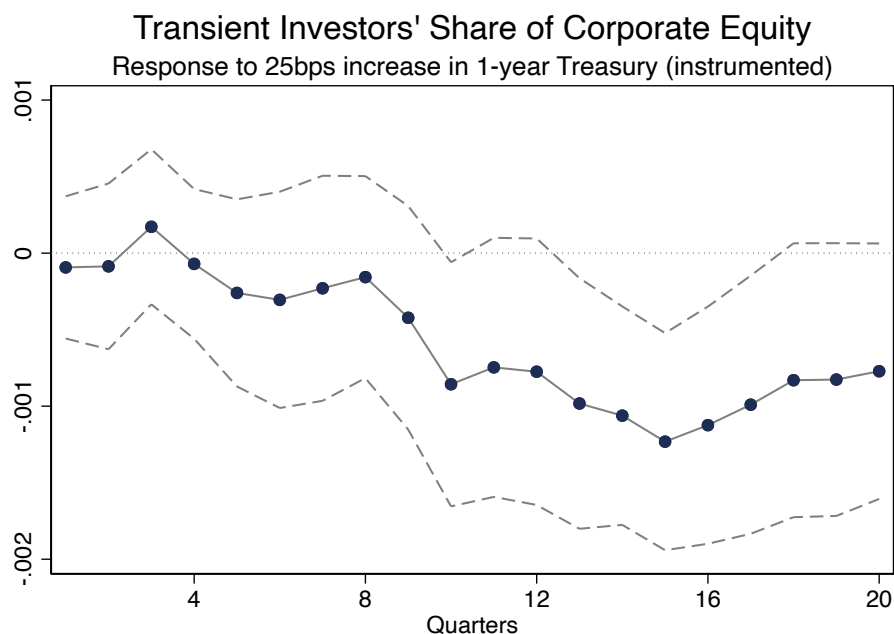
the transient share and the price-sensitive share are measured against the same base:

$$\text{TransientShare}_t = \frac{\sum_{m \in \text{TRA}} \sum_i \text{shares}_{m,i,t} \times p_{i,t}}{\text{Total corporate equities outstanding}_t},$$

where m indexes transient-classified institutions, i indexes stocks, and $p_{i,t}$ is the end-of-quarter price.

Response to monetary policy. Figure [I.1](#) reports the impulse response of the aggregate transient share to a 25 bps contractionary monetary policy shock, estimated from the same specification as Equation (19). The share declines steadily, reaching approximately -0.12 percentage points at $h = 15$ and statistically significant at the 90% level from $h \approx 10$. The profile mirrors the response of the price-sensitive equity share in Figure [2](#): both measures capture the same withdrawal of price-responsive investors from the equity market. Relative to its sample mean, the transient share contracts roughly twice as strongly as the broader price-sensitive share, consistent with transient institutions constituting the most price-sensitive segment of the marginal investor.

Appendix Figure I.1. Response of the Aggregate Transient Share of Corporate Equity to Monetary Policy



This figure shows the impulse response of the aggregate transient share of corporate equity to a 25 bps contractionary monetary policy shock, estimated from Equation (19) at horizons $h = 1, \dots, 20$ quarters. The dependent variable is the change in aggregate transient institutional holdings (LSEG 13F, Bushee classification) divided by total corporate equities outstanding at market value (Flow of Funds Table F.51.1.s), from quarter $t - 1$ to quarter $t + h$. The 1-year Treasury rate is instrumented by cumulative high-frequency monetary policy shocks from [Jarociński and Karadi \(2020\)](#). Controls include lagged log CPI, log industrial production, excess bond premium, and log employment ratio. Dashed lines are 90% confidence intervals based on robust Driscoll-Kraay standard errors.